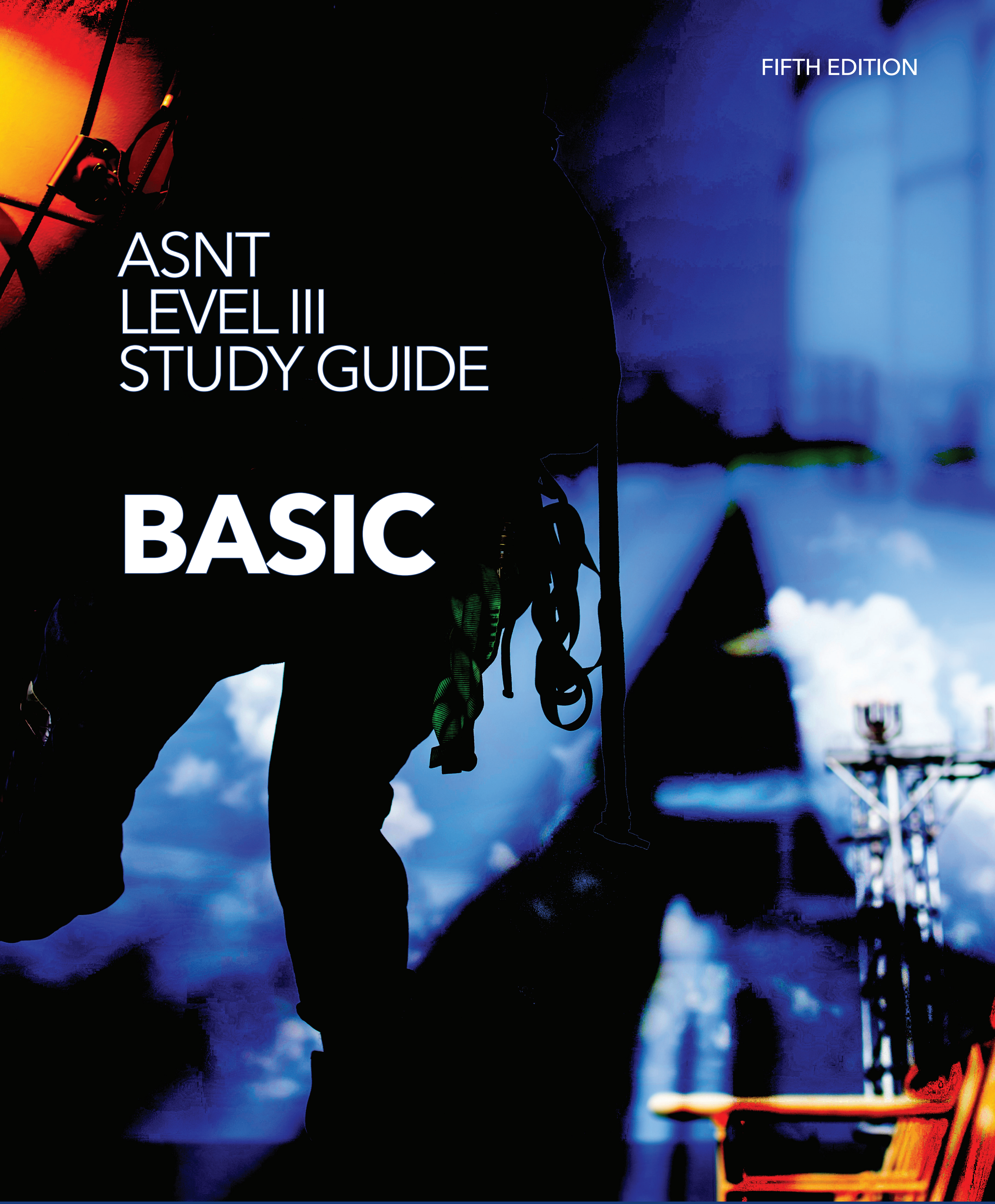

FIFTH EDITION

ASNT LEVEL III STUDY GUIDE

BASIC

The American Society for Nondestructive Testing

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ASNT Mission Statement:
ASNT's mission is to advance the field of nondestructive testing.

ASNT Code of Ethics:
The ASNT Code of Ethics was developed to provide members of the Society with broad ethical statements to guide their professional lives. In spirit and in word, each ASNT member is responsible for knowing and adhering to the values and standards set forth in the Society's Code. More information, as well as the complete version of the Code of Ethics, can be found on ASNT's website, asnt.org.

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FOREWORD

Purpose

This study guide is intended to aid individuals preparing to take the Basic examination as part of the process of becoming certified as an ASNT NDT Level III in one or more NDT methods. It is equally useful for persons preparing to take a Basic Level III examination under an employer's personnel qualification and certification program per ASNT Recommended Practice No. SNT-TC-1A: *Personnel Qualification and Certification in Nondestructive Testing* (2024). It is not intended to be their only source of preparation.

The material in this study guide addresses the body of knowledge included in the Basic examination administered under the ASNT NDT Level III certification program. The ASNT NDT Level III certification program is a service, offered by the American Society for Nondestructive Testing Inc., that gives NDT personnel an opportunity to have their familiarity with the principles and practices of NDT assessed by an independent body. The program uses an independent body to review credentials and uses comprehensive written examinations to identify those persons who meet the criteria for becoming an ASNT NDT Level III for each NDT method. Method examinations are offered in:

- acoustic emission testing
- electromagnetic testing
- leak testing
- liquid penetrant testing
- magnetic flux leakage
- magnetic particle testing
- radiographic testing
- thermal/infrared testing
- ultrasonic testing
- visual testing

NOTE: Six additional methods are listed in SNT-TC-1A but do not have ASNT examinations at present: ground penetrating radar, guided wave, laser methods, microwave technology, neutron radiographic testing, and vibration analysis.

The ASNT NDT Level III certification program requires satisfactory completion of a four-hour Basic examination. Eleven different NDT method examinations are given with durations of two to four hours, depending upon each method's complexity. To be eligible to take the ASNT NDT Level III examinations, persons must qualify by virtue of their documented education and experience.

The Basic examination covers:

- the administration of personnel qualification and certification programs based on the most recent editions of SNT-TC-1A and ANSI/ASNT CP-189: *ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel*,
- materials fabrication and product technology, and
- general principles and applications of common NDT methods.

Each method examination covers:

- fundamentals and principles of the method,
- applications and establishment of techniques and procedures, and
- interpretation of codes, standards, and specifications relating to the method.

NOTE: Reference to persons who have met the criteria for the Level III certifications issued by ASNT are called NDT Level III. This is in contrast to the person who has met the criteria of an employer and who is identified as the employer's Level III or just Level III.

How to Use the Study Guide

This study guide divides the subject matter into three main sections corresponding with the three sections of the Basic examination:

- **Section I:** certification and qualification of NDT personnel as outlined in SNT-TC-1A and CP-189.
- **Section II:** an overview of commonly used NDT methods.
- **Section III:** materials and processes in manufacturing and industry in relation to NDT technology.

Each section of this study guide begins with an introduction to the material to be covered. Chapters contain a set of representative multiple-choice questions covering the relevant portion of the Basic Topical Outlines in ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel* (2024).

Success in answering the questions will help the user to determine if more concentrated study in particular areas is needed. If the user can answer the questions confidently and correctly, additional study may be optional. Technical references are listed in Section II for individual methods if users require further study.

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Changes in This Edition

The fifth edition of the *ASNT Level III Study Guide: Basic* builds on the previous editions.

Section I has been updated to reflect the 2024 editions of two key ASNT documents: SNT-TC-1A and CP-189. This book also includes entire sections of both documents instead of small excerpts in previous editions.

Likewise, the topical outlines of the NDT methods presented in Section II have been drawn from the 2024 edition of CP-105.

Additional Information

In ASNT publications, the words discontinuity and defect have specific meanings. These words are used as follows:

- A *discontinuity* is an intentional (such as drilled holes) or unintentional interruption in the physical structure or configuration of a material or component. NDT techniques reveal indications of discontinuities. Discontinuities may exist without being detected.
- A *defect* is a discontinuity whose size, shape, orientation, or location make it detrimental to the useful service of the object or exceeds accept/reject criteria of an applicable specification. Some discontinuities do not exceed an accept/reject criterion and are therefore not defects.
- The process for determining rejectability is *interpretation* or *evaluation*. All defects are discontinuities, but not all discontinuities are defects.

In 2018 ASNT accepted the ASTM E1316 definitions of calibration and standardization for use in its publications.

- *Calibration* is the comparison (which may include adjustment) of a test instrument to a known reference that is normally traceable to some recognized authority (e.g., NIST). Calibration is typically performed by an organization considered qualified to do so (e.g., an accredited laboratory, or in some cases, the instrument manufacturer) at a determined, periodic interval. Calibration of electronic instrumentation typically involves verification of the linearity of the instrument's response over its usable range.
- *Standardization* is typically completed prior to performing an NDT test and may also be performed at times during the performance of the test and at the completion of the test as a validation of proper instrument operation. It is the adjustment of an NDT instrument using a reference standard (that contains a known condition) to obtain or establish a known and reproducible response.

Because ASNT is an International System of Units (SI) publisher, typically both SI and imperial units are used with SI units appearing first. However, in this study guide units will appear in the order used in the source document.

The ACCP program has been referenced for many years in editions of SNT-TC-1A, CP-189, and CP-105. At the time of publication of this study guide, the ACCP program was in the process of being replaced by ISO 9712-2021. With the introduction of the ASNT 9712 certification program, the ASNT Central Certification Program (ACCP) is being phased out. ACCP renewal is no longer available as of August 2023. ACCP Level III certificate holders who wish to maintain central certification must transition to the ASNT 9712 program by applying for ASNT 9712 certification before their ACCP certificate expires and completing any required examinations.

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CHAPTER 1

RECOMMENDED PRACTICE NO. SNT-TC-1A (2024)

History of SNT-TC-1A

Recommended Practice No. SNT-TC-1A: *Personnel Qualification and Certification in Nondestructive Testing* was first published in 1966 as a primary document comprising just four pages along with a series of supplements for each recognized method at the time: radiographic testing, magnetic particle testing, ultrasonic testing, liquid penetrant testing, and eddy current (now electromagnetic testing). Each supplemental document included recommended education, experience, and training requirements, as well as a recommended course outline, references, and general examination questions. The reasons for which SNT-TC-1A was created and the manner in which it evolved were described in an article published in *Materials Evaluation,* October 1968, Vol. 26, No. 10, pp. 12A–14A, by Harold Hovland and Carl B. Shaw, titled, "How to Qualify and Certify NDT Personnel." The following excerpts outline how this important recommended practice came into existence:

With the increasing complexity and number of nondestructive test methods in use and with greater reliance upon nondestructive testing, industry has not been able to recruit and train personnel in large enough numbers and in a timely fashion to fill the need. Many customers found that they were not receiving the nondestructive test examinations they felt they had paid to receive.

Large segments of the economy, such as the government and prime contractors, found it expedient to put the responsibility for the training and qualification, and verification of the qualifications of nondestructive test personnel upon the manufacturer. This resulted in the generation of a number of documents dealing with the question of the training, qualification, and certification of nondestructive test personnel. Although these documents all pertain to the same subject and have the same objective, in their details they were often conflicting, thereby causing undue cost to American industry.

Beginning about 1959, a few corporations, government agencies, and other technical societies inquired of ASNT as to whether the training and qualification of nondestructive test personnel was properly the domain of the Society.

In 1961, the Society, through its Technical Council, assigned a Task Group to study the feasibility of preparing a document that would deal with the training requirements and the documentation thereof for the qualification of nondestructive test personnel… [T]he Task Group endeavored to prepare a recommended practice which could contain the consensus of expert opinion regarding the training of personnel and an average of requirements for formal education, time for on-the-job training, and documentation.

The resulting SNT-TC-1A is a recommended practice for the qualification and certification of nondestructive test personnel. It provides a format which industry can follow in writing an individualized procedure for the qualification and certification of personnel which meets the needs and requirements unique to each segment of industry.

Contents and Intended Uses of SNT-TC-1A

The following information has been excerpted from the same article cited in the previous section. The text of this article has been modified to include changes made in subsequent editions of SNT-TC-1A.

It is intended that SNT-TC-1A serve as a guide for each employer user in preparing their specific personnel qualification and certification document (procedure or written practice).

SNT-TC-1A sets forth a scope and definitions, it lists the test methods, and it explains the various levels of qualifications. The primary document also details the manner in which personnel are examined and certified. The question arises, "how do I document certification in my particular procedure?" The point to remember here is that the procedure as written by any user of this document must satisfy the requirements of their company and must be acceptable to the company's customers. Within this context, certification may take any one of several forms. It may consist merely of the records of the training programs, the examinations, and the grades of the examinations. It may require a certificate that states explicitly that a particular person is qualified to a given level for a specific method. It may take the form of a card upon which an employer certifies the level of training and qualification of its personnel.

The employer has the same responsibility for the qualification and certification of its NDT personnel as it has for the product. For example, even though an employer may purchase a complete product and sell this product under the company's own name, the employer still bears full responsibility to the customer for the quality of that product and for the product being what the company purports it to be. The employer may elect to purchase components and assemble them and sell the result as a complete product; then again, the employer

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may manufacture all components of the product. In any of these cases, the employer bears a certain responsibility. The employer is responsible [for ensuring] that what is being sold is indeed what it is claimed to be, that it will meet certain requirements of quality, that it will not harm the user, and that it will not destroy the user's property.

The employer may choose to exercise any of these prerogatives in the training, qualification, and certification of its personnel. The company may employ persons previously trained and certify these people; however, the responsibility would be the same as if they had been trained and qualified by the employer. The employer may conduct a training program as recommended in the documents, examine these individuals, and certify them as to their qualification to do the work. The employer will exercise this responsibility through a person in the organization who is qualified as a Level III, and who is qualified to train or provide the training of the personnel under them and to exercise judgment as to the qualification of these personnel. The employer may elect to utilize the Level III services of an outside agency to provide the training, and develop, administer, and grade the examinations for certification. Regardless of the method used for training and examination of their personnel, the responsibility for ensuring that the program utilized complies with their written practice lies with the employer.

Questions & Answers Books for selected methods with sample questions for the Level I and Level II General examinations and Level III Method examination are available from the ASNT Store on the ASNT website (asnt.org). Each question cites the reference used to identify the correct answer. It is intended that the employer's Level III consider these questions as examples only and should not use them verbatim for qualification examinations. In addition to the General examination, a specific written examination is required. It is intended that the Specific examination reflect the equipment that is used by the employee being examined, that it reflect the requirements of the procedures normally used by the employer, and that it satisfy the specific requirements of any customer of the employer.

A Practical examination is also necessary. The Practical examination is primarily a hands-on test and an examination to determine that the examinee understands and knows how to use the written procedures and the equipment. Test objects should be representative of those that the inspector will most likely encounter. Critical points of reference should be predetermined for which the examinee will be graded in the Practical examination. This may include such things as close adherence to the procedure, action taken when the procedure cannot be followed, handling of the parts being tested, interpretation of test results, disposition, and the manner in which a report is written.

Note the emphasis in the foregoing text on the intent that SNT-TC-1A should be used as a guideline, not a fixed requirement. This is reaffirmed in paragraph 1.2 of the Scope section (p. 1) of SNT-TC-1A (2024) as follows: "This document provides guidelines for the establishment of a qualification and certification program." Likewise, in the Foreword, it states, "This recommended practice is not intended to be used as a strict specification. It is recognized, however, that contracts require programs, which meet the intent of this document. For such contracts, purchaser and supplier must agree upon acceptability of an employer's program."

From a factual point of view, very few employers will conform exactly to all the specific recommendations of SNT-TC-1A. Deviation from the recommendations of SNT-TC-1A should be documented in the employer's written practice.

Overview of SNT-TC-1A (2024)

This section provides a paragraph-by-paragraph overview of SNT-TC-1A including excerpts from the document, selected inquiries and official responses provided by the SNT-TC-1A Interpretation Panel, and more general comments.

The responses of the SNT-TC-1A Interpretation Panel are clarifications of intent and are subject to the statement of the Scope in each edition of SNT-TC-1A, paragraph 1.4: "It is recognized that these guidelines may not be appropriate for certain employers' circumstances and/or applications. In developing a written practice as required in Section 5.0, the employer should review the detailed recommendations presented herein and modify them, as necessary, to meet particular needs." The inquiries must be stated in general terms only, because the Interpretation Panel cannot serve as a referee between a buyer and seller or otherwise become involved in any specific case. Inquiries are numbered to include the year of the inquiry and refer to the edition of SNT-TC-1A in effect in that year, unless otherwise stated in the question. New editions of SNT-TC-1A were published for these years: 1968, 1975, 1980, 1984, 1988, 1992, 1996, 2001, 2006, 2011, 2016, 2020 and 2024. All inquiries reprinted below are taken from the most recent edition of *Interpreting SNT-TC-1A*, published annually by ASNT. Minor editorial changes have been made to keep the style consistent with the rest of this book. References to paragraphs in prior editions of SNT-TC-1A are the same as in the 2024 edition unless the corresponding paragraph number has been changed, as noted in brackets. As versions of SNT-TC-1A change, the older is replaced by the newer version with corrections to be addressed at recertification time.

The 2024 edition also includes an addendum, which was primarily developed and issued to address a very important issue. That issue is global misinterpretation/misuse of how to apply/formulate the Recommended Practice "should" standard into a "shall" "written practice" (or procedure) to qualify and certify employees for a given company. In many instances, the "misuse" includes the misuse of the ASNT name and logo on employer- and training agency–issued documents. ASNT will address these global issues by mandating in the 2028 edition two options to be decided on by the employer: (1) identify/name a "Responsible Level III," which must be an ASNT Level III certificate holder (ASNT NDT Level III or ASNT 9712 Level III), who will have overall responsibility for the employer's program, thus being bound by ASNT's ethics linked to their

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ASNT Level III certificate, OR (2) the employer must obtain the ASNT Employer-based Certification (EBC) which will then link the ASNT EBC program ethics to the employer. Violation of the applicable ethics requirements could result in revocation of ASNT Level III issued certificate(s) or the employer's EBC accreditation, as applicable.

Foreword

The Foreword (p. iii) of SNT-TC-1A (2024) states in part:

This Recommended Practice establishes the general framework for a qualification and certification program. In addition, the document provides recommended educational, experience, and training requirements for the different test methods.

This Recommended Practice is not intended to be used as a strict specification. It is recognized, however, that contracts require programs, which meet the intent of this document. For such contracts, purchaser and supplier must agree upon acceptability of an employer's program.

The verb "should" has been used throughout this document to emphasize the recommendation presented herein. It is the employer's responsibility to address specific needs and to modify these guidelines as appropriate in a written practice. In the employer's written practice, the verb "shall" is to be used in place of "should" to emphasize the employer's needs.

New ASNT Policy and Paragraph 1.5

Based on numerous examples over the years indicating improper implementation of SNT-TC-1A, the 2024 edition was modified to recommend that the employer designate the Responsible Level III (paragraph 4.3.4) to be fully responsible for all aspects of the employer's SNT-TC-1A program. This will be further modified in the next edition to mandate via ASNT policy that the Responsible Level III be an ASNT NDT Level III or an ASNT 9712 Level III, or the company must obtain accreditation under the ASNT Employer-Based Certification (EBC) program, where ASNT monitors the implementation of the employer's SNT-TC-1A program through periodic audits. Any one of these actions results in binding the ASNT NDT, ASNT 9712 Responsible Level III, or EBC-accredited company, to ASNT ethics requirements.

Sections 1.0, 2.0, and 3.0: Scope, Definitions, and Nondestructive Testing Methods

Sections 1.0, 2.0, and 3.0 (Scope, Definitions, and Nondestructive Testing Methods) on pp. 1–2 of SNT-TC-1A (2024) are reprinted below in full:

1.0 Scope

1.1 It is recognized that the effectiveness of nondestructive testing (NDT) applications depends upon the capabilities of the personnel who are responsible for, and perform, NDT. This Recommended Practice has been prepared to establish guidelines for the qualification and certification of NDT personnel whose specific jobs require appropriate knowledge of the technical principles underlying the nondestructive tests they perform, witness, monitor, or evaluate.

1.2 This document provides guidelines for the establishment of a qualification and certification program.

1.3 These guidelines have been developed by The American Society for Nondestructive Testing Inc. to aid employers in recognizing the essential factors to be considered in qualifying personnel engaged in any of the NDT methods listed in Section 3.0.

1.4 It is recognized that these guidelines may not be appropriate for certain employers' circumstances and/or applications. In developing a written practice as required in Section 5.0, the employer should review the detailed recommendations presented herein and modify them, as necessary, to meet particular needs. Such modification may alter but shall not eliminate basic provisions of the program such as training, experience, testing, and recertification. Supporting technical rationale for modification of detailed recommendations should be provided in an Annex to the written practice.

2.0 Definitions

2.1 Terms included in this document are defined as follows:

2.1.1 **Calibration, Instrument:** the comparison of an instrument with, or the adjustment of an instrument to, a known reference(s) often traceable to the applicable country's national institute or standards body. (See also Standardization, Instrument.)

2.1.2 **Certification:** written testimony of qualification.

2.1.3 **Certifying Authority:** the person or persons properly designated in the written practice to sign certifications on behalf of the employer.

2.1.4 **Certifying Agency:** the employer of the personnel being certified.

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2.1.5 **Closed-Book Examination:** an examination administered without access to reference material except for materials supplied with or in the examination. (See 8.1.3.)

2.1.6 **Comparable:** being at an equivalent or similar level of NDT responsibility and difficulty as determined by the employer's NDT Level III.

2.1.7 **Detection Rate:** expressed as a percentage, it represents the number of flaws or indications detected in a specimen compared to the number of flaws or indications that are actually in the specimen being examined.

2.1.8 **Documented:** the condition of being in written form.

2.1.9 **Employer:** the corporate, private, or public entity, which employs personnel directly or indirectly for wages, salary, fees, or other considerations. This would include employers who obtain their qualified supplemental workforce personnel through third-party agencies, providing the use and certification of those supplemental employees is addressed in the employer's written practice.

2.1.10 **Experience:** work activities accomplished in a specific NDT method under the direction of qualified supervision including the performance of the NDT method and related activities but not including time spent in organized training programs.

2.1.11 **False Call:** when an acceptable indication or unflawed grading unit is identified as being a defect or a flawed grading unit.

2.1.12 **Grading Unit:** a Qualification Specimen can be divided into sections called grading units, which do not have to be equal length or be equally spaced. Grading units are unflawed or flawed and the percentage of flawed/unflawed grading units required should be approved by the NDT Level III.

2.1.13 **Limited Certification:** nondestructive test methods may be further subdivided into limited disciplines or techniques to meet specific employers' needs; these are NDT Level II certifications, but to a limited scope.

2.1.14 **Method:** one of the disciplines of NDT; for example, ultrasonic testing, within which various test techniques may exist.

2.1.15 **Nondestructive Testing:** the development and application of technical methods to examine materials or components in ways that do not impair future usefulness and serviceability in order to detect, locate, measure, and evaluate flaws; to assess integrity, properties, and composition; and to measure geometrical characteristics.

2.1.16 **Outside Agency:** a company or individual who provides NDT Level III services and whose qualifications to provide these services have been reviewed and approved by the employer engaging the company or individual.

2.1.17 **Personalized Instruction**: may consist of blended classroom, supervised laboratory, and/or hybrid online competency-based course delivery. Modular content is covered through online presentations, in the classroom, and/or in small groups. Personalized instruction also enables students to achieve competency using strategies that align with their knowledge, skills, and learning styles.

2.1.18 **Predictive Maintenance (PdM):** evaluates the condition of equipment (typically in service) by performing periodic or continuous (online) equipment condition monitoring. Condition monitoring evaluates leading performance indicators of specified machinery, components, and assemblies (including structures in whole and in part such as buildings and bridges, etc.), detectable via the applied method. PdM uses principles of statistical process control to assess the condition by focusing on leading indicators that may signify deterioration in performance or potential equipment, structural, or material failure. The ultimate goal of PdM is to perform maintenance based upon the condition such that the maintenance activity is cost-effective and before the equipment loses optimum performance or fails.

2.1.19 **Qualification:** demonstrated skill, demonstrated knowledge, documented training, and documented experience required for personnel to properly perform the duties of a specific job.

2.1.20 **Recommended Practice:** a set of guidelines to assist the employer in developing uniform procedures for the qualification and certification of NDT personnel to satisfy the employer's specific requirements.

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2.1.21 **Standardization, Instrument:** the adjustment of an NDT instrument using an appropriate reference standard, to obtain or establish a known and reproducible response. (This is usually done prior to performing any NDT but can be carried out anytime there is concern about the performance of NDT or instrument response.) (See also Calibration, Instrument.)

2.1.22 **Technique:** a category within an NDT method; for example, ultrasonic thickness testing.

2.1.23 **Third-Party Agency:** a company or organization, without an established written practice, providing supplemental workforce to the employer; for example, a temporary staffing company.

2.1.24 **Training:** an organized program developed to impart the knowledge and skills necessary for qualification.

2.1.25 **Written Practice:** a written procedure developed by the employer that details the requirements for qualification and certification of their employees.

2.1.26 **Responsible Level III:** the Level III designated by the employer with the responsibility and authority to ensure that the requirements and recommendations of this Recommended Practice are met and to act on behalf of the employer. Also see paragraph 4.3.4.

2.1.27 **Service (In-Service):** a program of periodic tests during the lifetime of a system and/or component, designed to assess its structural integrity.

3.0 Nondestructive Testing Methods

3.1 Qualification and certification of NDT personnel in accordance with this Recommended Practice is applicable to each of the following methods:

Acoustic Emission Testing
Electromagnetic Testing
Ground Penetrating Radar
Guided Wave Testing
Laser Methods Testing
Leak Testing
Liquid Penetrant Testing
Magnetic Flux Leakage Testing
Magnetic Particle Testing
Microwave Technology Testing
Neutron Radiographic Testing
Radiographic Testing
Thermal/Infrared Testing
Ultrasonic Testing
Vibration Analysis
Visual Testing

Inquiries for Section 1.0

INQUIRY 80-4

NDT examiners in our employ who perform examinations using liquid penetrant use only the visible dye, solvent removable, penetrant technique. Since our examiners do not have need to be qualified in the other liquid penetrant techniques, is it permissible to modify the number of general and specific questions as well as the hours of training and work experience to satisfy requirements of SNT-TC-1A for Level I and Level II examiners?

RESPONSE:

... It is the intent that SNT-TC-1A is a recommended practice and that it is a guideline which should be modified by the employer as necessary to meet their particular needs. The employer should determine their needs, determine the necessary qualifications of their examiners to meet those needs, and describe those in their written practice.

INQUIRY 80-8

1. May an employer deviate from the guidelines to meet their specific needs?
2. Is our company's specific written practice acceptable?

RESPONSE:

1. Yes, in accordance with paragraph 1.4 of SNT-TC-1A, the employer should modify the guidelines to meet their needs.
2. It is against ASNT policy to judge the applicability of company documents. As a general comment, it is the employer's prerogative to establish their criteria for certification. It is then the customer's prerogative to accept or reject those criteria.

NOTE: Paragraph 1.4 very clearly states the intent that the detailed recommendations of SNT-TC-1A should be reviewed and modified by the user to satisfy unique needs. This allowance was originally incorporated in recognition of the fact that each manufacturer or each service organization has a different clientele. Customer or clientele requirements ultimately determine what is acceptable between buyer and seller, even to the details of qualifying NDT personnel.

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INQUIRY 04-1

1. Is it the intent of paragraph 1.4 that an employer can modify the "Guidelines" of SNT-TC-1A to the extent that Level II and Level III NDT personnel can be "certified" without any examinations if written this way in the written practice?
2. What are the limits that are intended regarding how much the employer can deviate from SNT-TC-1A as written? If no limits are given, the employer can change the entire context of SNT-TC-1A to eliminate all certification exams. Is that the intent?

RESPONSE:

1. No. The provisions of paragraph 1.4 allow modification of the detailed recommendations; it is not intended to allow elimination of the basic provisions of the document.
2. No. Paragraph 9.2 requires certification in accordance with "Section 8.0, Examinations" as described in the employer's written practice. [In the 2024 edition, this is rephrased as "in accordance with Sections 6.0, 7.0, and 8.0."]

GENERAL COMMENTS [REGARDING INQUIRY 04-1]:

SNT-TC-1A allows for modification of detailed recommendations as necessary to meet particular needs. The intent is that there be a technical rationale to support such modification. Elimination of requirements, such as training, experience, and examination, goes beyond modification of detailed requirements.

INQUIRY 13-3

Based on the background information provided below, is it correct to assume that "qualified supervision" may only be provided by a certified Level II or III individual, or may supervision be performed by other noncertified management personnel (such as a quality assurance manager) if they are so designated in the employer's written practice?

BACKGROUND:

Paragraph 2.1.8 of SNT-TC-1A (2011) [paragraph 2.1.10 in the 2024 edition] defines "experience" as "work activities accomplished in a specific NDT method under the direction of qualified supervision including the performance of the NDT method and related activities but not including time spent in organized training programs."

The only other place the word "supervision" is used is in paragraph 4.3.1, which says in part, "The NDT Level I should receive the necessary instruction and supervision from a certified NDT Level II or III individual."

RESPONSE:

Yes, the intent of "qualified supervision" is supervision by personnel who are certified Level II or III in the specific method.

INQUIRY 13-10

As per paragraph 2.1.12 of SNT-TC-1A (2006) [paragraph 2.1.19 in the 2024 edition], is it correct to say that knowledge is demonstrated when you clear the written examination (General and Specific) and skill is demonstrated when you clear the practical examination?

BACKGROUND:

Qualification is defined in paragraph 2.1.12 [2.1.19 in the 2024 edition] as demonstrated knowledge, demonstrated skill, documented training, and documented experience.

RESPONSE:

No. Paragraph 2.1.12 [2.1.19 in the 2024 edition] is a definition, not a statement of requirements. Paragraph 9.2 says that certification of NDT personnel shall be based on demonstration of satisfactory qualification in accordance with Sections 6.0, 7.0, and 8.0, as described in the employer's written practice.

INQUIRY 22-07

Section 1.4 [2024 edition] allows an employer to provide a limited certification as long as there is a technical rationale for such modification.

1. Does the intent of this statement allow limited certifications in UT shear wave of groove welds only?
2. Is this limited certification justifiable for only 210 hours of experience?

RESPONSE:

1. No.
2. No.

INQUIRY 24-16

Regarding section 2.1.13 (2024), does SNT-TC-1A allow limited certification in TOFD and or PAUT Level II? If yes, what are the requirements in terms of minimum training hours and prerequisites?

BACKGROUND:

ASME Section V does not allow limited certification.

RESPONSE:

Yes. Table 6.3.1 A, Note 10.0 first requires personnel seeking TOFD and/or PAUT certification to be fully certified as UT Level II personnel. That person could then have a limited certification issued in TOFD or PAUT, provided the description of the limited certification along with the training and experience requirements for that certification would be described in the employer's written practice.

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General Comments on Sections 1.0, 2.0, and 3.0

Paragraph 1.4 was modified in the 2011 edition to provide users with additional guidance to ensure that when users develop a written practice, the fundamental requirements of SNT-TC-1A are maintained and that any modifications are documented with a rationale for the deviations. This modification was retained in the 2016, 2020, and 2024 editions.

In Section 2.0 Definitions, each of these terms should be carefully studied. For example, some NDT practitioners may not clearly understand the difference between "qualification" and "certification." Considering each term as representing a process, the process of qualifying personnel involves assessment of the adequacy of the skills, training, and experience of personnel being considered for certain tasks. The process of qualifying may require that education and training be imparted to such personnel. Assessment of qualifications often requires that personnel be examined. Following the process of qualification, individuals meeting or exceeding the minimum qualification requirements of the employer can be certified. That is, the employer provides evidence that a qualifying process was followed. Such evidence backs up the employer's assertion that certain individuals are qualified to perform certain critical functions and is the employer's "certification" that the evidence of qualification exists.

Limited certifications for ultrasonic thickness measurement and radiographic film interpretation have been retained in this edition. These certifications are intended to be limited in the scope of qualification, but not the responsibilities of the Level II individual as defined in Section 4.0. There is no Level I limited certification.

Fundamental in free enterprise is the tenet that producers of goods and suppliers of services are ultimately responsible for the quality and effectiveness of such goods and services as well as bearing responsibility for shortcomings and failures. Therefore, regardless of the details of the processes by which individuals became qualified, the direct employer of the personnel being certified must bear the end responsibility for conferring a certification and, thereby, can be the only "certifying agency." Such absolute responsibility does not preclude the use of outside services by employers to assist in imparting and/or assessing qualifications of individuals being certified. The use of outside services does not relieve the employer from responsibility, nor can the employer abrogate or delegate this responsibility to an outside agency.

The definition of "Employer" used in SNT-TC-1A dates back to the 1975 edition. At that time, it was common for NDT personnel to work full time, exclusively for one employer. Over the years, the employment model of the 1970s has changed significantly. The use of supplemental personnel during peak periods has become a common model for many industries. The model has continued to evolve from using fully tested and certified personnel supplied by full-service inspection companies, to third-party organizations providing qualified but not certified personnel to companies who then certify them in accordance with their program and written practice. While companies rely on SNT-TC-1A interpretation 83-07 to support that model, later SNT-TC-1A interpretations 87-02 and 02-01 contradict that interpretation. Since companies continue to use and expand that model, the modified definition change is being submitted to more accurately reflect the role of the employer in the way supplemental personnel are used in the NDT workforce.

SNT-TC-1A has been referenced by certain codes and specifications to be used, in effect, as a model for employers to develop a written practice concerning some method of NDT not currently covered in SNT-TC-1A.

Review Questions for Sections 1.0, 2.0, and 3.0

Based on the foregoing discussion, answer the questions on pp. 30–31 relating to SNT-TC-1A and the administration of a qualification and certification program in NDT.

Section 4.0: Levels of Qualification

Section 4.0 (p. 3) of SNT-TC-1A (2024) is reprinted below in full:

4.0 Levels of Qualification

4.1 There are three basic levels of qualification. The employer may subdivide these levels for situations where additional levels are deemed necessary for specific skills and responsibilities.

4.2 While in the process of being initially trained, qualified, and certified, an individual should be considered a trainee. A trainee should work with a certified individual. The trainee should not independently conduct, interpret, evaluate, or report the results of any NDT test.

4.3 The recommended technical knowledge and skill sets for the three basic levels of qualification are as follows:

4.3.1 **NDT Level I.** An NDT Level I individual should have sufficient technical knowledge and skills to be qualified to properly perform specific standardizations, specific NDT, and specific evaluations for acceptance or rejection determinations according to written instructions and to record results. The NDT Level I should receive the necessary instruction or supervision from a certified NDT Level II or III individual.

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4.3.2 **NDT Level II.** An NDT Level II individual should have sufficient technical knowledge and skills to be qualified to set up and standardize equipment and to interpret and evaluate results with respect to applicable codes, standards, and specifications. The NDT Level II should be thoroughly familiar with the scope and limitations of the methods for which qualified and should exercise assigned responsibility for on-the-job training and guidance of trainees, NDT limited certified personnel, and NDT Level I personnel. The NDT Level II should be able to organize and report the results of NDT tests.

4.3.3 **NDT Level III.** An NDT Level III individual should have sufficient technical knowledge and skills to be capable of developing, qualifying, and approving procedures; establishing and approving techniques; interpreting codes, standards, specifications, and procedures; and designating the particular NDT methods, techniques, and procedures to be used. The NDT Level III should be responsible for the NDT operations for which qualified and assigned and should be capable of interpreting and evaluating results in terms of existing codes, standards, and specifications. The NDT Level III should have sufficient practical background in applicable materials, fabrication, service (in-service), and product technology to establish techniques and to assist in establishing acceptance criteria when none are otherwise available. The NDT Level III should have general familiarity with other appropriate NDT methods, as demonstrated by an ASNT Level III Basic examination or other means. The NDT Level III, in the methods in which certified, should have sufficient technical knowledge and skills to be capable of training and examining NDT Level I, II, and III personnel for certification in those methods.

4.3.4 **Responsible Level III.** The designated Level III that has overall responsibility for all aspects of the Employer's Personnel Qualification/Certification program. These responsibilities may be further delegated to other Level III personnel in the organization, including external Level III consultants. The delegations of specific responsibilities shall be documented.

Inquiries for Section 4.0

INQUIRY 03-2

May an NDT Level III develop, qualify, and approve NDT procedures, and establish and approve techniques in the methods in which not certified?

RESPONSE:

No, see the response to 90-1, Part 2 and Inquiry 10-5.

RESPONSE TO INQUIRY 90-1, PART 2:

Paragraph 4.3.3 of SNT-TC-1A (1980 edition), states ASNT's recommendations for an employer to consider when selecting their Level III. The last sentence specifies that a Level III should be familiar with other methods and should be qualified by experience and training. The intent is to train and examine Level I and Level II personnel for certification in the methods in which they are qualified. This is further clarified in SNT-TC-1A (1988 edition), paragraph 4.3.3, which states, "The NDT Level III, in methods in which certified, should [have sufficient technical knowledge and skills to] be capable of training and examining NDT Level I and II personnel for certification in those methods."

INQUIRY 09-1

1. With regard to NDT Level IIIs signing off on certification after successful completion of trainees' examinations, is it the intention that the NDT Level III must be certified in the method for which they are signing off as per SNT-TC-1A paragraph 4.3.3?
2. Is it commonly accepted in the industry that an NDT Level III can sign off on certifications for methods other than those in which they are certified?

RESPONSE:

1. This question refers to two separate issues.
   a. Can an NDT Level III sign off on certifications in methods for which they are not qualified? Yes. Signing off on certification is covered by Section 9.0, Certification Record, and paragraph 9.4.9 states that a person's certification record must have the signature of the Level III who verified the qualifications of a candidate for certification. It does not require that this Level III be certified in the applicable test methods.
   b. Can an NDT Level III develop and approve the training and examination program for NDT methods they are not qualified in? No. Paragraph 4.3.3 states that the NDT Level III should be capable of training and examining NDT Level I and II personnel for certification in those methods for which the Level III holds certification.
2. Determining industry practice is not within the scope of SNT-TC-1A.

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INQUIRY 09-2

Should NDT Level I technicians be allowed to make the final determination as to whether a production part is accepted or rejected and then sign off on the production order paperwork if they follow a step-by-step procedure developed by a Level III?

RESPONSE:

Level I personnel may accept or reject parts according to written instructions and record results. Reporting the final results requires a Level II or III.

INQUIRY 10-5

As stated in the SNT-TC-1A 2006 [and 2024] edition, paragraph 4.0 Level of Qualification, 4.3.3 NDT Level III: "An NDT Level III individual should be capable of developing, qualifying and approving procedures..., etc."

QUESTION:

1. Does it mean that all NDT procedures/standard operating procedures (SOP) should be prepared by an NDT Level III?
2. Could an NDT Level II prepare a specific procedure based on the related standard or specification?

EXAMPLE:

The NDT Level I prepares NDT procedures for ferromagnetic connection inspection based on API RP7G and API Spec 7 and all procedures actually adopted from the referenced specifications.

3. Could the director of the company (inspection agency) approve this SOP, or should this SOP be reviewed and approved by the NDT Level III?

RESPONSE:

1. No.

2., 3. The scope of SNT-TC-1A is limited to guidelines for the qualification and certification of NDT personnel. The capability of developing, qualifying, and approving procedures is a qualification for Level IIIs. This does not preclude individuals who are not Level IIIs from also having some or all of those skills. SNT-TC-1A does not restrict employers from using individuals they feel are qualified for doing those tasks.

INQUIRY 11-1

With regard to paragraphs 2.1.8 [2.1.10 in the 2020 edition], 4.2, and 4.3 of the 2006 edition of SNT-TC-1A:

1. May an NDT trainee gain experience hours while working under the direct supervision of a Level I in the method?
2. Are there any circumstances that would allow NDT trainees or NDT Level I technicians to progress to the next certification level without ever having gained actual field experience working with NDT Level II or Level III technicians in the method?

RESPONSE:

1. Yes, provided the Level I receives the necessary instructions or supervision from a certified Level II or III individual.
2. No. As stated in paragraph 2.1.8 [2.1.10 in the 2020 edition], experience can only be gained while working under the direction of qualified supervision.

INQUIRY 15-11

Can an employer create a Limited Level I certification and delete the NDT Level II or III supervision requirement described in paragraph 4.3.1 (2011 edition) if properly documented in the company's written practice?

RESPONSE:

No.

INQUIRY 16-2

The appendix of example questions (Level I and Level II) has a note in the opening paragraph that states, "All questions and answers should be referenced to a recognized source."

What is considered recognized?

RESPONSE:

Paragraph 4.3.3 makes the Level III responsible for training and examining NDT Level I and II personnel for certification. That implies that the Level III is responsible for selection of the reference material, approval of lesson plans, and so on. In designating student study material and handouts, the Level III is recognizing such material.

INQUIRY 24-13

1. Per paragraph 4.3.2, given that a Level II is authorized to deliver on-the-job training (OJT) to trainees, is it implied that a newly certified Level II is expected to commence providing OJT for trainees starting from the day following their certification?
2. Can limited Level II certified individuals provide OJT to trainees?

RESPONSE:

1. Yes, they would be qualified to provide, but not necessarily expected to provide OJT to trainees.
2. Yes, within the scope of their limited certifications.

General Comments on Section 4.0

Implicit in the definitions of Level I, Level II, and Level III as outlined in paragraphs 4.3.1, 4.3.2, and 4.3.3 is the concept that the qualifications for Level III equal and exceed those of Level II. The employer must be satisfied with the proficiency of any individual at any level to handle work tasks. SNT-TC-1A is not intended for use to determine an individual's proficiency. It is intended as a guideline to establish qualifications.

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Based on numerous examples over the years indicating improper implementation of SNT-TC-1A, the 2024 edition was modified to recommend that the employer designate the Responsible Level III (paragraph 4.3.4) to be fully responsible for all aspects of the employer's SNT-TC-1A program. This will be further modified in the next edition to mandate via ASNT policy, that the Responsible Level III be an ASNT NDT Level III or an ASNT 9712 Level III certificate holder or the company must obtain accreditation under the ASNT Employer-Based Certification (EBC) program, where ASNT monitors the implementation of the employer's SNT-TC-1A program through periodic audits. Either of these actions will result in binding the ASNT NDT or 9712 Level III Responsible Level III or EBC-accredited company to ASNT ethics requirements.

Review Questions for Section 4.0

Based on the foregoing discussion, answer the questions on pp. 31–32.

Section 5.0: Written Practice

Section 5.0 (p. 3) of SNT-TC-1A (2024) is reprinted below in full:

5.0 Written Practice

- 5.1 The employer shall establish a written practice for the control and administration of NDT personnel training, examination, and certification.
- 5.2 The employer's written practice should describe the responsibility of each level of certification for determining the acceptability of materials or components in accordance with the applicable codes, standards, specifications, and procedures.
- 5.3 The employer's written practice should describe the training, experience, and examination requirements for each level of certification by method and technique, as applicable.
- 5.4 The employer's written practice should identify the test techniques within each test method applicable to its scope of operations.
- 5.5 The employer's written practice shall be reviewed and approved by the employer's NDT Level III.
- 5.6 The employer's written practice shall be maintained on file.

Inquiries for Section 5.0

INQUIRY 16-3

1. Do paragraphs 5.3 and 5.4 of SNT-TC-1A (2011) require the written practice to clearly identify and describe each technique for certification for PT, MT, VT, and so on, whose techniques are not specified in Table 6.3.1A and 6.3.1B?
2. If yes, should NDE techniques for above be identified and described as specified in T-X50 of *ASME Boiler & Pressure Vessel Code (BPVC)*, Section V, 2015 edition?

RESPONSE:

1. SNT-TC-1A is intended to provide the employer with adequate flexibility to accommodate a variety of special needs, as documented in the written practice. The methods and techniques presented in SNT-TC-1A, along with the training outlines in ANSI/ASNT CP-105, are guidelines based on method committee input from multiple industries.
2. Application to other codes and standards is beyond the scope of the SNT-TC-1A Interpretation Committee.

INQUIRY 18-3

1. Paragraph 5.4 refers to listing specific test techniques to be used in the written practice. Should test techniques that are not explicitly listed in Table 6.3.1A or Table 8.3.4 (e.g., water washable vs. solvent removable PT) be differentiated and listed in the written practice?
2. Paragraph 5.3 states that the written practice should list the training, experience, and examination requirements by method and technique, as applicable. When the employer's written practice lists techniques not explicitly listed in Tables 6.3.1A or 8.3.4, do the cumulative training and experience hours apply to the method rather than to each technique to be used in the method?

RESPONSE:

1. Yes.
2. The training, experience, and examination requirements for techniques must be determined by the employer and specified in the employer's written practice.

INQUIRY 24-15

Sections: 5.0, 6.0, 7.0, 8.0 (2020 edition)

1. NDT personnel transferred from one site (plant) to another site of the same employer but different locations. Whether transferred NDT personnel can be certified to new site without further training and examination as both the sites are under the same employer?
2. Whether the certification issued by one site can be used as it is without any further certification by the new site, if the NDT personnel are working for different sites of the same employer.
3. If all sites of the same employer follow the same written practice, whether one qualification will be sufficient to perform NDT activities in all sites?

BACKGROUND:

The employer has multiple manufacturing sites/plants at different locations. NDT requirements of the components manufactured at different sites covers similar requirements/acceptance criteria. They are following different written practices for each of the manufacturing sites, but are managed by the same employer. In this case, the employee transferred from one location to another location, can they be recertified to the new location as per their written practice, without further training,

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TABLE 6.3.1 A

Recommended Initial Training and Experience Levels

Examination Method	NDT Level	Technique	Training Hours	Experience	
				Minimum Hours in Method or Technique	Total Hours in NDT
Acoustic Emission Testing	I		40	210	400
	II		40	630	1200
Electromagnetic Testing	I	AC Field Measurement	40	210	400
	II		40	630	1200
	I	Eddy Current Testing	40	210	400
	II		40	630	1200
	I	Remote Field Testing	40	210	400
	II		40	630	1200
Ground Penetrating Radar	I		8	60	120
	II		20	420	800
Guided Wave Testing	I		40	240	460
	II		40	240	460
Laser Methods Testing	I	Profilometry	8	70	130
	II		24	140	260
	I	Holography/Shearography	40	210	400
	II		40	630	1200
Leak Testing	I	Bubble Leak Testing	2	3	15
	II		4	35	80
	I	Pressure Change Leak Testing	24	105	200
	II		16	280	530
	I	Halogen Diode Leak Testing	12	105	200
	II		8	280	530
	I	Mass Spectrometer Leak Testing	40	280	530
	II		24	420	800
Liquid Penetrant Testing	I		4	70	130
	II		8	140	270
Magnetic Flux Leakage	I		16	70	130
	II		12	210	400
Magnetic Particle Testing	I		12	70	130
	II		8	210	400
Microwave Technology Testing	I		40	210	400
	II		40	630	1200
Neutron Radiographic Testing	I		28	420	800
	II		40	1680	2400
Radiographic Testing	I	Radiography	40	210	400
	II		40	630	1200
	I	Computed Radiography	40	210	400
	II		40	630	1200
	I	Computed Tomography	40	210	400
	II		40	630	1200
	I	Digital Radiography	40	210	400
	II		40	630	1200
Thermal/ Infrared Testing	I		32	210	400
	II	Building Diagnostics	34	1260	1800
	II	Electrical and Mechanical Testing	34	1260	1800
	II	Materials Testing	34	1260	1800
Ultrasonic Testing	I		40	210	400
	II		40	630	1200
	II	Full Matrix Capture	80	320	n/a
	II	Phased Array Ultrasonic Testing	80	320	n/a
	II	Time of Flight Diffraction	40	320	n/a
Vibration Analysis	I		24	420	800
	II		72	1680	2400
Visual Testing	I		8	70	130
	II		16	140	270

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TABLE 6.3.1 A NOTES:

1.0 A person may be qualified directly to NDT Level II with no time as a certified NDT Level I, providing the recommended training and experience consists of the sum of the hours recommended for NDT Level I and Level II.

2.0 For NDT Level III certification, the experience should consist of the sum of the hours for NDT Level I and Level II, plus the additional time in 6.3.2, as applicable. The formal training should consist of the NDT Level I and Level II training, plus any additional formal training as defined in the employer's written practice.

3.0 Listed training hours may be adjusted as described in the employer's written practice depending on the candidate's actual education level, e.g. grammar school, college graduate in engineering, etc.

4.0 Training should be outlined in the employer's written practice. MT training hours may be counted toward MFL training hours as defined in employer's written practice.

5.0 If an individual is currently certified in an ET technique and a full-course format was used to meet the initial qualifications in that technique, the minimum training hours to qualify in another ET technique at the same NDT level may be reduced up to 40% if so defined in the employer's written practice. If an individual is certified in an ET technique, the minimum experience to qualify for another ET technique at the same level or to the next level may be reduced by up to 50% if so defined in the employer's written practice.

6.0 While fulfilling total NDT experience requirement, experience may be gained in more than one method, however, the minimum hours must be met for each method.

7.0 If an individual is currently certified in an RT technique and a full-course format was used to meet the initial qualifications in that technique, the minimum additional training hours to qualify in another technique at the same level should be 24 hours (of which at least 16 hours should be equipment familiarization). The training outline should be as defined in the employer's written practice. If an individual is certified in a technique, the minimum additional experience required to qualify for another technique at the same level may be reduced by up to 50%, as defined in the employer's written practice.

8.0 Independent of the training recommended for Level I and Level II certification, a trainee is required to receive radiation safety training as required by the regulatory jurisdiction.

9.0 If an individual is currently certified in one IR technique and a full-course format was used to meet the initial qualifications in that technique, the minimum additional training hours to qualify in another technique at the same level should be 20 hours (of which at least 16 hours should be specific technique familiarization). The training outline should be as defined in the employer's written practice. If an individual is certified in a technique, the minimum additional experience required to qualify for another technique at the same level may be reduced by up to 50%, as defined in the employer's written practice.

10.0 Full Matrix Capture (FMC), Time of Flight Diffraction (TOFD), and Phased Array Ultrasonic Testing (PAUT) require Ultrasonic Testing Level II certification as a prerequisite.

11.0 In addition to the training recommended in this table for FMC, TOFD, and PAUT, supplemental specific hardware and software training should be required for automated or semiautomated technique applications. The employer's written practice should fully describe the nature and extent of the additional training required for each specific acquisition or analysis software and instrument/system used. The employer's written practice should also describe the means by which the examiner's qualification will be determined for automated and semiautomated techniques.

TABLE 6.3.1 B

Recommended Initial Training and Experience Levels for NDT Level II Limited Certifications

Examination Method	Limited Certification Technique	Technician's Starting Point	Formal Training	Minimum Work Experience in Technique (Hours)
Radiographic Testing	Film Interpretation	Non-Radiographer	40	220
	Film Interpretation	RT Level I	24	220
Ultrasonic Testing	Digital Thickness Measurement (numeric output only)	Trainee	8	40
	A-scan Thickness Measurement	Trainee	24	175

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and examination? Whether employee can work for multiple sites of the same employer, but certified by any one of the sites. Whether the same written practice is followed by all the sites, the person certified in one site can work in sites without any further training and certification?

RESPONSE:

1. Yes, provided the education, training, experience, and examination requirements of both written practices are the same and approved by the Responsible NDT Level III.
2. Yes. See Response 1.
3. Yes.

General Comments on Section 5.0

Implicit in the use of SNT-TC-1A is the requirement that the employer shall develop a written practice. SNT-TC-1A provides the guidelines; the written practice sets forth the details. When departures from SNT-TC-1A guidelines are made, the written practice should record the departure. It is good practice to record all departures, even when they represent situations in excess and/or with greater strength than the recommendations of SNT-TC-1A.

Review Questions for Section 5.0

Based on the foregoing discussion, answer the questions on p. 32.

Section 6.0: Education, Training, and Experience Requirements for Initial Qualification

Section 6.0 (pp. 3–6) of SNT-TC-1A (2024) is reprinted below in full:

6.0 Education, Training, and Experience Requirements for Initial Qualification

6.1 Candidates for certification in NDT should have sufficient education, training, and experience to ensure qualification in those NDT methods in which they are being considered for certification. Documentation of prior certification may be used by an employer as evidence of qualification for comparable levels of certification.

6.2 Documented training and/or experience gained in positions and activities comparable to those of Levels I, II, and/or III prior to establishment of the employer's written practice may be considered in satisfying the criteria of paragraph 6.3.

6.3 To be considered for certification, a candidate should satisfy one of the following criteria for the applicable NDT level:

6.3.1 NDT Levels I and II

6.3.1.1 Table 6.3.1A lists the recommended training and experience hours to be considered by the employer in establishing written practices for initial qualification of NDT Level I and Level II individuals.

6.3.1.2 Table 6.3.1B lists initial training and experience hours, which may be considered by the employer for specific limited applications as defined in the employer's written practice.

6.3.1.3 Limited certifications should apply to individuals who do not meet the full training and experience of Table 6.3.1A. Limited certifications issued in any method should be approved by the NDT Level III and documented in the certification records.

6.3.2 NDT Level III

6.3.2.1 Have a bachelor's degree (or higher) in engineering or science, plus one additional year of experience beyond the Level II requirements in NDT in an assignment comparable to that of an NDT Level II in the applicable NDT method(s), or:

6.3.2.2 Have completed with passing grades at least two (2) years of engineering or science study at a university, college, or technical school, plus two (2) additional years of experience beyond the NDT Level II requirements in NDT in an assignment at least comparable to that of NDT Level II in the applicable NDT method(s), or:

6.3.2.3 Have four (4) years of experience beyond the NDT Level II requirements in NDT in an assignment at least comparable to that of an NDT Level II in the applicable NDT method(s).

The above NDT Level III requirements may be partially replaced by experience as a certified NDT Level II or by assignments at least comparable to NDT Level II as defined in the employer's written practice.

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6.4 It is recognized that expanded training programs at educational institutions and colleges may include additional hands-on application and lab hours in applicable methods and/or techniques beyond the required minimum training hours. Those additional hours may be considered toward the required experience hours as approved by the NDT Level III and documented within the employer's written practice. Additionally, the maximum credit for these hours should not exceed 25% of the required experience hours. A minimum of 75% of the required experience hours shall be gained in the field.

Inquiries for Section 6.0

INQUIRY 77-4

May the experience requirements expressed as "certified NDT Level II" in paragraph 6.3.1(b), (c), and (d) of the 1975 edition of SNT-TC-1A [6.3.2.2 and 6.3.2.3 in the 2020 edition] be considered to include experience gained in "an assignment comparable to that of an NDT Level II" [as stated in paragraph 6.3.1(a)] in determining the prerequisite requirements of a candidate for certification as a Level III?

RESPONSE:

Paragraph 6.2 of the 1975 edition of SNT-TC-1A states, "Documented training and/or experience gained in positions and activities equivalent to those of Level II or Level III prior to establishment of the employer's written practice and a certification program in accordance with this document shall be considered as satisfying the criteria of paragraph 6.2.1 [6.3.1 in the 2020 edition] and 6.3." If documentation was not produced during such prior experience, an affidavit or other suitable testimony regarding such experience may be evaluated by the employer to aid in determining equivalence.

INQUIRY 83-8

We desire to have a member of management for our NDT Level III who, by virtue of their job duties, will not have worked as a Level II. Clarification is needed relative to SNT-TC-1A (1980 edition), paragraphs 6.3.2.1, 6.3.2.2, or 6.3.2.3. The employee may also not have any previous experience in the particular discipline. The questions are:

1. How can a person become a Level III if they have not had a Level II assignment?
2. What types of other experiences could be counted for the Level II requirements?

RESPONSE:

1. An individual may be given credit for work assignments comparable to that of an NDT Level II in the applicable test method.
2. The types of work assignments comparable to the Level II assignments should be defined by the employer in their written practice.

INQUIRY 94-1

Please clarify ASNT's intent regarding the phrase, "assignment comparable to that of an NDT Level II" as it [is] related to SNT-TC-1A, 1984 edition, paragraph 6.3.2.1. I interpret this statement to mean: even though an individual has a four-year degree in science or engineering, they should also have the "documented" NDT training and experience commensurate with that of a Level II. This documented training should follow the guidelines established in Table 6.3.1 of SNT-TC-1A (1984 edition) [Table 6.3.1A in the 2020 edition]. It also requires (in my opinion) this individual practice as a Level II for at least one year before being eligible for Level III certification.

RESPONSE:

Please reference Inquiry 83-8.

INQUIRY 13-5

According to SNT-TC-1A (2011), is it acceptable that an employer considers training and examination received prior to completion of experience requirements as evidence of qualification?

BACKGROUND:

Employer will permit a new employee as trainee only if the trainee has subject knowledge and ability to use equipment/technique safely is established, thus seeking organized training before experience.

RESPONSE:

Yes. Training and examination are partial requirements for the qualifications. A person cannot be considered qualified until all qualification requirements (training, examinations, experience, and visual acuity test) have been met.

INQUIRY 13-6

1. SNT-TC-1A (2011) allows for modification of detailed recommendations as necessary for the particular needs of the employer. In Table 6.3.1B where it requires practical review of 1000 radiographs for the limited certification of radiologic testing (RT) film interpretation, does SNT-TC-1A allow for a reduction in the practical review of the radiographs to suit a specific sector?
2. SNT-TC-1A requires a Level II qualification to allow for limited certification and qualification in a specific method (RT film interpretation). Is this a requirement that does not allow for change to suit the employer needs in its specific sector?

BACKGROUND:

1. This is to stipulate the requirements for industrial experience in a specific sector of the employer that needs to draw up a written practice for its needs.
2. Limited qualifications are obtained for authorized inspection authorities to sign off images, with the intent to accept or reject the images for quality assurance purposes.

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RESPONSE:

1. SNT-TC-1A requires the employer develop a written practice and, where necessary, modify the requirements of SNT-TC-1A to meet its needs, pending customer acceptance of those changes.
2. SNT-TC-1A requires the employer develop a written practice and, where necessary, modify the requirements of SNT-TC-1A to meet its needs, pending customer acceptance of those changes.

INQUIRY 14-4

When applying paragraph 6.3.2.1 (2006 edition) for Level III qualification in five methods (MT, PT, UT, RT, and VT) with graduation from a four-year college, is it acceptable to require in total one year of NDT experience beyond the Level II requirements (that is, total additional experience of 1600 hours in one or more of the five methods, as specified in our written practice), in excess of the total hours in NDT required for Level II, per Table 6.3.1A (that is, for RT and UT 1600 h), and documented work experience in each of the five methods fulfilling the required hours for Level I plus Level II as required per Table 6.3.1A for each of the five methods?

RESPONSE:

The candidate should meet 100% of the Level I and II written practice requirements for a method before beginning the acquiring of additional time in that method for Level III.

It is intended that 100% of the experience time specified in paragraph 6.3.2 be met in each applicable test method. The intent of the experience requirement of paragraph 6.3.2 is that the candidate should be actively working and acquiring progressive experience in the methods for which certification is sought. Just “maintaining” a valid certification does not meet this intent. The employer’s written practice should specify the requirements regarding qualifying work time experience.

INQUIRY 14-5

When requirements are changed on a new revision of an employer’s written practice, starting the effective date of the new revision, which requirements shall be followed for those personnel certified under the old revision? Should the new requirements in the new revision replace the older requirements in the previous edition for those personnel certified under the previous revision?

RESPONSE:

Paragraph 6.0, Table 6.3.1A, and Table 6.3.1B apply only to initial experience required for qualification and certification. New requirements are not intended to be retroactive to previously certified individuals. The current written practice as it relates to recertification would be applicable at the time of recertification.

INQUIRY 15-7

1. Per paragraph 6.3.2.2, does the requirement for two years of experience apply to initial certification only or does it apply to every method?
2. After passing my first NDT Level III method, can I sit for additional methods as long as I fulfill the requirement of Table 6.3.1 A, or do I need to document two years of experience for each method?

RESPONSE:

1. It applies to each applicable method.
2. The requirement for experience in any additional method would not be met until the two-year experience beyond NDT Level II requirement—as defined in Table 6.3.1A—in an assignment at least comparable to that of an NDT Level II in the additional method are met.

INQUIRY 17-5

Are the training and experience requirements of paragraph 6.3.2 of SNT-TC-1A satisfied if the person holds a current ASNT Level III certificate in the specific method?

RESPONSE:

SNT-TC-1A does not address the Level III certificate as a verification that the certificate holder has met the training and experience requirements.

INQUIRY 24-23

Is paragraph 6.4, in the 2024 edition, intended to be read beyond educational institutions, technical schools, and colleges to include training organizations or training done by an independent contractor to a training organization?

RESPONSE:

Yes, provided the expanded training has been formalized using flawed samples, and the results and additional lab hours are documented and approved by the Level III. The course of NDE study at educational institutions and colleges typically far exceeds (weeks not hours) the minimum training hours required by Table 6.3.1 A. That expanded class schedule permits additional lab hours for hands-on evaluation of flawed specimens representative of components to be examined, which can then be credited toward experience hours. That does not preclude Level III trainers or other professional training facilities (who typically schedule their course of study to meet the Table 6.3.1 A requirements) from expanding their program beyond the Table 6.3.1 A requirements, to provide additional lab hours to perform hands-on applications.

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General Comments on Section 6.0

Paragraph 6.3.1, like the rest of SNT-TC-1A, is only a guide. The employer's written practice should detail all such requirements, which may differ from the recommendations of paragraph 6.3.1.

Table note 5.0 provides for reductions in training hours and experience based on additional ET certifications. Similarly, in Table 6.3.1A of the 2011 edition of SNT-TC-1A, this same approach was used to divide radiographic testing (RT) into four techniques. Table note 7.0 provides for reductions in training hours and experience based on additional RT certifications. Additional reductions are allowed for the thermal/infrared method based on its three techniques in table note 9.0.

Inquiry 77-4 points to consideration given in SNT-TC-1A to the initial establishment of a formal qualification and certification program. Paragraph 6.2 recognizes that prior to establishing a program and a written practice, the employer may have provided training as well as other essentials of qualifying NDT personnel without formal procedures. Hence, documentation of such activities could provide evidence of the equivalence of prior activities with those recommended in SNT-TC-1A. Paragraphs 6.3.2.1, 6.3.2.2, and 6.3.2.3 use the phrase "in an assignment at least comparable to that of an NDT Level II." In other words, the experience can be "comparable to that of an NDT Level II" if documentation can be produced that substantiates the comparability.

In the 2011 edition of SNT-TC-1A, Table 6.3.1B from the 2006 edition was removed, thus eliminating the so-called "25% rule." This rule stated, "Initial experience may be gained simultaneously in two or more methods if the candidate spends a minimum of 25% of his [or her] work time on each method for which certification is sought." Table 6.3.1A requires that experience be accumulated by hours rather than months and both minimum and total experience hours must be satisfied. See note 6.0.

NOTE: Table 6.3.1B in the 2006 edition was titled "Alternate Initial Training and Experience Levels" and was deleted in the 2011 edition. Table 6.3.1C, "Initial Training and Experience Levels for Level II Limited Certifications," in the 2006 edition replaced Table 6.3.1B in the 2011 edition. This change was retained in the 2016, 2020, and 2024 editions; however, the title of Table 6.3.1B was amended to "Recommended Initial Training and Experience Levels for Level II Limited Certifications."

Review Questions for Section 6.0

Based on the foregoing discussion, answer the questions on pp. 33–34.

Section 7.0: Training Programs

Section 7.0 (pp. 6–7) of SNT-TC-1A (2024) is reprinted below in full:

7.0 Training Programs

7.1 Personnel being considered for initial certification should complete sufficient organized training. The organized training may include instructor-led training, personalized instruction, virtual instructor-led training, computer-based training, or web-based training. Computer-based training and web-based training should track hours and content of training with student examinations in accordance with paragraph 7.2. The sufficiently organized training shall be such as to ensure the student is thoroughly familiar with the principles and practices of the specified NDT method related to the level of certification desired, and applicable to the processes to be used and the products to be tested. All training programs should be approved by the NDT Level III responsible for the applicable method.

7.2 The training program should include sufficient examinations to ensure understanding of the necessary information and should include the assessment of manufacturing and service-induced (in-service) discontinuities, as applicable.

7.3 Recommended training course outlines and references for NDT Levels I, II, and III personnel, which may be used as technical source material, are contained in ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel*.

7.4 The employer who purchases outside training services is responsible for ensuring that such services meet the requirements of the employer's written practice.

Inquiries for Section 7.0

INQUIRY 84-4

1. Can home-study (correspondence) courses allow one to meet the training requirements of SNT-TC-1A? Would the course better meet SNT-TC-1A if it included quizzes which were mailed back to the responsible "Level III agency" for evaluation, allowing the agency to offer direction to the participant?
2. How many hours credit for training received would the participant be given? Would this be determined by the authors of the course?

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RESPONSE:

1. Yes. A home-study or correspondence course may be used toward satisfaction of SNT-TC-1A recommended training provided it is organized as a course with sufficient quizzes to ensure an understanding of the material and that it is specifically described in the employer's written practice. Quizzes associated with a home-study or correspondence course may be used toward compliance with [Section] 7.0 to ensure comprehension of training information, but not toward compliance with Section 8.0, Examinations.
2. The credit hours should be determined by the Level III responsible for the course as reflected in the employer's written procedure.

INQUIRY 07-1

Per the 2001 edition of SNT-TC-1A, paragraphs 7.1, 7.2, and 9.4.4, and Tables 6.3.1A and 6.3.1B:

1. Can computer- or web-based NDT training with associated electronic quizzes be used to satisfy the training requirements described in paragraphs 7.1 and 7.2?
2. If so, how should an employer document that training hours meet the recommended hours listed in Tables 6.3.1A and 6.3.1B so they can comply with the "satisfactory completion" requirements required by paragraph 9.4.4?

RESPONSE:

1. See Inquiry 84-4, response 1 [above]. Inquiry 84-4 addresses home-study (correspondence) courses, but the response is appropriate for both computer- and web-based training.
2. See Inquiry 84-4, response 2 [above].

General Comments on Section 7.0

In the 2006 edition of SNT-TC-1A, the course outlines have been moved to another publication, ANSI/ASNT CP-105: *Topical Outlines for Qualification of Nondestructive Testing Personnel*. Traditionally, both SNT-TC-1A and CP-189 have published the training course outlines as part of the respective documents. By moving the outlines from SNT-TC-1A and CP-189 and publishing them in CP-105, the problem of having two sets of outlines out of sync due to different publication dates is eliminated and provides a common set of outlines for both SNT-TC-1A and CP-189.

In the 2011 edition of SNT-TC-1A, paragraph 7.1 was modified to clarify the use of alternative means for training such as computer- and/or web-based training. Guidance was also included to ensure that such training provides the user with contact hours, correlation to the applicable training outline, and appropriate examinations. This change has been maintained in the 2024 edition.

Review Questions for Section 7.0

Based on the foregoing discussion, answer the questions on p. 34.

Section 8.0: Examinations

Section 8.0 (pp. 7–11) of SNT-TC-1A (2024) is reprinted below in full:

8.0 Examinations

8.1 Administration and Grading

8.1.1 All qualification examination questions shall be approved by the NDT Level III responsible for the applicable method.

8.1.2 An NDT Level III should be responsible for the administration and grading of examinations specified in paragraphs 8.3 through 8.6 for NDT Level I, II, or other Level III personnel. The administration and grading of examinations may be delegated to a qualified representative of the NDT Level III and documented. Training should be provided to the individual administering the examinations. This training and qualification shall be documented. A qualified representative of the employer may perform the actual administration and grading of NDT Level III examinations specified in paragraph 8.6.

8.1.2.1 To be designated as a qualified representative of the NDT Level III for the administration and grading of NDT Level I, Level II, and Level III personnel qualification examinations, the designee should have documented, appropriate instruction by the NDT Level III in the proper administration and grading of qualification examinations prior to conducting and grading independent qualification examinations for NDT personnel. The NDT Level III Practical examination specified in paragraph 8.6.4.1 shall be administered by a person certified in the applicable method as NDT Level III.

8.1.3 All NDT Level I, II, and III written examinations should be closed-book except that necessary data, such as graphs, tables, specifications, procedures, codes, etc., may be provided with or in the examination. Questions using such reference materials should require an understanding of the information rather than merely locating the appropriate answer.

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8.1.3.1 Sample questions are available in *ASNT Questions & Answers Books* which can be obtained from ASNT. These questions are intended as examples only and shall not be used verbatim for qualification examinations.

8.1.4 For NDT Level I and II personnel, a composite grade should be determined by simple averaging of the results of the General, Specific, and Practical examinations described in the following sections. For NDT Level III personnel, the composite grade should be determined by simple averaging of the results of the Basic, Method, Specific, and Practical (if applicable) examinations described in the following sections.

8.1.5 Examinations administered by the employer for qualification should result in a passing composite grade of at least 80%, with no individual written examination having a passing grade less than 70%. The Practical examination should have a passing grade of at least 80%.

8.1.6 When an examination is administered and graded for the employer by an outside agency and the outside agency issues grades of pass or fail only, on a certified report, then the employer may accept the pass grade as 80% for that particular examination.

8.1.7 The employer who purchases outside services is responsible for ensuring that the examination services meet the requirements of the employer's written practice and be documented.

8.1.8 In no case shall an examination be administered by oneself or by a subordinate.

8.2 Vision Examinations

8.2.1 Near-Vision Acuity. The examination should ensure natural or corrected (no pharmacological agents) near-distance acuity in at least one eye such that the applicant is capable of reading a minimum of Jaeger No. 2 or equivalent type and size letter at the distance designated on the chart but not less than 12 in. (30.5 cm) on a standard Jaeger test chart. The ability to perceive an Ortho-Rater minimum of 8 or similar test pattern is also acceptable. This should be administered annually.

8.2.1.1 Pharmacological agents (eye drops) that would improve or enhance visual acuity at any distance shall not be used.

8.2.2 Color-Contrast Differentiation. The examination should demonstrate the capability of distinguishing and differentiating contrast among colors or shades of gray used in the method as determined by the employer. This should be conducted upon initial certification and at five (5) year intervals thereafter.

8.2.3 Vision examinations expire on the last day of the month of expiration.

8.3 General (Written–for NDT Levels I and II)

8.3.1 The General examinations should address the basic principles of the applicable method.

8.3.2 In preparing the examinations, the NDT Level III should select or devise appropriate questions covering the applicable method and techniques described by the employer's written practice and the applicable elements of the outline in ANSI/ASNT CP-105.

8.3.3 The minimum number of questions that should be given is shown in Table 8.3.3.

8.3.4 A valid ACCP, ASNT NDT, or ASNT 9712 Level II certificate may be accepted as fulfilling the General examination criteria for each applicable method if the NDT Level III has determined that the ASNT examinations meet the requirements of the employer's written practice. This acceptance should be documented.

8.4 Specific (Written–for NDT Levels I and II)

8.4.1 The Specific examination should address the equipment, operating procedures, and NDT techniques that the individual may encounter during specific assignments to the degree required by the employer's written practice and the applicable elements of the outline in ANSI/ASNT CP-105.

8.4.2 The Specific examination should also cover the procedures, specifications or codes and acceptance criteria used in the NDT conducted by the employer.

8.4.3 The minimum number of questions that should be given is shown in Table 8.3.3.

8.4.4 A valid ACCP, ASNT NDT, or ASNT 9712 Level II certificate may be accepted as fulfilling the Specific examination criteria for each applicable method if the NDT Level III has determined that the ASNT examinations meet the requirements of the employer's written practice. This acceptance should be documented. If this assessment cannot be accomplished, an employer-administered Specific examination should be completed.

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TABLE 8.3.3

Minimum Number of Examination Questions

Method/Technique	General		Specific	
	Level I	Level II	Level I	Level II
Acoustic Emission Testing	40	40	20	20
Electromagnetic Testing				
Alternating Current Field Measurement	40	40	20	20
Eddy Current Testing	40	40	20	20
Remote Field Testing	30	30	20	20
Ground Penetrating Radar	30	40	20	20
Guided Wave Testing	40	40	20	20
Laser Methods Testing				
Profilometry	30	30	20	20
Holography/Shearography	30	30	20	20
Leak Testing				
Bubble Leak Testing	20	20	15	15
Pressure Change Leak Testing	20	20	15	15
Halogen Diode Leak Testing	20	20	15	15
Mass Spectrometer Leak Testing	20	20	20	40
Liquid Penetrant Testing	40	40	20	20
Magnetic Flux Leakage Testing	20	20	20	15
Magnetic Particle Testing	40	40	20	20
Microwave Technology Testing	40	40	20	20
Neutron Radiographic Testing	40	40	20	20
Radiographic Testing				
Radiography	40	40	20	20
Radiographic Film Interpretation – Non-Radiographer		40		20
Radiographic Film Interpretation – Radiographer (Certified RT NDT Level I)		20		15
Computed Radiography	40	40	20	20
Computed Tomography	40	40	20	20
Digital Radiography	40	40	20	20
Thermal/Infrared Testing	40		20	
Building Diagnostics		50		40
Electrical and Mechanical Testing		50		40
Materials Testing		50		40
Ultrasonic Testing	40	40	20	20
Full Matrix Capture		40		30
Phased Array Ultrasonic Testing		40		30
Time of Flight Diffraction		40		30
Digital Thickness Measurement (numeric output only)		20		10
A-scan Thickness Measurement		30		15
Vibration Analysis	40	40	20	60
Visual Testing	40	40	20	20

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8.5 Practical (for NDT Level I and II)

8.5.1 The candidate should demonstrate familiarity with and ability to operate the necessary NDT equipment, record, and analyze the resultant information to the degree required.

8.5.2 At least one flawed specimen or component should be tested, and the results of the NDT analyzed by the candidate. Specimens should include manufacturing or service-induced (in-service) discontinuities based on the needs of the employer.

8.5.2.1 Phased Array Ultrasonic Testing and Time of Flight Diffraction Practical Examination. Flawed samples used for practical examinations should be representative of the components and/or configurations that the candidates would be testing under this endorsement and approved by the NDT Level III.

8.5.2.2 Film Interpretation Limited Certification. The Practical examination should consist of review and grading of a sufficient number of radiographs to demonstrate satisfactory performance to the satisfaction of the NDT Level III. The number of radiographs should be addressed in the employer's written practice.

8.5.3 The description of the specimen, the NDT procedure, including checkpoints, and the results of the examination should be documented.

8.5.4 NDT Level I Practical Examination

8.5.4.1 Specimens. Proficiency should be demonstrated in performing the applicable NDT technique on one or more flawed specimens as appropriate for the method and approved and documented by the NDT Level III (Grading Key).

8.5.4.2 Evaluation. The NDT Level I should evaluate the results to the degree of responsibility as described in the employer's written practice. The candidate should detect all discontinuities and conditions specified and documented by the NDT Level III. The written practice should address the acceptable detection rate as well as the maximum number of false calls acceptable.

8.5.4.3 Grading. A checklist containing at least 10 different checkpoints requiring an understanding of test variables and the employer's procedural requirements should be included in this Practical examination. While it is normal to score the Practical on a percentile basis (80% required), the Practical examination checklist should also contain a single checkpoint or multiple checkpoints that failure to successfully complete will result in failure of the examination. This requirement should be clearly marked on the checkpoint(s).

8.5.5 NDT Level II Practical Examination

8.5.5.1 Specimens. Proficiency should be demonstrated in selecting and performing the applicable NDT technique within the method and interpreting and evaluating the results on one or more flawed specimens as appropriate for the method and approved and documented by the NDT Level III (Grading Key).

8.5.5.2 Evaluation. The candidate should detect all discontinuities and conditions specified and documented by the NDT Level III. The written practice should address the acceptable detection rate as well as the maximum number of false calls acceptable.

8.5.5.3 Grading. A checklist containing at least 10 different checkpoints requiring an understanding of NDT variables and the employer's procedural requirements should be included in this Practical examination. While it is normal to score the Practical on a percentile basis (80% required), the practical examination checklist should also contain a single checkpoint or multiple checkpoints that failure to successfully complete will result in failure of the examination. This requirement should be clearly marked on the checkpoint(s).

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8.5.6 A valid ACCP or ASNT 9712 Level II certificate may be accepted as fulfilling the Practical examination criteria for each applicable method if the NDT Level III has determined that the ASNT examinations meet the requirements of the employer's written practice. This acceptance should be documented. If this assessment cannot be accomplished, an employer-administered Practical examination should be completed.

8.5.7 An example of a Practical examination checklist is attached as Appendix A to this Recommended Practice. The example checklist has been provided as guidance on the development of practical examinations for any method and level.

8.6 NDT/PdM Level III Examinations

8.6.1 Basic Examinations

8.6.1.1 NDT Basic Examination (does not need to be retaken to add another test method as long as the candidate holds a current Level III certificate or certification). The minimum number of questions that should be given is as follows:

8.6.1.1.1 Fifteen (15) questions relating to understanding the SNT-TC-1A document.

8.6.1.1.2 Twenty (20) questions relating to applicable materials, fabrication, and product technology.

8.6.1.1.3 Twenty (20) questions that are similar to published NDT Level II questions for other appropriate NDT methods.

8.6.1.2 PdM Basic Examination (does not need to be retaken to add another test method as long as the candidate holds a current Level III certificate or certification). The minimum number of questions that should be given is as follows:

8.6.1.2.1 Fifteen (15) questions relating to understanding the SNT-TC-1A document.

8.6.1.2.2 Twenty (20) questions relating to applicable machinery technology.

8.6.1.2.3 Thirty (30) questions that are similar to published NDT Level II questions for other appropriate PdM methods.

8.6.2 Method Examination (for each method).

8.6.2.1 Thirty (30) questions relating to fundamentals and principles that are similar to published ASNT NDT Level III questions for each method,

8.6.2.2 Fifteen (15) questions relating to application and establishment of techniques and procedures that are similar to the published ASNT NDT Level III questions for each method, and

8.6.2.3 Twenty (20) questions relating to capability for interpreting codes, standards, and specifications relating to the method.

8.6.3 Specific Examination (for each method).

8.6.3.1 Twenty (20) questions relating to specifications, equipment, techniques, and procedures applicable to the employer's product(s) and methods employed and to the administration of the employer's written practice.

8.6.4 Practical Examination (for each method).

8.6.4.1 The candidate should prepare an NDT procedure appropriate to the employer's needs; however, if documented experience demonstrates that the candidate has previously prepared acceptable NDT procedures in the method using the specifications, codes, and standards that are applicable to that employer, a written Practical examination (for example, preparation of a procedure) is not required. If experience is substituted for the written Practical examination, the employer should document the pertinent practical experience of the NDT Level III candidate.

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8.6.4.2 If the NDT Level III will be required to perform NDT or evaluate NDT results, the Practical examination should include the same demonstrations of the candidate's ability to perform the required activity(ies) as required in paragraph 8.5.5. This Practical examination shall be administered by a person certified in the applicable NDT method as NDT Level III. For NDT Level III personnel certified under the employer's current written practice, this requirement should be met prior to the next recertification date, in the applicable method.

8.6.4.3 If the NDT Level III needs to meet the recommendations in paragraph 8.6.4.2, a valid ACCP or ASNT 9712 Level III certificate may be accepted as fulfilling the practical examination criteria in paragraph 8.5.5 for each applicable method if the NDT Level III has determined that the ASNT examinations meet the requirements of the employer's written practice. This acceptance should be documented. If this assessment cannot be accomplished, an employer-administered Practical examination should be completed.

8.6.5 A valid endorsement on an ASNT NDT, ACCP Professional, or ASNT 9712 Level III certificate fulfills the examination criteria described in paragraphs 8.6.1 and 8.6.2 for each applicable NDT method. A valid ACCP Professional or ASNT 9712 Level III certificate also fulfills the examination criteria described in paragraph 8.5.5.

8.7 Reexamination

8.7.1 Those failing to attain the required grades should wait at least thirty (30) days or receive suitable additional training as determined by the NDT Level III before reexamination.

Inquiries for Section 8.0

INQUIRY 99-3

Is it the intent of SNT-TC-1A that any composite examination grade less than 80% should be considered a failing grade?

RESPONSE:

Yes.

INQUIRY 05-1

Paragraph 8.1.3 states, "Examinations administered for qualification should result in a passing grade of at least 80%," but if the employer's written practice states that each examination must achieve 70% or more, and does not require an 80% composite, is this acceptable?

RESPONSE:

No. See Inquiry 99-3.

NOTE TO THE INQUIRER:

The SNT-TC-1A Interpretation Panel reviewed this question and agreed with the original response.

INQUIRY 08-2

Per the 2001 edition of SNT-TC-1A, a Level II examiner is certified in PT, MT, and VT and has been approved as capable of distinguishing and differentiating contrast among colors according to Ishihara standard plates, as determined by the written practice. Does this person also require approval by a shades of gray acuity examination to be qualified in RT to meet the requirements of paragraph 8.2.2?

RESPONSE:

Yes. … Paragraph 8.2.2 allows the employer to determine the method of testing for color differentiation or gray shade differentiation. Whether to test for color differentiation or gray shade differentiation is determined by which is appropriate for the method the individual is being certified in.

INQUIRY 10-4

May an employer appoint directly an "EN 473 Level III" as employer's NDT Level III after defined in employer's written practice to sign the certificates according to SNT-TC-1A in order to verify qualification of candidate for certification?

RESPONSE:

The employer may use the examination results that they have determined meet the requirement of the written practice. As stated in SNT-TC-1A, paragraph 8.1.5 [8.1.7 in the 2020 edition], the employer is responsible for ensuring the examination services meet the requirements of the employer's written practice. Certification can only be issued when it has been determined that all the requirements of the employer's written practice have been met.

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INQUIRY 14-7

Regarding paragraph 8.2 of SNT-TC-1A (2011), who may be responsible to carry out the vision acuity examination? We have in our written practice three possible options:

1. Performed by a nurse and approved by an NDT Level III; or
2. Performed by an ophthalmologist and approved by an NDT Level III; or
3. Performed by the NDT Level III.

Are all correct?

RESPONSE:

SNT-TC-1A does not address who should conduct the visual examination. All three of the options would be acceptable if addressed in the written practice.

INQUIRY 14-10

According to SNT-TC-1A (2011), if a person is being qualified directly to Level II, do they need to take both the Level I and Level II examinations, or can that person take just the Level II examinations?

RESPONSE:

Only the examinations for the level of certification being sought are required.

INQUIRY 17-7

Is it the intent of paragraphs 8.5.4 and 8.5.5 that the candidate has failed the examination if the candidate does not detect all discontinuities specified as mandatory and conditions specified by the NDT Level III?

RESPONSE:

Yes, if specified as mandatory by the NDT Level III.

General Comments on Section 8.0

In the 2011 edition of SNT-TC-1A, provisions were included to designate qualified individuals to administer and grade examinations. Such designations must be documented.

Also in the 2011 edition, employers were provided the option of using an ASNT NDT Level II certificate to satisfy the general examination requirement. This is an additional option to the ACCP Level II certificate used to satisfy this same requirement that was added in the 1996 edition of SNT-TC-1A in a 1998 addenda.

These changes have been retained in the 2020 and 2024 editions.

Review Questions for Section 8.0

Based on the foregoing discussion, answer the questions on pp. 34–35.

Section 9.0: Certification

Section 9.0 (pp. 11–12) of SNT-TC-1A (2024) is reprinted below in full:

9.0 Certification

9.1 Certification of NDT personnel to all levels of qualification is the responsibility of the employer.

9.2 Certification of NDT personnel should be based on demonstration of satisfactory qualification in accordance with Sections 6.0, 7.0, and 8.0, as described in the employer's written practice.

9.3 At the option of the employer, an outside agency may be engaged to provide NDT Level III services. In such instances, the responsibility of certification of the employees shall be retained by the employer.

9.4 Personnel certification records should be maintained on file by the employer for the duration specified in the employer's written practice and should include the following:

9.4.1 Name of certified individual.

9.4.2 Level of certification, NDT method and/or technique, as applicable, and limitations (if any), as applicable.

9.4.3 Educational background and experience of certified individuals.

9.4.4 Statement indicating satisfactory completion of training in accordance with the employer's written practice.

9.4.5 Results of the vision examinations prescribed in paragraph 8.2 for the current certification period.

9.4.6 Current examination copy(ies) or evidence of successful completion of examinations.

9.4.7 Composite grade(s) or suitable evidence of grades.

9.4.8 Signature of the NDT Level III that verified qualifications of candidate for certification.

9.4.9 Dates of certification and/or recertification.

9.4.10 Certification expiration date.

9.4.11 Signature of employer's certifying authority.

9.5 Dates of certification and expiration.

9.5.1 The date of certification should be a date after which the employer's requirements for education, training, experience, and examinations have been met for each method and/or technique, as specified in the employer's written practice.

9.5.2 The date of certification is the date when the Certifying Authority signs the certification record.

9.5.3 The date of expiration should be five (5) years from the date established in paragraph 9.5.2.

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9.6 Certification certificates and wallet cards. If the employer chooses to use certificates or wallet cards the following recommendations are considered best practice and should be followed:
- 9.6.1 The legal name of the employer should be clearly identified followed by the document number and revision/edition number of the written practice.
- 9.6.2 The certificate should indicate "SNT-TC-1A" followed by the four-digit year of the applicable edition of SNT-TC-1A that the written practice was developed to follow.
- 9.6.3 Full legal name of the certified individual in the ISO basic Latin alphabet. Additional characters may also be used where customary to accommodate local languages.
- 9.6.4 Level of certification, NDT method and/or technique, as applicable and limitations (if any).
- 9.6.5 Limitations of certification authorization should be clearly defined, when appropriate for the employer, which may include restrictions on product forms, industry sectors, codes, standards, and examination procedures.
- 9.6.6 The full printed legal name and signature of both the designated NDT Level III that verified the qualifications of the candidate for certification and the designated certifying authority when the Level III and certifying authority are identified as separate positions in the employer's written practice.
  - 9.6.6.1 If applicable, the ASNT issued Level III identification number, when either or both of the signers hold ASNT Level III certifications.
- 9.6.7 The certificate or wallet card should be fully completed at the time of signing by the designated NDT Level III and/or certifying authority so that no fill-in-the-blank data remains to be completed by others.
  - 9.6.7.1 Changes to certificates or wallet cards should require that a new certificate or wallet card be issued whenever changes or alterations are necessitated.
- 9.6.8 Dates of certification and expiration.

Inquiries for Section 9.0

INQUIRY 13-1

Paragraph 9.4.9 of SNT-TC-1A (2006) [paragraph 9.4.8 in the 2020 edition] says, "Signature of the Level III that verified qualifications of candidate for certification." If the candidate whose qualifications are being verified is an ASNT NDT Level III, must another Level III sign this verification?

RESPONSE:

Yes.

INQUIRY 14-1

We engage an outside agency to provide NDT Level III services. In SNT-TC-1A (2011), paragraph 9.3 states in part that the responsibility of certification of the employees should be retained by the employer. Paragraph 12.1.2 states that reexamination in those portions of the examinations in Section 8.0 are deemed necessary by the employer's NDT Level III.

Since we engage the services of an outside agency for Level III, does the outside agency's Level III determine which portions of the examinations in Section 8.0 are necessary?

RESPONSE:

Yes, as authorized by the contracting employer.

INQUIRY 15-5

In the 2006 edition of SNT-TC-1A, in paragraph 9.4.9, for the purposes of certification by an employer, must a Level III's qualifications be verified by another Level III prior to certification by the employer?

RESPONSE:

The employer's Level III must have documented verification that their Level III's qualifications meet the examination requirements of SNT-TC-1A. The documented verification should carry the signature of a Level III.

INQUIRY 15-9

Should certification documents contain dates of assignment to NDT as required by paragraph 9.4.10 (2011 edition)?

RESPONSE:

The intent of "dates of assignment to NDT" in paragraph 9.4.10 is that the employer, as a minimum, document the start and end date of employment assignments that affect the nature of the NDT work done by the individual.

General Comments on Section 9.0

Situations arise where employers, for various reasons, must resort to sources outside the employer's organization to provide services attendant to training, examination, and other activities that require qualified NDT personnel to conduct. Some companies cannot afford, or may not have time to develop, training sources and examinations to qualify their NDT personnel.

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Others simply prefer to contract such services as needed. For whatever reasons outside services may be needed or desired, the responses above merely reinforce the basic principles underlying SNT-TC-1A: "Certification of all levels of NDT personnel is the complete responsibility of the employer" and "the employer shall establish written practices covering all phases of certification including training as specified in Section 5.0."

Inquiries 90-4 and 96-3 address the issue of whether a Level III must take an examination in order to become certified. This is of special note in that versions prior to the 1988 edition clearly permitted examination waivers for the NDT Level III, leading to the concept of certification by "appointment." This latter practice has been strongly criticized throughout the industry since it is perceived as a loophole for employers to certify underqualified individuals to assume the key role of administering the NDT activities within an organization. In order to restrict this practice, the 1988 edition of SNT-TC-1A called for Level IIIs to be qualified by examination. But being a recommended practice that encourages employers to adapt their respective requirements based on each set of circumstances, the opportunity still remains open to employers to "appoint" their Level IIIs, as long as their written practice reflects this strategy as one of their permitted options.

Review Questions for Section 9.0

Based on the foregoing discussion, answer the questions on pp. 35–36.

Section 10.0: Technical Performance Evaluation

Section 10.0 (p. 12) of SNT-TC-1A (2024) is reprinted below in full:

10.0 Technical Performance Evaluation

10.1 NDT personnel may be reexamined anytime at the discretion of the employer and have their certificates extended or revoked.

10.2 Periodically, as defined in the employer's written practice, NDT Level I and II personnel should be reevaluated by the NDT Level III administering a Practical examination. The Practical examination should follow the format and guidelines described in paragraph 8.5.

10.2.1 If the NDT Level III needs to meet the recommendations in paragraph 8.6.4.2, the requirements in paragraph 10.2 should be met.

Inquiries for Section 10.0

INQUIRY 02-02

Is it the intent of SNT-TC-1A (2001) paragraph 10.2 to recommend periodic testing with test props with at least one flawed specimen or are other means of technical performance evaluation acceptable, for example, monitoring an inspector performing an official production inspection?

RESPONSE:

The means for periodic technical performance evaluation should be described in the employer's written practice. Note that the evaluation of technical performance should consist of not only monitoring the proper application of technique, but also the ability to recognize relevant indications and evaluate those indications against the employer's acceptance criteria.

INQUIRY 12-9

Is it correct that an Authorized Inspector could insist the employer must perform hands-on Practical examination every two years for the Level I and II personnel, in order to fulfill the requirements of paragraph 10.2 (periodical technical performance)?

As an employer, we would state on our written practice that the continuing NDT performance of the Level Is and IIs has been satisfied by the client and the appointed Level III, then he or she should be evaluated for good technical performance without taking any new hands-on Practical examination.

RESPONSE:

Additional customer or regulatory requirements are beyond the scope of Recommended Practice No. SNT-TC-1A.

INQUIRY 13-2

In SNT-TC-1A (2011), paragraph 10.2 states, "Periodically, as defined in the employer's written practice, NDT Level I and II personnel should be reevaluated by the NDT Level III administering a practical examination. The practical examination should follow the format and guidelines described in Section 8.5."

Is it the intent of 10.2 that technical performance review be performed between certification periods, or is it acceptable to perform them when a person's certification comes due for renewal if that is how the evaluation period is defined in the employer's written practice?

RESPONSE:

The intent is that the employer defines in the written practice when the technical performance review will be performed.

General Comments on Section 10.0

This section was not in editions of SNT-TC-1A before 2001; however, paragraph 10.1 was addressed in paragraph 9.5(3) of the 1996 edition. Paragraph 10.2 was added to provide a means whereby an employer could ensure that certified personnel are continuously and satisfactorily performing their responsibilities. It is assumed that, should an individual's performance evaluation indicate substandard performance, additional training would be provided and reexamination would be conducted.

Review Questions for Section 10.0

Based on the foregoing discussion, answer the questions on pp. 36.

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Section 11.0: Interrupted Service

Section 11.0 (p. 12) of SNT-TC-1A (2024) is reprinted below in full:

11.0 Interrupted Service

11.1 The employer's written practice should include rules covering the types and duration of interrupted service that requires reexamination and recertification.

11.2 The written practice should specify the requirements for reexamination and/or recertification for the interrupted service.

Inquiries for Section 11.0

INQUIRY 22-09

SNT-TC-1A (2020), Paragraph 11.0

1. What is meant by the term "interrupted service"?
2. Does interrupted service pertain to when an employee is still employed by the employer, but steps away from the employer for a period of time for a medical leave, military leave, leave of absence, etc.?
3. Does interrupted service pertain to when an employee leaves an employer, then has a gap in the time where the employee is not performing testing between the employment with the old employer and the employment with the new employer?
4. Does interrupted service pertain to when an employee is working for the employer, but not performing testing in a method for which certified for a period of time, such as six months? As an example, an employee is certified for RT, MT, PT, and VT, and they do not perform RT for six months, but are still performing testing in the remaining methods for which certified.
5. Does interrupted service pertain to when an employee is still working for the employer but is not performing testing or any other NDT-related activities in the methods for which certified?

RESPONSE:

1. To be defined in the employer's written practice.
2. If it is defined as such in the employer's written practice.
3. This would be covered under Section 13.0, Termination.
4. If it is defined as such in the employer's written practice.
5. If it is defined as such in the employer's written practice.

General Comments on Section 11.0

This section was not in editions of SNT-TC-1A before 2001; however, the subject of interrupted services was included in paragraph 9.5 (Recertification) of the 1996 edition.

Review Questions for Section 11.0

Based on the foregoing discussion, answer the question on pp. 36.

Section 12.0: Recertification

Section 12.0 (pp. 12–13) of SNT-TC-1A (2024) is reprinted below in full:

12.0 Recertification

12.1 All levels of NDT personnel shall be recertified periodically in accordance with one of the following criteria:

12.1.1 Evidence of continuing satisfactory technical performance.

12.1.2 Reexamination in those portions of the examinations in Section 8.0 deemed necessary by the employer's NDT Level III.

12.2 The recommended maximum recertification intervals are five (5) years from the date the Certifying Authority signs the certification record for all certification levels. Certifications expire on the last day of the month of expiration.

12.3 When new techniques are added to the employer's written practice and the NDT Level III personnel is assigned to perform examinations using these new techniques, the NDT Level III personnel should receive applicable training, take applicable examinations and obtain the necessary experience, such that the NDT Level III meets the requirements of the new techniques in Table 6.3.1. A, prior to their next recertification date, in the applicable method.

Inquiries for Section 12.0

INQUIRY 91-4, QUESTION 2

2. What constitutes "evidence of continuing satisfactory performance"?

RESPONSE:

2. The intent of "evidence of continuing satisfactory performance" in paragraph 9.7.1 [12.1.1 in the 2020 edition] of SNT-TC-1A is that the employer shall have objective evidence of an individual's satisfactory performance during the previous certification period of the appropriate duties outlined in the employer's written practice.

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INQUIRY 03-01

1. Regarding SNT-TC-1A (1984) paragraph 9.7.1 [12.1.1 in the 2020 edition], what is considered proper "evidence of continuing satisfactory performance"?
2. Does this mean the certified individual shall maintain documentation of satisfactory performance?
3. How often should performance be documented?
4. CP-189 (1991) indicates suspension should occur if duties are not performed during any consecutive 12-month period (Section 7.2). Is this implied in SNT-TC-1A?

RESPONSE:

1. Please reference Inquiry 91-4 Question 2 [above].
2. No.
3. SNT-TC-1A (1984) provides the user sufficient latitude under paragraph 9.7.1 [12.1.1 in the 2020 edition] to identify specifically how they would evaluate "evidence of continuing satisfactory performance." The requirements should be documented in the employer's written practice.
4. There are no specific provisions in the [1984 edition of] SNT-TC-1A regarding interruption of NDT duties while continuing to work for the same employer. However, the employer must be satisfied with the proficiency of any individual at any level to handle work tasks. The employer has direct knowledge of the employee's prior performance and can best judge the need for reexamination as a function of duration of interrupted NDT service.

INQUIRY 04-2

1. Is it the intent of paragraph 12.1.1 that individuals who are recertified based on "continuing satisfactory technical performance" must pass a new Practical examination?
2. May a Level I or II individual be recertified, based on "continuing satisfactory technical performance" without taking a new Practical examination?

RESPONSE:

1. No. See Inquiry 91-4, Question 2.
2. Yes. SNT-TC-1A provides the user sufficient latitude under paragraph 9.7.1 [12.1.1 in the 2020 edition] to identify specifically how they would evaluate "evidence of continuing satisfactory performance." The requirements should be documented in the employer's written practice.

INQUIRY 13-11

Does the technical performance evaluation mentioned in paragraph 12.1.1 of SNT-TC-1A (2006) refer to the periodic technical performance evaluation as mentioned in paragraph 10.2?

BACKGROUND:

Paragraph 12.1 states that "all levels of NDT personnel shall be recertified periodically in accordance with one of the following criteria..."

Paragraph 12.1.1 mentions one of the criteria as "Evidence of continuing satisfactory technical performance."

RESPONSE:

No. Paragraph 10.2 says, "The [technical performance] evaluation and documentation should follow the format and guidelines described in Section 8.5." Section 8.5 describes the Practical examination. The "continuing satisfactory technical performance" mentioned in 12.1.1 refers to evidence of having satisfactorily performed those tasks that are relevant to the level of qualification of the person being evaluated.

INQUIRY 14-1

We engage an outside agency to provide NDT Level III services. In SNT-TC-1A (2011), paragraph 9.3 states in part that the responsibility of certification of the employees should be retained by the employer. Paragraph 12.1.2 states that reexamination in those portions of the examinations in Section 8.0 are deemed necessary by the employer's NDT Level III.

Since we engage the services of an outside agency for Level III, does the outside agency's Level III determine which portions of the examinations in Section 8.0 are necessary?

RESPONSE:

Yes, as authorized by the contracting employer.

INQUIRY 18-5

If a person has previously qualified as NDT Level II that required recertification in according with Recommended Practice SNT-TC-1A, edition 2016, Section 12.2 (due to certification expiration date) to be performed by outside agency and the person has met all requirements of continuous performance satisfactory to the method referred, physical aptitude, level education, formal training, and experience, and has passed the method examination then the new recertification interval shall be 5 years (not less)?

RESPONSE:

The maximum recertification interval shall not exceed 5 years. The employer's written practice may specify a shorter period.

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INQUIRY 18-11

Our firm provided training to our inspectors according to SNT-TC-1A (2011 version). It has been 5 years since the certifications were issued to them. Recertification would be necessary now. However, Recommended Practice SNT-TC-1A itself has been updated to the 2016 version. In such a situation, is it still valid to reissue certificates to SNT-TC-1A (2011)? Or are we entitled to issue certificates to SNT-TC-1A (2016) instead? Is there any specific requirement for the inspectors to update from the 2011 to the 2016 version?

RESPONSE:

The version utilized by the employer is determined by the employer's written practice unless the purchasing requirements indicate otherwise.

General Comments on Section 12.0

This section was not in editions of SNT-TC-1A before 2001; however, this information was in paragraph 9.5 (Recertification) of the 1996 edition. The only change beginning with the 2006 edition is the change in the recommended maximum certification interval to five (5) years.

Review Question for Section 12.0

Based on the foregoing discussion, answer the questions on p. 36.

Section 13.0: Termination

Section 13.0 (p. 12) of SNT-TC-1A (2024) is reprinted below in full:

13.0 Termination

13.1 The employer's certification shall be deemed revoked when employment is terminated.

13.2 An NDT Level I, Level II, or Level III whose certification has been terminated may be certified to the former NDT level by a new employer based on examination, as described in Section 8.0, provided all of the following conditions are met to the new employer's satisfaction:

13.2.1 The employee has proof of prior certification.

13.2.2 The employee was working in the capacity to which certified within six months of termination.

13.2.3 The employee is being recertified within six (6) months of termination.

13.2.4 Prior to being examined for certification, employees not meeting the above requirements should receive additional training as deemed appropriate by the NDT Level III.

Inquiries for Section 13.0

INQUIRY 87-5

A Level II is terminated from Company A and is employed by Company B as a trainee. Eight months after termination, Company B certifies the employee to Level II. May an employee working at a lower NDT level be upgraded to their former NDT level without retraining?

RESPONSE:

Inquiry 84-3 established the precedent that once an individual has met the certifying employer's required training, even if the training were attained while working for another employer, it is not necessary to repeat the training. Based on this, the trainee could be certified to the former level. However, this assumes that all of the certifying employer's written practice requirements are satisfied. It should further be noted that based on paragraph 10.2.3 of SNT-TC-1A (which requires recertification within six months of termination), certification eight months after termination does not comply with the stated conditions. The employer's written practice should address this matter.

Review Questions for Section 13.0

Based on the foregoing discussion, answer the questions on pp. 37.

Section 14.0: Reinstatement

Section 14.0 (p. 13) of SNT-TC-1A (2024) is reprinted below in full:

14.0 Reinstatement

14.1 An NDT Level I, Level II, or Level III whose certification has been terminated may be reinstated to the former NDT level, without a new examination, provided all of the following conditions are met:

14.1.1 The employer has maintained the personnel certification records required in paragraph 9.4.

14.1.2 The employee's certification did not expire during termination.

14.1.3 The employee is being reinstated within six (6) months of termination.

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Inquiries for Section 14.0

INQUIRY 78-10

NOTE: This inquiry relates to paragraph 14.0 beginning with the 2001 edition.

3. When an employee returns to work for a former employer where he or she was certified, may this employee's certification(s) be reinstated without examination if the provisions of paragraphs 10.2(b) and 10.2(c) (1980 edition) are met?
4. If an employee has been continuously working for another employer certified in the same capacities, may their certification(s) be reinstated for the remainder of the original three-year period of certification in accordance with paragraph 9.7 [12.0 in the 2020 edition]?

RESPONSE:

3. The provisions of paragraph 10.2 apply only to a new employer. For this part of the inquiry, the provisions of paragraph 9.7.3 [paragraph 11.1 in the 2020 edition] would prevail.
4. For this part of the inquiry, the receiving employer would be considered a new employer and the provisions of paragraph 10.2 would apply. If, however, the employee in question was previously employed by the receiving employer, paragraph 9.7.3 [paragraph 11.1 in the 2020 edition] would prevail.

INQUIRY 23-15

An NDT Level III whose certification has been terminated in Company A; may the certification be reinstated to the former NDT level in Company B, without a new examination, when all of the paragraph 14.1.1, 14.1.2, and 14.1.3 [SNT-TC-1A 2020 edition] conditions are met?

RESPONSE:

No. Reinstatement pertains to the employee's previous employer who issued the certification, not a new employer.

General Comments on Section 14.0

This section was not in editions of SNT-TC-1A before 2001. This section was added to establish the guidelines for reinstatement of an individual's certification that has been terminated. This specifically relates to a former employee, not a new employee (see paragraph 13.2). This would cover those situations where a certified employee had their certification terminated or suspended due to a temporary layoff, or a temporary non-NDT job assignment.

NOTE: There are no review questions for Section 14.0.

Section 15.0: Referenced Publications

Section 15.0 (p. 13) of SNT-TC-1A (2024) is reprinted below in full:

15.0 Referenced Publications

15.1 The following documents contain provisions which, through reference in this text, constitute provisions of this Recommended Practice. Copies may be obtained from The American Society for Nondestructive Testing Inc., 1201 Dublin Rd., Suite #G04, Columbus, OH 43215, USA, or asnt.org.

15.1.1 ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel*, latest edition.

15.1.2 ANSI/ASNT CP-9712 (ISO 9712:2021): 2023 *Standard Nondestructive Testing - Qualification and Certification of NDT Personnel*.

15.1.3 ASNT Central Certification Program, *ASNT Document CP-1*, latest edition.

15.1.4 ASNT Policy G-14 - *Use of the ASNT Name and ASNT Marks*, latest edition.

15.2 The following document contains specific NDT terms which, though referenced in this text, constitute provisions of this Recommended Practice. Copies may be obtained from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, USA.

15.2.1 ASTM E 1316 (latest edition) - *Standard Terminology for Nondestructive Examinations,* Section A – Common NDT Terms.

NOTE: There are no review questions for Section 15.0.

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REVIEW QUESTIONS

Sections 1.0, 2.0, and 3.0

1. Which of the following statements is true concerning the usage of SNT-TC-1A?
   a. SNT-TC-1A is intended for use by a limited set of industrial segments.
   b. SNT-TC-1A was generated to satisfy the specification requirements of ASME.
   c. SNT-TC-1A was last revised in 1988.
   d. SNT-TC-1A is not intended to be used as a strict specification.

2. The SNT-TC-1A Interpretation Panel will respond to inquiries about SNT-TC-1A. The inquiry must be:
   a. written and stated in general terms.
   b. written including specific details of the case, such as names, places, dates, and other pertinent facts.
   c. considered by the ASNT Board of Directors if it involves decisions that would place ASNT in a position between buyer and seller.
   d. ruled upon by the Technical and Education Council of ASNT before the SNT-TC-1A Interpretation Panel prepares a response.

3. Which of the following statements is true?
   a. ASNT has been providing certification examinations for Level III personnel since the late 1960s.
   b. SNT-TC-1A was first published in the late 1960s.
   c. SNT-TC-1A requires that Level III personnel be qualified and certified by ASNT.
   d. SNT-TC-1A was developed to satisfy requirements for NDT personnel qualification and certification set forth in parts of the *ASME Boiler & Pressure Vessel Code*.

4. With regard to the training of NDT personnel, the employer:
   a. must conduct all of the training on the premises.
   b. may engage an outside service who must conduct the training on the employer's premises.
   c. should not conduct the training, being too close to the company's problems.
   d. may engage an outside service but is, nevertheless, responsible for the certification of the company's NDT personnel.

5. SNT-TC-1A is intended as a guideline for employers:
   a. to establish their own written practice that must be used as a strict specification.
   b. to establish their own written practice for the qualification and certification of their NDT personnel.
   c. to define training course requirements for contractors performing outside training services.
   d. who are corporate members of ASNT.

6. The following is a statement in SNT-TC-1A: "It is recognized that these guidelines may not be appropriate for certain employers, circumstances, and/or applications." What should be done if the guidelines are not appropriate?
   a. The employer must change its operations to conform to the guidelines.
   b. The employer must seek relief from ASNT for inappropriate applications.
   c. The employer should review the detailed recommendations and modify them, as necessary, to meet its particular needs with justification documented in an annex to the written practice.
   d. The employer shall submit an inquiry in writing directed to the SNT-TC-1A Interpretation Panel.

7. Which of the following statements is true?
   a. Qualification is the sum total of training and experience.
   b. The employer may not certify Level III personnel.
   c. A certifying agency is any organization used by an employer in training NDT personnel.
   d. Certification is written testimony of qualification.

8. Who should be responsible to assess whether or not an individual should be qualified and certified who does not perform NDT, but monitors and evaluates NDT?
   a. ASNT.
   b. The individual's employer.
   c. A government regulatory agency.
   d. The customer's auditor.

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9. In SNT-TC-1A (2024) holographic testing is a technique under which of the following NDT methods?

   a. Visual testing.
   b. Guided wave testing.
   c. Acoustic emission testing.
   d. Laser testing.

10. ASNT intended that the recommendations of SNT-TC-1A be applied:

   a. with flexibility and reason.
   b. precisely as written.
   c. as minimum requirements.
   d. as maximum requirements.

11. Of the following, which is most appropriate to determine the needs to qualify and certify personnel whose only NDT function is to operate digital thickness equipment?

   a. ASNT.
   b. ASME.
   c. The employer.
   d. The customer.

12. Certification is:

   a. the skill, training, and experience required for personnel to properly perform the duties of a specific job.
   b. written testimony of qualification.
   c. intended to be conferred by an organization independent from the employer.
   d. intended to be conferred by an organization hired by the employer.

13. Use of SNT-TC-1A is mandatory when:

   a. the material being tested is for a US Air Force contract.
   b. the material being tested is for a US Navy contract.
   c. it has been specified by the customer.
   d. the material being tested is for a US Army contract that requires certification of NDT personnel.

Section 4.0

14. The basic levels of qualification recommended by SNT-TC-1A are:

   a. trainee, Level I, Level II, and Level III.
   b. trainee, apprentice, Level I, Level II, and Level III.
   c. Level I, Level II, and Level III.
   d. Level I, Level II, Level III, and instructor.

15. Should personnel who operate ultrasonic digital thickness measurement equipment be qualified and certified?

   a. Yes, because SNT-TC-1A requires that all personnel performing NDT be qualified and certified.
   b. No, because SNT-TC-1A does not cover that specific operation.
   c. Only if required by industry codes, standards, and specifications.
   d. Whether any NDT personnel should be qualified and certified depends solely upon the needs of the employer and the requirements of the employer's customers or clientele.

16. According to the recommendations of SNT-TC-1A, which of the following is true concerning a trainee's activities?

   a. The trainee may not conduct nondestructive tests independently and may not report test results.
   b. The trainee may not conduct nondestructive tests independently, but may interpret test results if acting under written instructions.
   c. The trainee should work along with a certified individual and may be considered a Level I, provided that the certified individual cosigns any test reports.
   d. Once the trainee has worked along with a certified individual, the trainee may then independently perform any activities as directed by the certified individual.

17. According to SNT-TC-1A (2024), which of the following is mandatory for certification of a Level III?

   a. Shall be capable of assisting in establishment of acceptance criteria.
   b. Shall meet the requirements of the employer's written practice.
   c. Shall be familiar with other commonly used NDT methods.
   d. Shall be capable of establishing techniques and selecting test methods.

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18. According to written instructions, an NDT Level I may:

    a. develop and establish techniques.
    b. perform specific calibrations, specific nondestructive tests, and specific evaluations.
    c. perform ultrasonic phased array tests.
    d. provide direction to trainees and other Level I personnel.

19. May a Level I independently perform, evaluate, and sign for results of nondestructive tests with supervision and guidance from a Level II or III?

    a. Yes, the intent in SNT-TC-1A is that the Level I may perform the above functions provided that they are in accordance with written instruction of a Level II or III.
    b. No, the Level I may perform the above functions in accordance with written procedures, but must be under constant supervision and guidance of a Level II or Level III.
    c. No, the Level I may not sign for test results.
    d. No, the Level I is not allowed to take any independent action.

20. Which of the following statements is true concerning the definitions of trainee, Level I, II, and III in SNT-TC-1A?

    a. The qualifications for Level III equal and exceed those of Level II.
    b. SNT-TC-1A was intended for use by employers to determine the proficiency of individuals at each level.
    c. Except for a provision for a trainee, Levels I, II, and III may not be further subdivided. Level IIIs may not perform Level II functions unless they pass Level II examinations.
    d. A trainee may perform all of the functions of a Level I if following written instructions.

21. Which of the following is within the scope of activities of a Level II individual, as recommended in SNT-TC-1A?

    a. Calibrate equipment.
    b. Establish techniques.
    c. Standardize equipment.
    d. Administer and score exams.

22. As recommended in SNT-TC-1A, a Level II:

    a. may conduct on-the-job training and guidance of Level I personnel, with the Level I training and guiding trainees.
    b. is responsible for the training and examination of Level I personnel for certification.
    c. must be capable of and responsible for establishing techniques.
    d. may conduct on-the-job training and guidance of Level I personnel.

23. According to SNT-TC-1A, which of the following is true for a Level III?

    a. A Level III must have successfully completed at least two years of science or engineering study at a college or university.
    b. It is desirable that a Level III be a registered professional engineer.
    c. The Level III is responsible for establishing all acceptance criteria.
    d. The Level III shall be capable of evaluating test results in terms of codes, standards, and specifications.

Section 5.0

24. In accordance with SNT-TC-1A, who is responsible for establishing a written practice for the control and administration of NDT personnel training, examination, and certification?

    a. The NDT Level III.
    b. The employer.
    c. ASNT.
    d. An appropriate regulatory authority.

25. The responsibility of each level of certification for determining the acceptability of materials or components:

    a. need not be described in the employer's written practice since those responsibilities are defined specifically in SNT-TC-1A.
    b. should be described in the employer's written practice only if they are different from SNT-TC-1A recommendations.
    c. should be described in the employer's written practice under all circumstances.
    d. should be described in the employer's written practice if dictated by customer requirements.

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Section 6.0

26. What factors are to be considered to ensure that a candidate for certification in NDT understands the principles and procedures involved?

    a. Training, experience, and education.
    b. Training, experience, and prior certifications held.
    c. Education, experience, and percentage of time on the job doing NDT.
    d. Training, experience, and professional credentials.

27. As recommended in SNT-TC-1A, which of the following is true?

    a. Overtime cannot be considered in meeting the minimum experience.
    b. Overtime can only be considered if the candidate is being qualified in more than one method simultaneously.
    c. Overtime can be credited based on total hours.
    d. SNT-TC-1A does not currently provide a recommendation regarding overtime.

28. High school education is recommended as a minimum requirement for:

    a. Level III only.
    b. Level II and Level III only.
    c. none of the levels.
    d. all three levels.

29. In leak testing, recommended work time experience and training:

    a. differ for each of four major techniques.
    b. do not consider different techniques.
    c. are listed for Levels II and III only.
    d. are significantly greater than for ultrasonic testing.

30. It is recommended that the education and experience of a Level III candidate include:

    a. graduation from a four-year university or college with a degree in engineering or science, plus one additional year of experience beyond the Level II requirements, with that assignment being comparable to that of an NDT Level II in the applicable method.
    b. graduation from a four-year college or university with a degree in NDT plus three months' experience in NDT comparable to that of a Level II.
    c. two years' experience in NDT comparable to that of a Level II if a high school graduate.
    d. six years' experience in NDT if the candidate did not graduate from high school.

31. Records substantiating training and experience for qualification are recommended to be kept on a(n):

    a. daily or weekly basis.
    b. hourly basis.
    c. fractional yearly basis.
    d. fractional hourly basis.

32. For a person being qualified directly to Level II with no time at Level I, the recommended experience consists of:

    a. the time recommended for Level II.
    b. the time recommended for Level III.
    c. the sum of the times recommended for Level I and Level II.
    d. not less than six months for any method.

33. In some cases, the training times recommended for Level I are greater than for Level II. Why?

    a. In preparing for qualification at Level I, the candidate should always receive more training than for Level II, regardless of the NDT method.
    b. Some numbers in the table are erroneous.
    c. Candidates for Level II generally have more formal education than those for Level I.
    d. Some methods require more initial training at Level I because of differences in complexity and manipulative skills.

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34. The recommended number of training hours in a particular method are:

  a. listed as a function of the candidate's education.
  b. the same regardless of the candidate's education.
  c. listed as a function of the candidate's experience.
  d. reduced if the candidate is being qualified in more than one method simultaneously.

Section 7.0

35. It is recommended that a training program for qualification and certification purposes should include:

  a. one-on-one practical instruction by the Level III.
  b. training applicable to all industries where the method is used.
  c. examinations to verify that the training material has been comprehended.
  d. a practical examination to verify that the training material has been comprehended.

36. Recommended training course outlines:

  a. are included in CP-105 and must not be modified.
  b. are included in the most recent editions of SNT-TC-1A.
  c. are not available for visual and leak testing.
  d. are included in CP-105 for the methods listed.

37. Recommended training reference material:

  a. is available only through ASNT.
  b. is available from a variety of sources.
  c. can only include those references listed in CP-105.
  d. must be made available to each trainee.

38. The recommended training course outline includes:

  a. technical principles of the method.
  b. review of API 1104 requirements.
  c. review of ASTM guidelines.
  d. review of interpretation requirements in ASME B31.3.

Section 8

39. In accordance with SNT-TC-1A, the NDT Level III should be responsible for:

  a. ensuring Level II personnel examine Level I personnel.
  b. interpretation of all test results obtained by Level II personnel.
  c. all questions to be used on examinations for Level I and Level II.
  d. writing all company standard operating procedures.

40. Which of the following may conduct and grade examinations for Level I and Level II personnel?

  a. An NDT Level II.
  b. A qualified representative of the NDT Level III.
  c. ASNT personnel because they offer examinations on a regular basis.
  d. The company president.

41. How often should the near-vision acuity examination be administered?

  a. Semiannually.
  b. At five-year intervals.
  c. Annually.
  d. Once only, upon initial certification.

42. As recommended in SNT-TC-1A, physical examination requirements are intended to be:

  a. the same for all methods.
  b. the same for all employers.
  c. related to each employer's specific needs.
  d. as specified in applicable sections of the *ASME Boiler & Pressure Vessel Code*.

43. The General examination is intended to cover:

  a. basic test principles of the method.
  b. only the material included in the training course outlines of SNT-TC-1A.
  c. the equipment operational capabilities of the candidate.
  d. the operating procedures that the candidate may encounter in the job.

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44. What is the proper use of ASNT *Questions & Answers Books*?

a. To create Practical exam questions.
b. To develop questions for ASNT certification tests.
c. To compile the General and Specific examinations.
d. To help prepare for exams, not to write them.

45. Which of the following parts of Level I and Level II examinations should be written?

a. The General and Specific examinations.
b. The General and Practical examinations.
c. The General, Specific, and Practical examinations.
d. Only the General examination.

46. The Practical examination is recommended to include operational familiarity with test equipment and analysis of test results for:

a. Levels I, II, and III.
b. Levels I and II.
c. Levels II and III.
d. trainee, Level I, and Level II.

47. Which of the following statements is true with regard to the administration of written examinations?

a. The examinee should not be permitted access to any reference material.
b. Reference data may be supplied for the Practical examination only.
c. Level III personnel should be required to memorize everything except codes, specifications, and procedures.
d. Codes, specifications, and procedures may be provided to examinees for reference during examinations provided that they do not contain data that can be used to answer questions in the General examination.

48. For the written examinations, tables, graphs, and charts may be used:

a. only during Level I examinations.
b. only during Level II examinations.
c. during all examinations.
d. only for the Level II Practical examination.

49. For Level I and Level II Practical examinations, one or more test specimens are recommended, and the examinee should perform tests and evaluations using the appropriate equipment and test specimens. The minimum number of different checkpoints recommended is:

a. 5
b. 10
c. 15
d. 20

50. For Level I and II examinations, the recommended minimum composite score is:

a. 60%.
b. 90%.
c. 70%.
d. 80%.

51. For Level III Basic examinations, it is recommended that Level II questions also be included. These questions should be:

a. Level II questions based on other applicable NDT methods.
b. based on Level II tasks for the particular method examination.
c. selected at random from questions previously used on Level II examinations.
d. given only if the candidate is being examined in more than one method.

Section 9.0

52. If an outside agency is engaged to provide Level III services, the:

a. written practice of the outside agency pertains, and it is not necessary for the employer that uses the outside services to have a written practice.
b. responsibility of certification must be retained by the employer using outside services.
c. employer using outside services must nonetheless have a Level III in direct employment.
d. outside agency may certify the personnel of the employer using outside services.

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53. What purpose is best served by maintaining certification records and the written practice?

    a. Periodic approval by ASNT.
    b. To determine the effectiveness of outside services.
    c. To provide documentation for review by customers, clients, and regulatory agencies.
    d. To protect against product liability claims.

54. The employer is responsible for certification of:

    a. Level I and Level II NDT personnel.
    b. Level III personnel only.
    c. outside services.
    d. all levels of NDT personnel.

55. The employer can consider an individual to be qualified to Level III, but only if they:

    a. take a comprehensive written examination.
    b. have in excess of five (5) years' experience comparable to a Level II.
    c. meet the requirements of the employer's written practice.
    d. have taken the ASNT NDT Level III Basic examination and at least one of the method examinations.

Section 10.0

56. According to SNT-TC-1A (2024), the employer shall:

    a. reexamine Level I and Level II personnel annually.
    b. reexamine Level I and Level II personnel every two (2) years.
    c. reexamine Level I and Level II personnel every five (5) years.
    d. reexamine Level I and Level II personnel anytime at the discretion of the employer.

57. The technical performance evaluation:

    a. must be performed at the mid-point of the certification interval.
    b. consists of a Practical examination following the format and guidelines of the Practical examination used for initial qualification.
    c. is not required if the performance of Level II duties has been satisfactory over the current certification period.
    d. may be performed on actual product during the course of normal testing.

Section 11.0

58. Regarding personnel service, SNT-TC-1A (2024) states:

    a. if Level I or Level II personnel service is interrupted by more than three (3) months, they must recertify by examination.
    b. if Level I or Level II personnel service is interrupted by more than three (3) months (except for military duty or maternity leave), they must recertify by examination.
    c. if Level I or Level II personnel service is interrupted by more than six (6) months, they must recertify by examination.
    d. the employer's written practice should include rules covering the types and duration of interrupted service that requires reexamination and recertification.

Section 12.0

59. Which of the following statements is true concerning recertification?

    a. Recertification can be accomplished only by reexamination at least once every three (3) years.
    b. Recertification can be based upon evidence of continuing satisfactory performance.
    c. Once certified to a particular level, certification can only be terminated if the certified individual terminates employment with the certifying employer.
    d. Reexamination of a certified individual can be accomplished only after three (3) years at a particular level.

60. According to SNT-TC-1A (2024), when a new technique is added to the employer's written practice, the NDT Level III personnel:

    a. may be "grandfathered" into the technique.
    b. should first meet the requirements of Level II before being certified as a Level III in the new technique.
    c. should be qualified to the new technique in accordance with the employer's written practice.
    d. should receive applicable training, take applicable examinations, and obtain the necessary experience, such that the NDT Level III meets the requirements of the new techniques in Table 6.3.1A, prior to their next recertification date, in the applicable method.

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Section 13.0

61. Automatic termination of certification is recommended when the certified individual:

    a. terminates employment with the employer where certified.
    b. is temporarily assigned to a different job function.
    c. takes a leave of absence greater than 30 days.
    d. achieves a higher level of certification.

62. A certified Level III individual terminates their employment with employer A and is immediately employed by employer B. Employer B may certify the individual as Level III based upon which of the following?

    a. The employer must examine the individual if he or she has not been working as a Level III during the past six (6) months.
    b. The employer may recertify the individual to Level III, but only after six (6) months of satisfactory service.
    c. The employer must examine the individual.
    d. The employer may certify the individual to Level III in accordance with the written practice.

Section 14.0

63. An NDT Level II individual terminates their employment with their employer and starts employment with another employer outside of NDT for three (3) months. Is it possible for the individual to return to their previous employer and be reinstated to their former Level II status after being removed from NDT for three (3) months?

    a. Yes.
    b. Not unless the employer meets all relevant recertification requirements in the employer's written practice.
    c. The employer must retrain the employee.
    d. Yes, provided the previous employer maintained relevant certification records for the employee.

ANSWERS

1d	2a	3b	4d	5b	6c	7d	8b	9d	10a	11c	12b	13c	14c
15d	16a	17b	18b	19a	20a	21c	22d	23d	24b	25c	26a	27d	28c
29a	30a	31b	32c	33d	34b	35c	36d	37b	38a	39c	40b	41c	42c
43a	44d	45a	46b	47d	48c	49b	50d	51a	52b	53c	54d	55c	56d
57b	58d	59b	60d	61a	62c	63d							

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CHAPTER 2

ANSI/ASNT CP-189 (2024)

Overview

The Abstract of ANSI/ASNT CP-189: *ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel* (2024) contains the following statements: "This standard applies to personnel whose specific tasks or jobs require appropriate knowledge of the technical principles underlying nondestructive testing (NDT) methods for which they have responsibilities within the scope of their employment. These specific tasks or jobs include, but are not limited to, performing, specifying, reviewing, monitoring, supervising, and evaluating NDT work. …

"Employers or other persons utilizing nondestructive testing services are cautioned that they retain full responsibility for ultimate determination of the qualifications of NDT personnel and for the certification process. The process of personnel qualification and certification as detailed in the standard does not relieve the employer of the ultimate legal responsibility to ensure that the NDT personnel are fully qualified for the tasks being undertaken."

These statements define who is under the jurisdiction of the standard and state that the ultimate responsibility of the qualification and certification process remains with the employer. The fact that CP-189 is an American National Standard signifies that the requirements for due process, consensus, and other criteria as defined by the American National Standards Institute (ANSI) have been met by the standard developer, in this case, the American Society for Nondestructive Testing.

Section 1.0: Scope

Section 1.0 of CP-189 (2024) reads as follows:

1.0 Scope

1.1 This standard establishes the minimum requirements for the qualification and certification of nondestructive testing (NDT) and predictive maintenance (PdM) personnel.

1.2 This standard details the minimum training, education, and experience requirements for NDT personnel and provides criteria for documenting qualifications and certification.

1.3 This standard requires the employer to establish a procedure for the certification of NDT personnel.

1.4 This standard requires that the employer incorporate any unique or additional requirements in the certification procedure.

It is clear that the tone of this document is distinctly more regimented than the scope of SNT-TC-1A, which states that "this document provides guidelines for the establishment of a qualification and certification program" and "it is recognized that these guidelines may not be appropriate for certain employers' circumstances and/or applications." The standard demands that a minimum set of requirements be met by all employers who claim to have a program that embraces the standard.

The following comments are intended to amplify the differences and similarities between the standard, CP-189, and the recommended practice, SNT-TC-1A.

Section 2.0: Definitions

In Section 2.0, "Definitions," of CP-189 (2024), 25 definitions are listed to remove any ambiguity about terms used throughout the body of the standard. The term "Practical Examination" is clarified in its use on behalf of the employer and the fact that observations and results must be documented. "Grading Unit" is a term used in both SNT-TC-1A and CP-189 to explain that a qualification specimen can be divided into sections that do not have to be equal length or equal spacing, unflawed or flawed. "Test Technique" (a category within an NDT method—for example, immersion ultrasonic testing) is distinguished from "Method" (one of the disciplines of NDT—for example, ultrasonic testing—within which various test techniques may exist).

In paragraph 2.2.1, an "NDT Level III" is identified as "an individual possessing a currently valid ASNT NDT, PdM, ACCP Professional, or ASNT 9712 Level III certificate and certified in accordance with this standard." Thus, in order to become an employer's NDT Level III, CP-189 mandates that the individual hold a current, valid ASNT Level III certificate. Other requirements must be customized to the needs of the employer in accordance with the employer's written NDT personnel qualification and certification procedure.

Section 2.0 of CP-189 (2024) reads as follows:

2.0 Definitions

2.1 Purpose. These definitions are intended to clarify the meanings of terms used in this standard, as they apply to this standard, and only to this standard. No broader application of these definitions is implied. For specific definitions related to NDT, see paragraph 10.2.

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2.1.1 **Calibration, Instrument:** the comparison of an instrument with, or the adjustment of an instrument to, a known reference(s) often traceable to the applicable country's national institute or standards body. (See also *Standardization, Instrument*.)

2.1.2 **Certification:** written testimony that an individual has met the applicable requirements of this standard.

2.1.3 **Certification Procedure:** a written procedure, developed by the employer, that details the requirements for qualification and certification of an employee to this national standard.

2.1.4 **Certifying Authority:** the person or persons properly designated in the certification procedure to sign certifications on behalf of the employer.

2.1.5 **Closed-book Examination:** an examination administered without access to reference material except that supplied with or in the examination.

2.1.6 **Documented:** the condition of being in written form.

2.1.7 **Education:** an institutionalized program, prescribed by appropriate authorities, that is offered by schools, institutes, organizations, colleges, or universities established for the sole purpose of providing instruction in an orderly, planned, and systematic fashion.

2.1.8 **Employer:** the corporate, private, or public entity that employs personnel directly or indirectly for wages, salary, fees, or other considerations. This would include organizations that obtain their qualified supplemental workforce personnel through third-party agencies, providing the use and certification of those supplemental employees is addressed in the employer's certification procedure.

2.1.9 **Examination:** a formal, controlled, documented assessment of knowledge or skills conducted in accordance with a procedure.

2.1.10 **Experience:** actual performance of an NDT method conducted in the work environment resulting in the acquisition of knowledge and skill. This does not include formal classroom training but may include laboratory and on-the-job training as defined by the employer's certification procedure.

2.1.11 **Grading Unit:** a qualification specimen can be divided into sections called grading units, which do not have to be equal length or be equally spaced. Grading units are unflawed or flawed and the percentage of flawed/unflawed grading units required shall be approved by the NDT Level III.

2.1.12 **Formal Training:** an organized and documented program of activities designed to impart the knowledge and skills required to be qualified to this standard. Formal training may be a mix of classroom, practical, and programmed self-instruction as approved by the responsible NDT Level III.

2.1.13 **General Examination:** a written examination addressing the basic principles of the applicable NDT method.

2.1.14 **Method:** one of the disciplines of NDT: for example, ultrasonic testing, within which various test techniques may exist.

2.1.15 **Nondestructive Testing (NDT):** the development and application of technical methods to examine materials or components in ways that do not impair future usefulness and serviceability in order to detect, locate, measure, and evaluate flaws; to assess integrity, properties, and composition; and to measure geometrical characteristics.

2.1.16 **NDT Instructor:** an individual qualified and designated in accordance with this standard to train or educate NDT personnel. (See also paragraph 3.7.)

2.1.17 **NDT Procedure:** a written instruction for conducting a nondestructive test.

2.1.18 **Outside Organization:** an agency or individual who provides NDT Level III services. (See also paragraph 4.5.)

2.1.19 **Personalized Instruction:** personalized instruction may consist of blended classroom, supervised laboratory, and/or hybrid online competency-based course delivery. Modular content is covered through online presentations, in the classroom, and/or in small groups. Personalized instruction also enables students to achieve competency using strategies that align with their knowledge, skills, and learning styles.

2.1.20 **Practical Examination:** an examination used to demonstrate an individual's ability in conducting the NDT methods that will be performed for the employer. For practical examinations, questions and answers need not necessarily be written, but observations and results must be documented.

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2.1.21 **Predictive Maintenance (PdM):** evaluates the condition of equipment (typically in service) by performing periodic or continuous (online) equipment condition monitoring. Condition monitoring evaluates leading performance indicators of specified machinery, components, and assemblies (including items such as buildings; e.g., structures in whole and in part), detectable via the applied method. PdM uses principles of statistical process control to assess the condition by focusing on leading indicators that may signify deterioration in performance or potential equipment, structure, or material failure. The ultimate goal of PdM is to perform maintenance based upon the condition such that the maintenance activity is cost-effective and before the equipment loses optimum performance or fails.

2.1.22 **Qualification:** the education, skills, training, knowledge, and experience required for personnel to properly perform to a specified NDT level.

2.1.23 **Specific Examination:** a written examination to determine an individual's understanding of procedures, codes, standards, specifications, and equipment or instrumentation for an NDT method used by the employer.

2.1.24 **Standardization, Instrument:** the adjustment of an NDT instrument using an appropriate reference standard to obtain or establish a known and reproducible response. (This is usually done prior to performing any NDT, but can be carried out anytime there is concern about the performance of the examination or instrument response.) (See also *Calibration, Instrument*.)

2.1.25 **Test Technique:** a category within an NDT method: for example, immersion ultrasonic testing.

2.2 NDT Levels

2.2.1 NDT Level III. An individual possessing a currently valid ASNT NDT, PdM, ACCP Professional (see paragraphs 10.1.2, 10.1.3, or 10.1.5), or ASNT 9712 Level III certificate and certified in accordance with this standard. (See also Section 3.0.) Reference to ASNT Level III throughout this standard implies the individual holds one of these certificates.

NOTE: As of the printing of this standard, the ASNT NDT Level III neutron radiography (NR) and vibration analysis (VA) examinations are no longer being offered by ASNT. Personnel who hold these certificates are still certified as long as they maintain the certification with ASNT through the recertification by points process.

2.2.2 NDT Level I, NDT Level II. An individual certified in accordance with this standard. (See also Section 3.0.)

Section 3.0: Levels of Qualification

In Section 3.0, "Levels of Qualification," the six levels of qualification are defined in terms of the skills and knowledge required in a given method or methods to perform specified NDT activities. These are Level III, Level II, Level II Limited, Level I, trainee, and NDT instructor. Level III, Level II, and Level I are nominally the same as those identified in SNT-TC-1A. The last two formalize the status of the trainee and create a category of NDT instructor who can function as course organizer and presenter, but under the cognizance of the employer's authorized (certified) Level III.

Section 3.0 of CP-189 (2024) reads as follows:

3.0 Levels of Qualification

3.1 Classification. Six levels of qualification are defined in terms of the skills and knowledge required in a given method or methods to perform specified NDT activities.

3.2 NDT Level III. An NDT Level III shall have the skills and knowledge to establish techniques; to interpret codes, standards, and specifications; to designate the particular technique to be used; and to verify the adequacy of procedures. The individual shall also have general familiarity with the NDT methods covered in Appendix A of this standard. The NDT Level III shall be capable of conducting or directing the training and examining of NDT personnel in the methods for which the NDT Level III is qualified.

3.3 NDT Level II. An NDT Level II shall have the skills and knowledge to set up and standardize equipment, to conduct tests, and to interpret, evaluate, and document results in accordance with procedures approved by an NDT Level III. The NDT Level II shall be thoroughly familiar with the scope and limitations of the method to which certified and should be capable of directing the work of trainees and NDT Level I personnel. The NDT Level II shall be able to organize and report nondestructive test results.

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3.4 NDT Level II Limited. An NDT Level II Limited shall have the skills and knowledge to set up and standardize equipment, to conduct tests, and to interpret, evaluate, and document results in accordance with procedures approved by an NDT Level III in the techniques listed in Appendix B. The NDT Level II Limited shall be thoroughly familiar with the scope and limitations of the technique to which certified and should be capable of directing the work of trainees and NDT Level I personnel. The NDT Level II Limited shall be able to organize and report nondestructive test results.

3.5 NDT Level I. An NDT Level I shall have the skills and knowledge to properly perform specific standardizations, specific tests, and, with prior written approval of the NDT Level III, perform specific interpretations and evaluations for acceptance or rejection and document the results, in accordance with specific approved procedures. The NDT Level I shall be able to follow approved nondestructive testing procedures and shall receive the necessary guidance or supervision from a certified NDT Level II or NDT Level III individual.

3.6 Trainee. A person who is not yet certified to any level shall be considered a trainee. Trainees shall work with a certified person, under the direction of an NDT Level II or NDT Level III and shall not independently conduct any tests or write a report of test results.

3.7 NDT Instructor. An NDT instructor shall have the skills and knowledge to plan, organize, and present classroom, laboratory, demonstration, and/or on-the-job NDT instruction, training, and/or education programs in accordance with course outlines approved by an NDT Level III.

Section 4.0: Qualification Requirements

Section 4.0, "Qualification Requirements," addresses training, experience, instructor criteria, and the use of outside NDT Level III services. The training is to be performed in accordance with a course outline approved by an NDT Level III and must include the topics contained in ANSI/ASNT CP-105 *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel* for the appropriate NDT method. The training program may include other topics deemed necessary by the NDT Level III. "Organized training" is expanded in CP-189, paragraph 4.1.1.1, to include "instructor-led training, personalized instruction, virtual instructor-led training, computer-based training, or web-based training."

Computer-based training and web-based training shall track hours and content of training with student examinations. The training programs shall include sufficient examinations to demonstrate that the necessary information has been comprehended. A satisfactory passing score on a final examination covering the topics contained in the training program is necessary in order to receive credit for the training hours.

Recognizing that NDT is a unique application of the concepts of physics, electronics, and chemistry, CP-189 requires that all training shall be presented by an NDT instructor designated by the NDT Level III individual. The NDT instructor is an individual who not only has the skills and knowledge for conducting training programs but also is required to develop and conduct such courses in accordance with the course outlines approved by the NDT Level III. The NDT Level III in all cases is responsible for the content of each completed course. In order to qualify, an NDT instructor must satisfy at least one of the four criteria. (See paragraphs 4.4.1.1–4.4.1.4.)

Training and experience qualifications of the NDT Level III are automatically met when an individual holds a valid ASNT Level III certificate.

Both SNT-TC-1A and CP-189 call for a minimum number of hours worked in the specific method as well as a minimum total number of hours worked in NDT. The total experience requirement can be satisfied by working in two or more methods as well as doing other activities that support the NDT program of the employer. (SNT-TC-1A and CP-189 have equivalent training and experience requirements.)

When using CP-189 Appendix A or SNT-TC-1A Table 6.3.1A, it is important to note that the minimum experience hours must be documented by method and by hour. A candidate's previous training and experience may be accepted by the employer if verified and documented in writing by the previous employer(s) or training agency(ies).

Although employers often have their own NDT Level III to administer the various aspects of the employer's NDT personnel qualification and certification program, an outside organization may be engaged to perform the duties of an NDT Level III. In such instances, the employer is responsible for verifying that the organization complies with the employer's certification procedure and CP-189.

Section 4.0 of CP-189 (2024) reads as follows:

4.0 Qualification Requirements

4.1 Training. Candidates for certification as NDT Level I or Level II shall complete sufficient organized training to become familiar with the principles of the method and the practices of the applicable test technique. This training shall be conducted in accordance with a course outline approved by an NDT Level III.

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4.1.1 NDT Level I and Level II. The minimum number of training hours required for NDT Level I and Level II candidates is described in Appendixes A and B. The course shall include the topics contained in ANSI/ASNT CP-105 for the appropriate NDT method, plus such additional topics as deemed necessary by the NDT Level III. The sequence, content, amount of time spent, and depth of coverage for each topic shall be approved by the NDT Level III. Training programs shall include sufficient examinations to demonstrate that the necessary information has been comprehended.

4.1.1.1 The organized training may include instructor-led training, personalized instruction, virtual instructor-led training, computer-based training, or web-based training. Computer-based training and web-based training shall track hours and content of training with student examinations in accordance with paragraph 4.1.2.

4.1.2 Credit. To receive credit for training hours, the individual shall pass a final examination covering the topics contained in that program.

4.1.3 Presentation of Training. All training shall be presented by an NDT instructor. However, the NDT instructor may use personnel with specialized expertise (for example, metallurgists, welding engineers, etc.) who are not qualified in accordance with this standard to assist in presentation of specific information. The NDT Level III shall in all cases be responsible for the content of the completed course.

4.1.4 NDT Level III. Training requirements for NDT Level III are satisfied as a result of holding a current ASNT Level III certificate in the specific NDT method.

4.2 Experience. Candidates for certification shall have acquired the practical experience to ensure they are capable of performing the duties of the level in which certification is being sought.

4.2.1 NDT Level I and Level II. The minimum number of hours of experience required for NDT Level I and Level II candidates is described in Appendix A.

4.2.2 NDT Level II Limited. The minimum number of hours of experience required for NDT Level II Limited candidates is described in Appendix B.

4.2.3 NDT Level III. Experience requirements for NDT Level III are satisfied as a result of holding a current ASNT Level III certificate in the specific NDT method.

4.3 Previous Training and Experience

4.3.1 NDT Level I and Level II. A candidate's previous training and experience may be accepted by the employer's NDT Level III if documented and verified. Any claimed training or experience that is not documented and cannot be verified shall be considered invalid.

4.3.2 NDT Level III. The employer shall verify and document the current validity of a candidate's ASNT Level III certificate.

4.4 NDT Instructor

4.4.1 Criteria. An NDT instructor shall meet at least one of the following criteria:

4.4.1.1 possess a current ASNT Level III certificate in the NDT method to be taught; or

4.4.1.2 have academic credentials at least equivalent to a B.S. in engineering, physical science, or technology, and possess adequate knowledge in the NDT method to be taught; or

4.4.1.3 be a graduate of a two-year school of science, engineering, or NDT and have five or more years of experience as an NDT Level II, or equivalent, in the NDT method to be taught; or

4.4.1.4 have 10 or more years of NDT experience as an NDT Level II, or equivalent, in the NDT method to be taught.

4.4.2 Designation. The NDT instructor shall be designated by an NDT Level III individual. The designation shall become part of the individual's qualification records.

4.5 Outside Services. At the option of the employer, an outside organization may be engaged to perform the duties of an NDT Level III. For organizations other than ASNT, the employer shall be responsible for evaluating the organization to ensure the services are in accordance with the employer's certification procedure and this standard, and be so documented. An NDT Level III of the engaged outside organization shall be responsible for the services provided.

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Section 5.0: Qualification and Certification

Section 5.0, "Qualification and Certification," addresses the certification procedure for qualifying and certifying NDT personnel. The employer's certification procedure describes the minimum requirements for certifying personnel in each NDT method and is approved by the designated NDT Level III. The certification procedure includes personnel duties and responsibilities, as well as the required training, experience, examinations, records, and recertification processes to be followed.

Section 5.0 of CP-189 (2024) reads as follows:

5.0 Qualification and Certification

5.1 **Certification Procedure.** The employer shall develop and maintain a procedure detailing the program that will be used for qualification and certification of NDT personnel in accordance with this standard.

5.2 **Certification Procedure Requirements.** The procedure shall describe the minimum requirements for certifying personnel in each NDT method and the levels of qualification desired. The procedure shall satisfy the requirements of this standard. The procedure shall include, as a minimum, the following:

5.2.1 personnel duties and responsibilities;
5.2.2 training requirements;
5.2.3 experience requirements;
5.2.4 examination requirements;
5.2.5 records and documentation requirements, including control, responsibility, and retention period; and
5.2.6 recertification requirements.

5.3 **Approval.** The employer's certification procedure shall be approved by an NDT Level III designated by the employer.

Section 6.0: Examinations

Section 6.0, "Examinations," addresses vision requirements and the administration of written and practical examinations for the qualification of NDT Level I, II, and III personnel. The near-distance vision examination calls for an individual to be capable of reading a Jaeger No. 1 test chart at a distance of not less than 12 in. (30.5 cm), rather than the Jaeger No. 2 of SNT-TC-1A, and to be administered in accordance with a procedure, and by personnel, approved by the designated NDT Level III. In addition to color differentiation, NDT personnel must be able to distinguish shades of gray used in a given method.

Level I and Level II personnel are to receive a closed-book General examination approved by the designated NDT Level III over a cross section of the body of knowledge applicable to each method and NDT level. Level I and Level II personnel are to receive a closed-book Specific examination, supported by NDT Level III approved reference materials, addressing various examples of equipment, procedures, and test techniques that the candidate may use in the performance of assigned duties.

The Level I Practical examination requires that the candidate demonstrate proficiency in using the applicable NDT method to examine at least one representative test sample for each technique to be used in the candidate's job. This includes the documentation of the results of the test(s).

The Level II Practical examination requires that the candidate demonstrate proficiency using the applicable NDT method to examine two or more representative test samples for each method and at least one sample for each technique to be used in the candidate's job. This includes the interpretation, evaluation, and documentation of the examination's results.

The Level I Practical examination and the Level II Practical examination are similar to those recommended by SNT-TC-1A, with the exception that the test samples used are to be representative of the products that will be encountered when performing their job functions. This difference ensures that the CP-189 examinations must be representative of the product typical of that which candidates will likely encounter in performing their job function.

More explicit examination requirements have been placed on the NDT Level III due to the many variations in background and work activities found in the field. As a prerequisite, a candidate for the position of an employer's Level III must hold an ASNT Level III certificate with a currently valid endorsement for each method for which employer certification is sought, and he or she must also satisfactorily complete a Specific examination comprising 30 questions on the employer's specifications and standards for each method. A valid endorsement on an ASNT Level III certificate fulfills the examination criteria for only the Basic examination and the method examination for each applicable NDT method. (NOTE: Table 1. Minimum Number of Examination Questions, in CP-189 is comparable to Table 8.3.4 in SNT-TC-1A with the exception that only the 12 methods for which ASNT administers NDT Level III examinations are listed. Vibration analysis, although listed, is considered a predictive maintenance [PdM] method in CP-105 and thus falls outside the scope of this study guide. At the time of this printing, the neutron radiographic testing examination was temporarily suspended, but remains in the table. The following four methods are not represented: ground penetrating radar, guided wave testing, laser methods testing, and microwave technology testing.)

In addition, candidates may have to show the ability to prepare an NDT procedure appropriate to the employer's needs, if they do not have documented experience demonstrating that they had previously prepared similar procedures in the method using the specifications, codes, and standards that are applicable to that employer.

If the NDT Level III will be required to perform tests or evaluate test results, the practical examination must include the same demonstrations of ability to perform the required activities as that of the Level II.

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The employer's NDT Level III is responsible for the administration and grading of the examinations, but the grading and administration of multiple-choice objective questions can be delegated when properly documented. The practical examination is to be administered by an NDT Level III in the respective test method. The employer is responsible for having an ASNT Level III develop, administer, and grade the Level III Specific and Practical examinations. Employer examinations require a minimum 70% on individual tests and a minimum 80% on the overall average grade.

For a Level I or Level II candidate to pass the practical examination, discontinuities or conditions previously specified by the Level III are to be located and evaluated. A written checklist is to be used addressing equipment and technique proficiency, proper adherence to the procedure, test sequence, calibrations, materials, documentation, and extent of examination. If the candidate is required to perform interpretation or evaluation of results, these are also to be part of the checklist. In addition, the Level II checklist includes proper extent of examination, accuracy, and completeness of interpretations, evaluations, and documentation of the activities and test results.

If a candidate fails an examination, the requirements for reexamination are similar to SNT-TC-1A (receive additional training or wait 30 days), except that a candidate is not to be reexamined using the previous failed examination and/or specimen.

The employer's representative who administers the Level III examinations must possess a current ASNT Level III certificate in the method and be familiar with the standards used and the products made by the employer. Self-examination and examination by subordinates are prohibited.

Section 6.0 of CP-189 (2024) reads as follows:

6.0 Examinations

6.1 Vision

6.1.1 Near Distance. Prior to certification, NDT personnel shall be examined to ensure that they have natural or corrected (no pharmacological agents) near-distance acuity in at least one eye such that each individual is capable of reading Jaeger No. 1 test chart or equivalent* at a distance of not less than 12 in.
*Equivalent eye examination results for Jaeger J-1 visual acuity are: Snellen 20/22; Times Roman 3.5 point text; OrthoRater #9; or Titmus (SAB-1, SAL-1 or SAR-1) #10.

6.1.1.1 Pharmacological agents (eye drops) shall not be used which would improve or enhance visual acuity at any distance.

6.1.2 Color Vision. NDT personnel for all methods shall demonstrate the ability to differentiate among the colors or shades of gray used in the method.

6.1.3 Frequency. Vision examinations shall be administered annually, except that color differentiation examinations need be repeated only at each recertification. Vision examinations shall expire on the last day of the month of expiration.

6.1.4 Administration. Vision examinations shall be administered in accordance with a procedure, and by personnel approved by an NDT Level III designated by the employer.

6.2 NDT Level III Examinations

6.2.1 Initial Requirement. Prior to the employer's certification examinations, the candidate shall hold a current ASNT Level III certificate with a currently valid endorsement for each method for which employer certification is sought.

6.2.2 Specific Examination (for each method). The employer shall administer a written examination consisting of at least 30 questions relating to the comprehension of the NDT-related requirements of specifications or standards used by the employer. Copies of the applicable specifications or standards shall be available as reference material during the examination.

6.2.3 Practical Examination. The candidate shall prepare an NDT procedure appropriate to the employer's needs; however, if documented experience demonstrates that the candidate has previously prepared acceptable NDT procedures in the method using the specifications, codes, and standards that are applicable to that employer, a written practical examination (for example, preparation of a procedure) is not required. If experience is substituted for the written practical examination, the employer shall document the pertinent practical experience of the NDT Level III candidate.

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6.2.3.1 If the NDT Level III will be required to perform tests or evaluate test results, the practical examination shall include the same demonstrations of the candidate's ability to perform the required activity(ies) as required in Section 6.3.3.2.

6.3 NDT Level I and Level II Examinations

6.3.1 General. A General examination shall be approved by an NDT Level III. Administration of the examination shall be in accordance with paragraph 6.6 of this standard and shall be closed book. Reference material, such as charts, formulas, tables, and graphs, may be provided by the NDT Level III. Questions shall be developed which represent a cross section of the body of knowledge contained in ANSI/ASNT CP-105 applicable to each method and NDT level. The minimum number of questions required for each method and level is listed in Table 1. Questions used in General examinations for NDT Level I and Level II personnel shall be similar in type and difficulty to those published by ASNT.

6.3.1.1 A valid ACCP, ASNT NDT, or ASNT 9712 Level II certificate may be accepted as fulfilling the General examination criteria.

6.3.2 Specific. A Specific examination shall be approved by an NDT Level III. Administration shall be in accordance with paragraph 6.6 of this standard. The NDT Level III shall determine whether appropriate procedures, specifications, standards, or code sections will be provided. The examination shall address various examples of equipment, procedures, and test techniques that the candidate may use in the performance of assigned duties. The minimum number of questions required for each method and level is listed in Table 1.

6.3.2.1 A valid ACCP, ASNT NDT, or ASNT 9712 Level II certificate may be accepted as fulfilling the Specific examination criteria for each applicable method if the NDT Level III has determined that the ASNT examinations meet the requirements of the employer's certification procedure. This acceptance shall be documented. If this assessment cannot be accomplished, an employer-administered Specific examination shall be completed.

6.3.3 Practical. A practical examination shall be approved by an NDT Level III. Administration shall be in accordance with paragraph 6.6 of this standard. The practical examination shall consist of the following:

6.3.3.1 NDT Level I. The candidate shall demonstrate proficiency using the applicable nondestructive test method to examine at least one test sample for each technique to be used in the candidate's job and by documenting the results of the test. The test samples shall be representative of the products that the candidate will encounter in performing the job functions.

6.3.3.2 NDT Level II. The candidate shall demonstrate proficiency by performing the applicable nondestructive test method in examining at least one sample per technique and a minimum of two samples per method and by interpreting, evaluating, and documenting the results of the examination. The test samples shall be representative of the product that the candidate will encounter in performing the job functions.

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For purposes of the practical examinations described in paragraph 6.4.3, the techniques required to be demonstrated are in Appendix A. Where no techniques are listed in Appendix A, the employer must determine the techniques required for their particular needs in that method and specify those techniques required for certification in their qualification and certification procedure.

6.3.3.3 A valid ACCP, ASNT NDT, or ASNT 9712 Level II certificate may be accepted as fulfilling the practical examination criteria for each applicable method if the NDT Level III has determined that the ASNT examinations meet the requirements of the employer's certification procedure. This acceptance shall be documented. If this assessment cannot be accomplished, an employer-administered practical examination shall be completed.

6.4 Administration and Grading

6.4.1 Responsibilities

6.4.1.1 NDT Level I and II Examinations. An NDT Level III shall be responsible for the development, approval, administration, and grading of examinations for NDT Level I and Level II personnel for those methods in which the NDT Level III has a valid ASNT Level III certificate. For the practical examination, the individual administering the examination must be an NDT Level III in the respective test method.

6.4.1.2 NDT Level III Examinations. The employer shall be responsible for having an individual possessing a current ASNT Level III certificate, in the respective test method, develop, administer, and grade NDT Level III Specific and practical examinations.

6.4.1.3 The administration and grading of multiple-choice questions may be delegated by the NDT Level III, if so documented. Training shall be provided to the individual administering the examinations. This training and qualification shall be documented.

6.4.2 Employer Examinations. For each employer-administered certification written examination, each candidate shall achieve a grade of at least 70%. For practical examinations, each candidate shall achieve a grade of at least 80%. A composite average grade of 80% is required to be eligible for certification. All certification examinations shall have equal weight in determining the average grade.

6.4.3 Practical Examinations

6.4.3.1 For all practical examinations, all applicable discontinuities and/or conditions which the candidate is expected to find and/or evaluate must be specified and documented by the NDT Level III prior to administration of the examination.

6.4.3.2 A requirement for passing the NDT Level I, II, or III practical examination shall be the detection of the discontinuities and/or conditions previously specified by the NDT Level III responsible for preparing that examination.

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6.4.4 NDT Level I Practical Examinations. The NDT Level III shall use a written checklist in administering and grading NDT Level I practical examinations. This checklist shall address at least the following items: proficiency in use of equipment and technique, proper adherence to procedure, test sequence, standardizations, materials, documentation, and extent of examinations. If the NDT Level I will interpret or evaluate results, the checklist shall include these item(s).

6.4.5 NDT Level II Practical Examinations. The NDT Level III shall use a written checklist in administering and grading NDT Level II practical examinations. The checklist shall address at least the following items: proficiency in use of techniques and equipment; proper adherence to procedure, test sequence, standardizations, and materials; satisfactory detection and location of discontinuities; proper extent of examination; and the accuracy and completeness of interpretations, evaluations, and documentation of the activities and test results.

6.4.5.1 Film Interpretation Limited Certification. The practical examination shall consist of a sufficient number of radiographs to demonstrate satisfactory performance to the satisfaction of the NDT Level III. The number of radiographs shall be addressed in the employer's certification procedure.

6.4.5.2 Phased Array Ultrasonic Testing (PAUT) and Time of Flight Diffraction (TOFD). Flawed samples used for practical examinations should be representative of the components and/or configurations that the candidates would be testing under this endorsement and approved by the NDT Level III.

6.4.6 NDT Level III Practical Examinations. Persons administering NDT Level III practical examinations shall use a written checklist. The checklist shall address items relating to the technical and practical adequacy of the NDT procedure(s) prepared by the candidate. When applicable to the candidate's job responsibilities, the checklist shall also address the items listed in paragraph 6.4.5.

6.5 Reexamination. Candidates who fail to attain the required passing grade must receive additional documented training or wait at least 30 days for reexamination. This training shall address the deficiencies which caused failure. A candidate shall not be reexamined using the examination or specimen previously failed or both.

6.6 Administration of Examinations. In no case shall an examination be prepared or administered by the individual being examined or by a subordinate.

6.7 Administration of NDT Level III Examinations. The employer's representative who administers the NDT Level III Specific and practical examinations shall possess a current ASNT Level III certificate in the method for which the examination is administered and shall be knowledgeable and familiar with the standards, specifications, and products used or made by the employer.

6.8 Certification. Certification of NDT personnel to all levels of qualification is the responsibility of the employer.

6.8.1 Certification of NDT personnel is based on demonstrations of satisfactory qualification in accordance with paragraphs 4.1, 4.2, 6.1, 6.2, and 6.3 of this document, as described in the employer's certification procedure.

6.8.2 Satisfactory completion of the qualification requirements shall be documented by the issuance of a certification record which has been signed by the NDT Level III who verified the qualifications have been met and the certifying authority issuing the certification record.

6.8.3 Personnel certification records shall be maintained on file by the employer for the duration specified in the employer's written practice and shall contain those records identified in Section 9.0.

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Section 7.0: Expiration, Suspension, Revocation, and Reinstatement of Employer Certification

Section 7.0 addresses the expiration, suspension, revocation, and reinstatement of employer certification. As with SNT-TC-1A, an individual's certification(s) expire when employment with the employer is terminated or on the last day of the month at the end of five years for Levels I and II. The Level III certification expires when the ASNT Level III certificate has expired.

Certifications can be suspended for exceeding the one-year vision recheck period, for inactivity in a method for more than 12 months, or for deficient performance as determined by the Level III. Level IIIs are suspended if their ASNT Level III certificate is not renewed.

Certifications are revoked if inactivity in a method exceeds 24 months, if the ASNT Level III certificate is revoked, or if there is unethical or incompetent conduct.

Reinstatement for Level I and Level II is determined by the Level III. Reinstatement for the Level III is determined by the employer as long as the Level III holds a valid ASNT Level III certificate. Expired or revoked certifications may only be reinstated by complying with Section 6.0, paragraph 7.4.3, or Section 8.0.

Section 7.0 of CP-189 (2024) reads as follows:

7.0 Expiration, Suspension, Revocation, and Reinstatement of Employer Certification

7.1 Expiration. Individual certifications shall expire:

7.1.1 when employment with the employer is terminated;

7.1.2 on the last day of the month of expiration at the end of five years for NDT Level I and NDT Level II individuals;

7.1.3 for NDT Level III individuals, when the ASNT Level III certificate has expired.

7.2 Suspension. The employer shall suspend an individual's certification if:

7.2.1 the near-distance vision examination interval exceeds one year. Certification is reinstated concurrently with passing the vision reexamination; or

7.2.2 the individual has not performed the duties in the method(s) for which certified during any consecutive 12-month period; or

7.2.3 the individual's performance is determined to be deficient in the required method or technique for specific documented reasons; or

7.2.4 for NDT Level III personnel, when the ASNT Level III certificate has not been renewed.

7.3 Revocation. The employer shall revoke an individual's certification when:

7.3.1 the individual has not performed the duties in the method(s) for which certified during any consecutive 24-month period; or

7.3.2 for NDT Level III personnel, the ASNT Level III certificate has been revoked; or

7.3.3 an individual's conduct is deemed by the employer to be or have been unethical or incompetent.

7.4 Reinstatement

7.4.1 Suspended Certifications. Reinstatement of suspended certifications for NDT Level I or NDT Level II shall be determined by the NDT Level III. Reinstatement of suspended NDT Level III certifications shall be determined by the employer, except that the requirement for ASNT Level III certification may not be waived.

7.4.2 Expired or Revoked Certifications. Certifications which have expired or have been revoked may only be reinstated by complying with Section 6.0, paragraph 7.4.3, or paragraph 8.1. Reinstatement per paragraph 7.4.3 is only allowed when the employee's certification expired due to termination of employment. Such reinstatement of certification shall only be for the period of time remaining on the original certification.

7.4.3 Expired for Termination of Employment. An NDT Level I, Level II, or Level III whose certification has expired due to termination of employment may be reinstated to the former NDT level, without a new examination, providing all of the following conditions are met:

7.4.3.1 the employer has maintained the personnel certification records required in Section 9.0;

7.4.3.2 the employee certification did not expire during termination; and

7.4.3.3 the employee is being reinstated within 12 months of termination.

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Section 8.0: Employer Recertification

Section 8.0, "Employer Recertification," has been divided into two parts. For Level I and Level II personnel, every five years recertification may be based on evidence of experience in the method of at least two months or 350 hours (over that five-year period) and successfully passing a Specific examination that meets the requirements of paragraph 6.3.2. Every 10 years, examinations in Section 6.0 must be repeated. For Level III personnel, recertification relies on verification of the currency of the Level III's ASNT certificate every five years.

Section 8.0 of CP-189 (2024) reads as follows:

8.0 Employer Recertification

8.1 NDT Level I and NDT Level II. NDT Level I and NDT Level II personnel shall be recertified by reexamination in accordance with Section 6.0 or by meeting the requirements of paragraph 8.1.1, when applicable.

8.1.1 Every five years, the individual may be recertified if the individual has at least 350 hours of documented experience using the applicable method over the five-year certification interval and has successfully passed a Specific examination which complies with paragraph 6.3.2.

8.1.2 At least every 10 years, the individual must repeat the examinations in Section 6.0.

8.2 NDT Level III. NDT Level III personnel shall be recertified by the employer every five years by verifying that the individual's ASNT Level III certificate is current in each method for which recertification is sought.

Section 9.0: Records

In Section 9.0, "Records," a minimum set of documents that address the qualifications of each NDT individual is required. Included are the employer's certification documentation, an experience record, a record of previous experience (if applicable), the employee's current examinations, and a vision examination record.

The certification record includes level, method, and technique(s), results, and copies of recent examinations; a copy of the current ASNT Level III certificate for Level III personnel only; dates of certification, expiration, suspension, revocation, and reinstatement; and the signature, printed name, and title of the employer's certifying representative.

A training record is required that includes the training received, the name of the training organization, the date when training was completed, the hours involved, evidence of satisfactory completion, and the instructor's name.

A record that identifies the individual's experience performing various nondestructive tests shall be maintained for the purpose of verifying initial certification experience and continuing experience. Previous experience shall also be documented if it is used to satisfy any part of the qualification requirements.

Section 9.0 of CP-189 (2024) reads as follows:

9.0 Records

9.1 Responsibility for Documentation. The employer shall document certifications in accordance with this standard.

9.2 Contents of Documentation. The employer's certification documentation shall include at least a training record, a certification record, an experience record, a record of previous experience (if applicable), employee's current examinations, and a vision examination record.

9.2.1 Certification Record. The certification record shall include at least the following information:

9.2.1.1 name of certified individual;

9.2.1.2 level of certification and NDT method, including the test technique covered; limitations (if any), as applicable;

9.2.1.3 results of current employer examinations that the individual has taken;

9.2.1.4 for Level III personnel, a copy of the candidate's ASNT Level III certificate;

9.2.1.5 dates of certification, expiration, suspension, revocation, and reinstatement; and

9.2.1.6 the signature, printed name, and title of both the designated NDT Level III that verified the qualifications of candidate for certification and the signature of employer's Designated Certifying Authority, when the Level III and Certifying Authority are identified as separate positions in the certification procedure.

9.2.2 Dates of Certification and Expiration.

9.2.2.1 The date of certification shall be a date after which the employer's requirements for education, training, experience, and examinations have been met for each method and/or technique, as specified in the employer's certification procedure.

9.2.2.2 The date of certification is the date when the Certifying Authority signs the certification record.

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9.2.2.3 The date of expiration should be five years from the date established in paragraph 9.2.2.2.

9.2.3 NDT Training Record. A documented history of the employee's training shall be maintained that identifies NDT training received by the individual, the organization providing the training, dates of the training, hours of training, evidence of satisfactory completion, and the instructor's name.

9.2.4 NDT Experience Record. A record that identifies the individual's experience performing various nondestructive tests shall be maintained for purposes of verifying initial certification experience and continuing experience.

9.2.5 Record of Previous Experience. Documented evidence of the individual's previous NDT training and experience shall be maintained if previous training and experience are used to satisfy any part of the requirements of this standard.

9.2.6 Visual Examination Records. Current records of vision examinations required by paragraph 6.1 shall be maintained.

9.2.7 Certification Certificates and Wallet Cards. The following requirements shall be followed if certification certificates and/or wallet cards are issued by the employer:

9.2.7.1 When certificates or wallet cards are issued and provided by the employer, the requirements of ASNT Policy G-14 (asnt.org) shall be followed to ensure appropriate usage of the ASNT name, acronym, and logo.

9.2.7.2 The certificate shall indicate that the certification is in accordance with the requirements of the employer's certification procedure.

9.2.7.3 The legal name of the employer shall be clearly identified followed by the document number and revision/edition number of the certification procedure.

9.2.7.4 The certificate shall indicate "CP-189" followed by the four-digit year of the applicable edition of CP-189 that the certification procedure was developed to follow.

9.2.7.5 Full legal name of the certified individual in the ISO basic Latin alphabet. Additional characters may also be used where customary to accommodate local languages.

9.2.7.6 Level of certification, NDT method, and/or technique, as applicable and limitations (if any).

9.2.7.7 Limitations of certification authorization should be clearly defined, when appropriate for the employer, which may include restrictions on product forms, industry sectors, codes, standards, and examination procedures.

9.2.7.8 The full printed legal name and signature of both the designated NDT Level III that verified the qualifications of the candidate for certification and the designated Certifying Authority when the Level III and Certifying Authority are identified as separate positions in the employer's certification procedure.

9.2.7.8.1 The ASNT-issued Level III identification number, when either or both of the signers hold ASNT Level III certifications.

9.2.7.9 The certificate or wallet card should be fully completed at the time of signing by the designated NDT Level III and/or Certifying Authority so that no fill-in-the-blank data remains to be completed by others.

9.2.7.10 When changes or alterations to certificates or wallet cards are necessitated, this requires the issuing of new certificates or wallet cards.

9.2.7.11 Dates of certification and expiration.

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Section 10.0: Referenced Publications

Section 10.0 lists documents relevant to the provisions of CP-189 (2024) as follows:

10.0 Referenced Publications

10.1 The following documents contain provisions which, through reference in this text, constitute provisions of this standard. Copies may be obtained from The American Society for Nondestructive Testing Inc., 1201 Dublin Road, Suite #G04, Columbus, OH 43215, USA, asnt.org.

10.1.1 ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel,* latest edition.

10.1.2 ASNT application package for certification of nondestructive testing personnel.

10.1.3 ASNT Central Certification Program, ASNT Document CP-1, latest edition.

10.1.4 ASNT 9712 *Certification Program, Qualification and Certification of NDT Personnel to ISO 9712*, latest edition

10.1.5 ASNT Policy G-14 - *Use of the ASNT Name and ASNT Marks*, latest edition.

10.2 The following document contains specific NDT terms which, though referenced in this text, constitute provisions of this standard. Copies may be obtained from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, USA, astm.org.

10.2.1 ASTM E 1316 (latest edition) - Standard Terminology for Nondestructive Examinations, Section A – Common NDT Terms.

In their fall 1974 meeting, the ASNT Board of Directors authorized implementation of a voluntary program for certification of NDT Level III personnel. The Board decided that use of the term "certification" as applied in the ASNT NDT Level III program was appropriate, indicating that personnel holding ASNT NDT Level III certificates had met certain education, training, experience, and examination requirements. It is important to note that the word "certification" as used in the Level III program indicates a record of achievement and/or qualification. As used in Recommended Practice No. SNT-TC-1A: *Personnel Qualification and Certification in Nondestructive Testing*, "certification" indicates that employers have authorized NDT Level III personnel to perform work on their behalf. The first ASNT NDT Level III examinations were given in early 1977.

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REVIEW QUESTIONS

1. In accordance with CP-189 (2024), an NDT Level III:

   a. shall be responsible for grading all Level I and Level II certification examinations.
   b. must prepare all training materials for NDT Level I and Level II personnel.
   c. must approve all questions to be used on examinations for Level I and Level II.
   d. must administer all NDT Level I and II written examinations.

2. Which of the following is true regarding NDT Level III Practical examinations?

   a. This examination requirement may be waived if the candidate holds a currently valid ASNT NDT Level III certificate.
   b. The exam administrator must hold a valid ASNT Level III certification in the applicable test method.
   c. The use of a written checklist may be used at the discretion of the administrator.
   d. Exam administration may be delegated to any authorized representative by the employer.

3. When the near-distance visual acuity examination is given using a Jaeger reading card, it must be given:

   a. in accordance with a procedure.
   b. by any company NDT Level III.
   c. by medical personnel only.
   d. biannually if Jaeger J-1 letters are used.

4. As required in CP-189, NDT Level I Practical examinations are intended to be:

   a. given by the employer's Level III.
   b. passed only if all predefined discontinuities are detected and evaluated.
   c. given using one or more test samples for each technique.
   d. given using one or more test samples for each method.

5. The Practical examination shall address the technical and practical competency of the examinee when following prepared NDT procedures for:

   a. NDT Level I, NDT Level II, and NDT instructor.
   b. corporate Level IIIs.
   c. all levels of qualification including the trainer or instructor.
   d. NDT Levels I and II.

6. CP-189 recognizes the following levels of qualification:

   a. Level I, Level II, and Level III.
   b. trainee, Level I, Level II, and Level III.
   c. trainee, Level I, Level II limited, Level II, and Level III.
   d. trainee, Level I, Level II limited, Level II, Level III, and NDT instructor.

7. The Level I Practical examination and the Level II Practical examination are similar to those recommended by SNT-TC-1A, with the exception that:

   a. two test samples must be used for each technique.
   b. two test samples must be used for each different product that will be encountered in their job functions.
   c. the test samples used are to be representative of the products that will be encountered when performing their job functions.
   d. two test samples are to be used for each different product that will be encountered in their job functions.

8. In CP-189, as a prerequisite, a candidate for the position of an employer's Level III must hold an ASNT Level III certificate with a currently valid endorsement for each method for which employer certification is sought, and:

   a. must also satisfactorily complete a Specific examination comprising 15 questions on the employer's specifications and standards.
   b. must also satisfactorily complete a Specific examination comprising 30 questions on the employer's specifications and standards for each method.
   c. must also satisfactorily complete a Specific examination comprising 30 questions on the employer's specifications and standards.
   d. must also satisfactorily complete a Specific examination comprising 15 questions on the employer's specifications and standards for each method.

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9. Under CP-189, an individual's certification expires when employment with the employer is terminated or on the last day of the month at the end of five (5) years for Levels I and II. The Level III certification expires:
   a. when the ASNT Level III certificate has expired.
   b. when the ASNT Level III's employment with the employer is terminated.
   c. when the ASNT Level III is accused of unethical or incompetent conduct.
   d. when the ASNT Level III retires from employment with the employer.

10. In accordance with CP-189 (2024), the requirements for an NDT Level III include:
    a. college degree.
    b. 10 years as a Level II.
    c. certification by the employer as a company Level III.
    d. ASNT NDT Level III or PdM Level III certificate.

11. What is the difference between "calibration" and "standardization" in CP-189 (2024)?
    a. Calibration is the adjustment of an instrument to a known reference.
    b. Calibration is the adjustment of an instrument to the dead weight tester or master gauge.
    c. Standardization is the adjustment of an instrument to an NIST traceable national standard.
    d. Standardization is the adjustment of an instrument to test blocks supplied by the customer.

12. How many levels of qualification are there in CP-189 (2024)?
    a. 3
    b. 4
    c. 5
    d. 6

13. In accordance with CP-189 (2024), the minimum passing score on the Practical examination is:
    a. 70% following manufacturer's instructions.
    b. 75%.
    c. 80%.
    d. 85%.

14. In accordance with CP-189 (2024), the administration and grading of the NDT Level I and NDT Level II multiple-choice examinations may be performed by the:
    a. NDT Level III only.
    b. NDT Level III or delegate.
    c. NDT Level II.
    d. employer's quality assurance manager.

15. What are the requirements for vision examinations in accordance with CP-189 (2024)?
    a. Jaeger No. 1
    b. Jaeger No. 2
    c. Jaeger No. 2 and color vision
    d. Jaeger No. 1 and color vision

16. To be certified as an NDT Level III, what examinations must be administered by the employer in accordance with CP-189 (2024)?
    a. General, Specific, and Practical exams
    b. Practical exam only
    c. General and Specific exams
    d. Specific and Level II Practical exams

ANSWERS

1c 2b 3a 4c 5d 6d 7c 8b 9a 10d 11a 12d 13c 14b 15d 16d

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CHAPTER 3

THE ASNT NDT LEVEL III CERTIFICATION PROGRAM

In their fall 1974 meeting, the ASNT Board of Directors authorized implementation of a voluntary program for certification of NDT Level III personnel. The Board decided that use of the term "certification" as applied in the ASNT NDT Level III program was appropriate, indicating that personnel holding ASNT NDT Level III certificates had met certain education, training, experience, and examination requirements. It is important to note that the word "certification" as used in the Level III program indicates a record of achievement and/or qualification. As used in Recommended Practice No. SNT-TC-1A: *Personnel Qualification and Certification in Nondestructive Testing*, "certification" indicates that employers have authorized NDT Level III personnel to perform work on their behalf. The first ASNT NDT Level III examinations were given in early 1977.

The following is a description of the ASNT Level III Certification Programs. Additional information is available on the ASNT website at asnt.org under the Certification menu option. The program information document, CP-ASNT-1D, which describes the ASNT NDT Level III Program in detail, is reprinted from the ASNT website as Appendix B in this publication.

ASNT NDT Level III

The ASNT NDT Level III program initially offered certification in five NDT methods—it currently offers certification examinations in 10 methods. More than 8900 personnel in 86 countries currently hold nearly 28 000 certifications in NDT methods, making the ASNT NDT Level III program the largest NDT certification program in the world. ASNT is an independent, third-party certification body accredited by the American National Standards Institute (ANSI) in accordance with ISO/IEC 17024: *Conformity Assessment – General Requirements for Bodies Operating Certification of Persons*. All ASNT certification examinations are developed and maintained using psychometric principles that comply with the ISO 17024 requirements.

The ASNT method examinations are given in multiple-choice format in the following NDT methods. Where reference information is needed, to complete the test, that information is included in the body of the examination.

- Acoustic Emission Testing (AE)
- Electromagnetic Testing (ET)
- Leak Testing (LT)
- Liquid Penetrant Testing (PT)*
- Magnetic Flux Leakage Testing (MFL)*
- Magnetic Particle Testing (MT)*
- Radiographic Testing (RT)
- Thermal/Infrared Testing (IR)*
- Ultrasonic Testing (UT)
- Visual Testing (VT)*

*These examinations include 90 questions; the other examinations have 135 questions. For examinations, MFL is designated ML.

Methods under development (at the time of publication) include microwave technology testing, guided wave testing, and the reintroduction of neutron radiographic testing.

What Is an ASNT NDT Level III?

ASNT NDT Level III personnel are individuals who have demonstrated that they are sufficiently knowledgeable to pass the Basic and method qualification examinations developed and administered by ASNT. To gain initial ASNT NDT Level III certification, eligible candidates must pass the Basic examination and at least one method examination. Once certified, additional certifications can be added by passing the applicable method examination as long as the candidate holds one valid ASNT Level III certification. If all certifications expire, the Basic and method examinations must be passed to regain that certification.

The Basic examination is a 135-question test that covers the administration of certification programs developed in accordance with SNT-TC-1A and ANSI/ASNT CP-189: *ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel*; general knowledge of other NDT test methods; and knowledge of materials, fabrication, and production technology. Method examinations may be 90 or 135 questions long and address in-depth knowledge of the theory and practices of the applicable NDT method.

ASNT Predictive Maintenance Level III

ASNT began offering the Predictive Maintenance (PdM) Level III certification program in 2000 and now offers PdM Level III certification in IR. Together, the ASNT NDT Level III program and the ASNT PdM Level III program are known as the ASNT Level III Certification Program.

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ASNT PdM certification was developed as a result of industry requests for a third-party certification that focused on PdM knowledge and test methods instead of the traditional NDT methods used for NDT Level III certification. In response to this request, ASNT developed a 90-question PdM Basic examination that covers the same certification requirements as the NDT Basic examination. However, instead of addressing knowledge of materials, fabrication, production technology, and other traditional NDT test methods, it covers the basics of common PdM test methods and knowledge of machinery technology. PdM certification is currently offered in IR only, and successful completion of the PdM Basic and IR method examinations result in the issuance of an ASNT PdM Level III certificate.

Eligibility for ASNT Level III Examinations

Candidates must have met the eligibility requirements specified in paragraph 6.3.2 (p. 6) of SNT-TC-1A (2024):

> 6.3.2 NDT Level III
>
> 6.3.2.1 Have a bachelor's degree (or higher) in engineering or science, plus one additional year of experience beyond the Level II requirements in NDT in an assignment comparable to that of an NDT Level II in the applicable NDT method(s), or:
>
> 6.3.2.2 Have completed with passing grades at least two (2) years of engineering or science study at a university, college, or technical school, plus two additional years of experience beyond the NDT Level II requirements in NDT in an assignment at least comparable to that of NDT Level II in the applicable NDT method(s), or:
>
> 6.3.2.3 Have four years of experience beyond the NDT Level II requirements in NDT in an assignment at least comparable to that of an NDT Level II in the applicable NDT method(s).
>
> The above NDT Level III requirements may be partially replaced by experience as a certified NDT Level II or by assignments at least comparable to NDT Level II as defined in the employer's written practice.

Upon successful completion of the necessary qualification examinations, ASNT will issue the candidate an ASNT NDT Level III certificate and wallet card that are valid for five years.

NDT/PdM or PdM/NDT Conversion

1. Personnel with a currently valid NDT certificate in the IR test method may attain PdM certification in IR by successfully completing the PdM Basic examination.
2. Personnel with a currently valid PdM certificate in the IR test method may attain NDT certification in IR by successfully completing the NDT Basic examination.
3. This applies to the IR method only.

Recertification

All ASNT Level III certificate holders must be recertified at five-year intervals by one of the following methods:

1. By examination, prior to their certification expiration date in the applicable method. As long as at least one method remains current, the Basic examination does not have to be taken again; or
2. By application, using a "points" system. Recertification by points requires the following three items:
   - reaffirmation of the ASNT Level III Code of Ethics;
   - demonstration of continued NDT activity and Level III employment; and
   - submittal of documentation showing that 25 recertification points have been earned within the applicant's current five-year certification period. Activities that earn recertification points are shown in the ASNT NDT Program Renewal Requirements document downloadable from the ASNT website (asnt.org) under the Certification menu.

The same recertification requirements apply to PdM Level III certificate holders.

SNT-TC-1A Certification Options

SNT-TC-1A offers the employer several options for fulfilling certification responsibilities:

- incorporate acceptance of ASNT NDT Level III certification into the employer's written practice; or
- incorporate acceptance of Level III qualification examinations administered by an outside agency that meet the requirements of the employer's written practice; or
- incorporate the employer's examinations as defined in the employer's written practice.

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Use of the ASNT NDT Level III Certification

ASNT NDT Level III certification provides an internationally recognized way for individuals, companies, and industry sectors to take advantage of standardized examinations developed by qualified subject matter experts and administered by an ISO/IEC 17024 accredited third-party certification body. Under SNT-TC-1A, employers may accept valid ASNT NDT Level III certification as proof that the certificate holder has met the Basic and method examination requirements for the test methods listed on the ASNT certificate. However, the employer must still determine if the Specific examination requirements have been met and has the sole responsibility for authorizing (certifying) their Level III NDT personnel to perform NDT tasks on behalf of their company.

NOTE: In the case of CP-189, an individual must hold an ASNT NDT Level III certificate prior to becoming an organization's Level III. Under SNT-TC-1A, an individual can use the ASNT NDT Level III certificate as one of the many optional approaches for becoming an organization's Level III. In both cases, the ASNT NDT Level III certificate must be supported by a written practice that details how the organization's Level III is qualified and certified for the specific needs of the organization.

ACCP Professional Level III

With the introduction of the ASNT 9712 certification program, the ASNT Central Certification Program (ACCP) is being phased out. ACCP renewal is no longer available as of August 2023. Level III certificate holders under ACCP who wish to keep their central certification must transition to the ASNT 9712 program. To do this, ACCP Level III holders must apply for ASNT 9712 certification before their ACCP certificate expires and complete any required examinations. If the individual already holds an ASNT NDT Level III certification, the Basic and method examinations are waived during this transition. However, additional training or experience requirements may apply.

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REVIEW QUESTIONS

1. ASNT NDT Level III certification is a requirement of:
   a. SNT-TC-1A.
   b. CP-189.
   c. CP-106.
   d. both SNT-TC-1A and CP-189.

2. A currently valid ASNT NDT Level III certificate indicates:
   a. an ASNT Level III is authorized to supervise but not perform NDT tasks on behalf of the employer.
   b. the employer can accept a currently valid ASNT NDT Level III certificate as proof of qualification.
   c. an individual is certified in both NDT and PdM.
   d. that the certificate holder has satisfied the Basic and method examination requirements as set forth in SNT-TC-1A.

3. ASNT NDT Level III certification:
   a. immediately qualifies a certificate holder to act as the employer's Level III.
   b. makes an individual potentially eligible to become an employer's Level III.
   c. permits the individual to function as an independent Level III without further documentation.
   d. only applies when recognized by some requirements document.

4. The ASNT NDT Level III Basic examination covers all of the following except:
   a. general knowledge of other NDT test methods.
   b. knowledge of SNT-TC-1A and CP-189 certification programs.
   c. knowledge of materials, fabrication, and production technology.
   d. knowledge of international certification programs.

5. To recertify by application, an ASNT NDT Level III must:
   a. demonstrate continued NDT activity and Level III employment.
   b. reexamine prior to their current expiration date.
   c. participate in the ASNT Central Certification Program.
   d. acquire 25 recertification points since the beginning of their NDT employment.

6. To be eligible to sit for the ASNT NDT Level III examinations, a candidate with two years of passing grades in engineering or science study at a university or technical school must have:
   a. one additional year of experience beyond the Level II requirements in NDT in an assignment comparable to that of an NDT Level II in the applicable NDT method(s).
   b. two additional years of experience beyond the Level II requirements in NDT in an assignment at least comparable to that of NDT Level II in the applicable NDT method(s).
   c. four years of experience beyond the Level II requirements in NDT in an assignment at least comparable to that of an NDT Level II in the applicable NDT method(s).
   d. one additional year of experience beyond the Level II requirements in NDT in an assignment comparable to that of an NDT Level II and an additional 20 hours of classroom training in the applicable test method(s).

ANSWERS

1b 2d 3c 4d 5a 6b

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CHAPTER 4

GENERAL APPLICATIONS OF NDT METHODS

Overview of the ASNT Basic Examination

The second part of the ASNT Basic examination for Level III certification covers the examinee's knowledge and comprehension of basic and fundamental applications of various NDT methods. While an employer's current requirements for a Level III may only involve one, two, or a few NDT methods, it is not envisioned that a Level III as described in paragraph 4.3.3 of Recommended Practice No. SNT-TC-1A can function adequately without some basic knowledge of the existence and applications of commonly used NDT methods.

In light of rapidly expanding technology, NDT Level III personnel have an obligation to continually review current practices, recommend and develop new techniques where applicable, and seek more effective methods where applicable. Without some basic knowledge of the broad applications of NDT technology, the NDT Level III could not be considered as meeting the Level III qualifications as defined in paragraph 4.3.3 (p. 3) of SNT-TC-1A (2024):

> An NDT Level III individual should have sufficient technical knowledge and skills to be capable of developing, qualifying, and approving procedures; establishing and approving techniques; interpreting codes, standards, specifications, and procedures; and designating the particular NDT methods, techniques, and procedures to be used. The NDT Level III should be responsible for the NDT operations for which qualified and assigned and should be capable of interpreting and evaluating results in terms of existing codes, standards, and specifications. The NDT Level III should have sufficient practical background in applicable materials, fabrication, service (in-service), and product technology to establish techniques, and to assist in establishing acceptance criteria when none are otherwise available. The NDT Level III should have general familiarity with other appropriate NDT methods, as demonstrated by an ASNT Level III Basic examination or other means. The NDT Level III, in the methods in which certified, should have sufficient technical knowledge and skills to be capable of training and examining NDT Level I, II, and III personnel for certification in those methods.

Topical Outlines, Reference Resources, and Review Questions

This section contains topical outlines, reference resources, and review questions for selected NDT methods recognized in SNT-TC-1A. The reader is advised to use the reference resource material if difficulty is encountered in answering the following questions. Note especially that the questions are not difficult for those with practical exposure in the use of the method. The questions are devised to cover fundamentals, basic techniques, and applications.

Topical outlines for the methods included in this section are from "General Familiarity with Other NDT Methods" (pp. 111–113) in the Basic examination Level III unit of ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel* (2024). A general reference for each method presented below is the *Nondestructive Testing Handbook, Vol. 10: Nondestructive Testing Overview,* third edition, published by ASNT in 2012. Another comprehensive reference is *Materials and Processes for NDT Technology*, second edition: Chapter 12, "Nondestructive Testing Methods," published by ASNT in 2016.

Acoustic Emission Testing

Topical Outline

2.1 Acoustic Emission Testing (AE)

- 2.1.1 Fundamentals
  - 2.1.1.1 Principles/theory of AE
  - 2.1.1.2 Sources of acoustic emissions
  - 2.1.1.3 Equipment and material
- 2.1.2 Proper selection of acoustic emission technique
  - 2.1.2.1 Instrumentation and signal processing
  - 2.1.2.2 Cables (types)
  - 2.1.2.3 Signal conditioning
  - 2.1.2.4 Signal detection
  - 2.1.2.5 Noise discrimination
  - 2.1.2.6 Electronic technique
  - 2.1.2.7 Attenuation materials
  - 2.1.2.8 Data filtering techniques
- 2.1.3 Interpretation and evaluation of test results

REFERENCES

ASNT, 2008, *Questions & Answers Book: Acoustic Emission Testing Method*, American Society for Nondestructive Testing, Columbus, OH.

ASNT, 2005, *Nondestructive Testing Handbook, Vol. 6: Acoustic Emission Testing*, third edition, American Society for Nondestructive Testing, Columbus, OH.

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AE Review Questions

1. In AE testing, what is the most common frequency range used for detecting signals?

    a. 10 to 15 kHz
    b. 100 to 300 kHz
    c. 500 to 750 kHz
    d. 1 to 5 MHz

2. Which type of discontinuity is least likely to be detected by AE testing?

    a. Leaks
    b. Plastic deformation
    c. Growing cracks
    d. Rounded inclusions

3. The total energy loss of a propagating wave is called:

    a. scatter.
    b. dispersion.
    c. diffraction.
    d. attenuation.

4. The Kaiser effect refers to:

    a. velocity changes due to temperature changes.
    b. low amplitude emissions from aluminum structures.
    c. the behavior where emission from a source will not occur until the previous load is exceeded.
    d. emissions from dissimilar material interfaces.

5. The Felicity effect is useful in evaluating:

    a. fiber-reinforced plastic components.
    b. high-alloy castings.
    c. large structural steel members.
    d. ceramics.

6. The Kaiser effect is useful in distinguishing:

    a. electrical noise from mechanical noise.
    b. electrical noise from growing discontinuities.
    c. mechanical noise from growing discontinuities.
    d. electrical noise from continuous emissions.

7. The term "counts" refers to the:

    a. number of times a signal crosses a preset threshold.
    b. number of events from a source.
    c. number of transducers required to perform a test.
    d. duration of hold periods.

8. The AE signal amplitude is related to:

    a. the preset threshold.
    b. the intensity of the source.
    c. the bandpass filters.
    d. background noises.

9. Threshold settings are determined by the:

    a. graininess of the material.
    b. attenuation of the material.
    c. test duration.
    d. background noise level.

10. Background noise can be reduced by:

    a. electronic filtering.
    b. using flat response amplifiers.
    c. using in-line amplifiers.
    d. using heavier gauge coaxial cable.

Electromagnetic Testing

Topical Outline

2.2 Electromagnetic Testing (ET)
- 2.2.1 Sensors
- 2.2.2 Basic types of equipment, types of readout
- 2.2.3 Reference standards
- 2.2.4 Applications and test result interpretation
    - 2.2.4.1 Flaw detection
    - 2.2.4.2 Conductivity and permeability sorting
    - 2.2.4.3 Thickness gauging
    - 2.2.4.4 Process control

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ET Review Questions

1. Eddy currents are circulating electrical currents induced in conductive materials by:

    a. continuous direct current.
    b. gamma rays.
    c. an alternating magnetic field.
    d. a piezoelectric force.

2. The method used to generate eddy currents in a test specimen by means of a coil can most closely be compared with the action of a:

    a. transformer.
    b. capacitor.
    c. storage battery.
    d. generator.

3. ET relies on the principle of:

    a. magnetostriction.
    b. electromagnetic induction.
    c. piezoelectric energy conversion.
    d. magnetomotive force.

4. What happens when the electrical current in an eddy current coil reverses direction?

    a. The direction of the eddy currents in the test part remains the same.
    b. The eddy currents in the test part will change phase by 45°.
    c. The direction of the eddy currents in the test part also reverses.
    d. The eddy currents in the test part will change phase by 90°.

5. In order to generate measurable eddy currents in a test specimen, the specimen must be:

    a. an electrical conductor.
    b. an electrical insulator.
    c. a ferromagnetic material.
    d. a nonmagnetic material.

6. The magnetic field generated by eddy currents induced in a test specimen:

    a. reinforces the magnetic field that induced the eddy currents.
    b. cancels the magnetic field that induced the eddy currents.
    c. opposes the magnetic field that induced the eddy currents.
    d. has no effect on the magnetic field that induced the eddy currents.

7. In ET, what is IACS a recognized abbreviation for?

    a. Induced Alternating Current System
    b. Inductively Activated Comparison System
    c. Internal Applied Current System
    d. International Annealed Copper Standard

8. In ET, what is the specimen coupled to the test coil by?

    a. Core coupling
    b. Magnetic saturation
    c. The coil's electromagnetic fields
    d. Magnetic domains

9. When is the penetration of eddy currents in a conductive material decreased?

    a. When the test frequency or conductivity of the specimen is decreased.
    b. When the test frequency is decreased or conductivity of the specimen is increased.
    c. When the test frequency, conductivity of the specimen, or permeability of the specimen is increased.
    d. When the permeability of the specimen is decreased.

10. At a fixed test frequency, in which of the following materials will the eddy current penetration be greatest?

    a. Aluminum (35% IACS conductivity)
    b. Brass (15% IACS conductivity)
    c. Copper (95% IACS conductivity)
    d. Lead (7% IACS conductivity)

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11. A term used to describe the effect observed due to a change in the coupling between a test specimen and a flat probe coil when the distance of separation between them is varied is:

    a. liftoff.
    b. fill factor.
    c. edge effect.
    d. end effect.

12. When performing eddy current testing, under what condition are discontinuities most easily detected?

    a. When the eddy currents are coplanar with the major dimension of the discontinuity.
    b. When the eddy currents are perpendicular to the major plane of the discontinuity.
    c. When the eddy currents are parallel to the major dimension of the discontinuity.
    d. When the eddy currents are 90° out of phase with the current in the coil.

13. Which of the following discontinuities is easiest to detect with an electromagnetic test? (Assume that the area of the discontinuity is equal in all four choices listed.)

    a. A subsurface crack which lies parallel to the direction of the eddy current.
    b. A discontinuity located in the center of a 51 mm (2 in.) diameter bar.
    c. A radial crack that extends to the outer surface of a 51 mm (2 in.) diameter bar.
    d. A subsurface radial crack located at a depth of 13 mm (0.5 in.) in a 51 mm (2 in.) diameter bar.

14. What is a term used to define the timing relationships involved in alternating current signals?

    a. Magnitude
    b. Phase
    c. Impedance
    d. Time-gain correction

15. The impedance of a test coil can be represented by the vector sum of:

    a. inductive reactance and resistance.
    b. capacitive reactance and resistance.
    c. inductive reactance and capacitive reactance.
    d. inductive reactance, capacitive reactance, and resistance.

16. What does the term "fill factor" apply to?

    a. A surface coil
    b. Coaxial cable
    c. An encircling coil
    d. The ability to null an eddy current instrument

17. Which of the following materials would be more likely used as a mounting material for a probe coil?

    a. Aluminum
    b. Plastic
    c. Copper
    d. Nonferromagnetic steel

18. Which heat exchanger tube material CANNOT be inspected by standard eddy techniques using an internal coil probe?

    a. Copper
    b. Austenitic stainless steel
    c. Brass
    d. Carbon steel

19. Reference standards used for ET:

    a. must contain artificial discontinuities such as notches and drilled holes.
    b. must contain natural discontinuities such as cracks and inclusions.
    c. must be free of measurable discontinuities, but may contain artificial or natural discontinuities, or may be free of discontinuities, depending on the test system and the type of test being conducted.
    d. must be constructed from the same material of the object being inspected.

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20. Which of the following conditions would be the most difficult to detect when testing a rod using an encircling coil?

    a. A short surface crack that has a depth of 10% of the rod diameter.
    b. A small inclusion in the center of the rod.
    c. A 5% change in diameter.
    d. A 10% change in conductivity.

21. The thickness of nonconductive coatings on a conductive base can be most simply measured by:

    a. observing the liftoff effect caused by the coating.
    b. testing both sides of the specimen.
    c. varying the test frequency over a given range during the test.
    d. using a specially shaped encircling coil.

22. Some of the products commonly tested using encircling coils are:

    a. rods, tubes, and wire.
    b. interior of hollow tubes.
    c. sheets and metal foil.
    d. square billets and plates.

23. It is often possible to sort various alloys of a nonmagnetic metal using ET when:

    a. there is a unique range of permeability values for each alloy.
    b. there is a unique range of conductivity values for each alloy.
    c. the direction of induced eddy currents varies for each alloy.
    d. the magnetic domains for each alloy are different.

24. When conducting ET on tubing with a system that includes a frequency discriminating circuit, which of the following variables would be classified as a high-frequency variable?

    a. Conductivity changes
    b. Diameter changes
    c. Wall thickness variations
    d. Small discontinuities

25. Tubing is generally inspected using:

    a. U-shaped coils.
    b. gap coils.
    c. encircling coils.
    d. a sliding probe.

26. An out-of-phase condition between current and voltage:

    a. can exist in both the primary and secondary windings of an eddy current coil.
    b. can exist only in the secondary winding of an eddy current coil.
    c. can exist only in the primary winding of an eddy current coil.
    d. exists only in the test specimen.

27. For age hardenable aluminum and titanium alloys, changes in hardness are indicated by changes in:

    a. retentivity.
    b. permeability.
    c. conductivity.
    d. magnetostriction.

28. In performing inspection of heat exchanger tubing using an internal differential probe, which type of defect would be difficult to detect?

    a. Longitudinally oriented crack
    b. Localized corrosion pit
    c. Circumferentially oriented crack
    d. Dent

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Leak Testing

Topical Outline

2.3 Leak Testing (LT)
- 2.3.1 Fundamentals
  - 2.3.1.1 Bubble leak testing
  - 2.3.1.2 Pressure leak testing
  - 2.3.1.3 Halogen detector leak testing
  - 2.3.1.4 Mass spectrometer leak testing
- 2.3.2 LT procedures, and techniques
  - 2.3.2.1 System factors
  - 2.3.2.2 Relative sensitivity
  - 2.3.2.3 Evacuated systems
  - 2.3.2.4 Pressurized systems; ambient fluids, tracer fluids
  - 2.3.2.5 Locating leaks
  - 2.3.2.6 Standardization
- 2.3.3 Test result interpretation
- 2.3.4 Essentials of safety
- 2.3.5 Test equipment
- 2.3.6 Applications
  - 2.3.6.1 Piping and pressure vessels
  - 2.3.6.2 Evacuated systems
  - 2.3.6.3 Low-pressure fluid containment vessels, pipes, and tubing
  - 2.3.6.4 Hermetic seals
  - 2.3.6.5 Electrical and electronic components

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LT Review Questions

1. Which of the following systems or components would require the greatest precautions and special considerations when preparing for a leak test?

   a. Piping and pressure vessels
   b. Refrigeration piping
   c. Vacuum chambers
   d. Sintered material components

2. Pressure change leak testing is dependent on three variables. A change to any of these three will have a direct effect on at least one of the other two variables. These three variables are:

   a. pressure, volume, and pumping speed.
   b. pressure, volume, and temperature.
   c. volume, temperature, and ambient atmospheric pressure.
   d. temperature, pumping speed, and fluid viscosity.

3. Which of the following is an LT technique?

   a. Static
   b. Gaseous diffusion
   c. Dynamic
   d. Detector probe

4. Which of the following LT techniques can reliably quantify a leak $\leq 1.0 \times 10^{-7}$ atm cc/sec in a normal industrial environment?

   a. Bubble leak testing
   b. Pressure change leak testing
   c. Mass spectrometer leak testing
   d. Liquid penetrant testing

5. Establishing differential pressure between the test object and the environment is an essential element in which of the following NDT methods?

   a. X-ray diffraction
   b. Leak testing
   c. Neutron radiography
   d. Electromagnetic testing

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6. Which of the following best describes the LT technique in which the interior of the test object is continuously evacuated during the test, a tracer gas is applied to the exterior, and a leak detector is connected to the vacuum system?

   a. Static leak test
   b. Helium leak test
   c. Dynamic leak test
   d. Halogen leak test

7. Assuming no significant leakage in a constant-volume system, if the temperature increases during a pressure-drop leak test, the pressure in the system will:

   a. increase.
   b. remain the same.
   c. decrease.
   d. first decrease, then increase to its former level.

8. If the sensitivity of a halogen leak detector remains constant throughout a test, and the pretest and posttest calibrations are acceptable, which of the following conclusions can be made at the end of the test?

   a. No leaks smaller than a certain size have gone undetected.
   b. The total leakage rate of the test object is less than a certain amount.
   c. The instrument and the test procedure were capable of detecting leakage of a certain size during the test.
   d. The instrument and test procedure were only capable of detecting leakage of a certain size upstream of the tracer gas during the test.

9. All leak detection techniques are dependent upon:

   a. barometric pressure.
   b. oxygen content of the surrounding air.
   c. all of the other gases in the mass spectrometer.
   d. establishing a differential pressure across the test boundary.

10. In high-vacuum systems, the sensitivity of a pressure change leak test depends not only on the observed pressure change but also on the degree of outgassing. Outgassing is best defined as:

   a. the release of gas from materials in a vacuum.
   b. directly proportional to the temperature of the gas.
   c. the viscosity of the pressurizing gas.
   d. the drop in test pressure due to leakage from the vacuum manifold.

11. What is the essential information that must be recorded to perform a pressure change leak test?

   a. Helium concentration, leak response, background
   b. Barometric pressure, start time, stop time, temperature
   c. Temperature, elapsed time, volume, pressure
   d. Mean free path of helium, fluid viscosity, pressure differential

12. What is the easiest way to increase sensitivity in a pressure change leak test?

   a. Change instrumentation.
   b. Turn down the thermostat.
   c. Put the test item in an environmental chamber.
   d. Increase test time.

13. When performing a helium mass spectrometer hood test, using a 50% helium concentration (instead of 100% helium) will ________ the final measured leak rate for the test item.

   a. increase
   b. decrease
   c. produce no change in
   d. invalidate

14. In a pressure-change "rate-of-rise" test (vacuum decay test), which test variable is NOT recorded?

   a. Pressure
   b. Volume
   c. Temperature
   d. Elapsed time

15. In a helium mass spectrometer leak test using the hood technique, if the inspector has a final calibration 40% greater than the preliminary calibration, what would they do?

   a. Nothing—the difference is negligible.
   b. Recalibrate and repeat the test on the item.
   c. Increase the test duration.
   d. Increase the helium concentration and continue testing.

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16. Which of the following safety precautions is most important when performing leak testing on pressurized vessels or piping?

    a. Ensure the test area is free of any flammable materials.
    b. Always calibrate the leak detector with a known leak standard.
    c. Stand clear of potential pressure release paths and wear appropriate protective equipment.
    d. Use a secondary containment chamber around the test object.

17. Which leak test technique is well-suited for locating leaks in low-pressure fluid containment vessels (such as low-pressure tanks or piping)?

    a. Bubble leak testing (soap solution applied over slight pressure or vacuum)
    b. Helium mass spectrometer testing in vacuum mode
    c. Pressure change leak testing
    d. Mass spectrometer leak testing

18. What technique is typically used to detect fine leaks in hermetically sealed components (such as electronic packages)?

    a. Helium mass spectrometer leak testing
    b. Bubble immersion (underwater) leak testing
    c. Halogen diode leak testing
    d. Simple pressure gauge monitoring

Liquid Penetrant Testing

Topical Outline

2.4 Liquid Penetrant Testing (PT)
- 2.4.1 Fundamentals
  - 2.4.1.1 Interaction of penetrants and discontinuity openings
  - 2.4.1.2 Fluorescence and contrast
- 2.4.2 PT
  - 2.4.2.1 Penetrant processes
  - 2.4.2.2 Test equipment and systems factors
  - 2.4.2.3 Test result interpretation, discontinuity indications
  - 2.4.2.4 Applications
    - 2.4.2.4.1 Castings
    - 2.4.2.4.2 Welds
    - 2.4.2.4.3 Wrought metals
    - 2.4.2.4.4 Machined parts
    - 2.4.2.4.5 Leaks
    - 2.4.2.4.6 Field inspections

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PT Review Questions

1. What is the tendency of a liquid penetrant to enter a discontinuity primarily related to?

    a. The viscosity of the penetrant
    b. Capillary action
    c. The chemical inertness of the penetrant
    d. The specific gravity of the penetrant

2. PT is a nondestructive test method that can be used for:

    a. locating and evaluating all types of discontinuities in a test specimen.
    b. locating and determining the length, width, and depth of discontinuities in a test specimen.
    c. determining the tensile strength of a test specimen.
    d. locating discontinuities open to the surface.

3. Which of the following processes is best situated to detect shallow discontinuities?

    a. Postemulsifiable
    b. Solvent-removable
    c. Aqueous developer
    d. Water-washable

4. What is it called when a dye in penetrant materials gives off visible light after being exposed to ultraviolet light?

    a. Emissivity
    b. Irradiation
    c. Spectrum blocking
    d. Fluorescence

5. When using a fluorescent, postemulsifier penetrant, the length of time the emulsifier is allowed to remain on the part is critical for detecting shallow discontinuities. For high viscosity emulsifiers, the optimum dwell time is typically:

    a. 10 seconds.
    b. 5 seconds.
    c. determined by experimentation.
    d. 2 to 3 minutes.

6. A red penetrant indication against white background is most likely to be seen when:

    a. dry developers are used.
    b. visible dye penetrants are used.
    c. fluorescent postemulsified penetrants are used.
    d. ultraviolet light is used with visible dye penetrants.

7. The most widely accepted technique for removing excess water-washable penetrant from the surface of a test specimen is by:

    a. using a wet rag.
    b. using a coarse water spray rinse.
    c. washing the part directly under water running from a tap.
    d. immersing the part in water.

8. Which of the following PT systems is generally considered the least sensitive?

    a. Water-washable; visible dye
    b. Solvent-removable; visible dye
    c. Water-washable; fluorescent dye
    d. Postemulsified; visible dye

9. When performing PT using solvent-removable visible dye penetrant, there are several ways to remove excess penetrant from the surface of the part. Which of the techniques is generally regarded as most suitable for giving accurate test results?

    a. Squirting solvent over the surface with no more than 69 kPa (10 psi) pressure.
    b. Wiping with a solvent-soaked cloth, then wiping with a dry cloth.
    c. Wiping with a solvent-dampened cloth, then wiping with dry cloths.
    d. Wiping with dry cloths, then wiping with a solvent-dampened cloth, and finally wiping with a dry cloth.

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10. A problem with retesting a specimen that has been previously tested using PT is that the:

    a. penetrant residue left in discontinuities may not readily dissolve, and the retest may be misleading.
    b. penetrant may form beads on the surface.
    c. penetrant will lose a great deal of its color brilliance.
    d. added penetrant will intensify the penetrant residue, making indications larger than normal.

11. A commonly used method of checking the overall performance of a liquid penetrant material system is to:

    a. determine the viscosity of the penetrant.
    b. measure the wettability of the penetrant.
    c. compare two sections of artificially cracked specimens.
    d. check the penetrant contaminant levels.

12. The function of emulsifier in the postemulsified penetrant process is to:

    a. influence the rate of penetration into deep, tight cracks.
    b. react with the surface penetrant to make the penetrant water washable.
    c. add fluorescent dye or pigment to the penetrant.
    d. emulsify surface oils and greases to facilitate their removal.

13. Which of the following statements most applies to developers used during PT?

    a. Developers are normally highly fluorescent.
    b. Some developers create a fluctuating background during inspection.
    c. Developers absorb or blot the penetrant that remains in discontinuities after the excess penetrant has been removed.
    d. Nonaqueous developers are better suited for very humid environments.

14. The penetrant indication for a cold shut on the surface of a casting will normally be:

    a. a dotted line.
    b. a large bulbous indication.
    c. a smooth continuous line.
    d. undetectable since cold shuts are closed over on the surface.

15. A crack-type discontinuity will generally appear as:

    a. a rounded indication.
    b. a continuous line, either straight or jagged.
    c. a broad, fuzzy indication.
    d. random round or elongated holes.

16. In PT, scattered round indications on the surface of a part could indicate:

    a. fatigue cracks.
    b. porosity.
    c. weld laps.
    d. hot tears.

17. Which of the following are typical nonrelevant indications found in PT?

    a. Indications due to part geometry or part design configurations.
    b. Nonmagnetic indications.
    c. Indications due to contact with another test object.
    d. Indications on low-stressed areas of the part.

18. Which of the statements best states the effects of sandblasting for the cleaning of surfaces to be tested with PT?

    a. Surface discontinuities may be closed.
    b. Oil contaminants might be sealed in the discontinuities.
    c. The sand used in the sandblasting operation may be forced into the discontinuity.
    d. The sandblasting operation may introduce discontinuities into the part.

19. The penetrant indication of a forging lap will normally be a:

    a. round or nearly round indication.
    b. cluster of indications.
    c. thin continuous line.
    d. dotted line.

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20. Aluminum alloy test specimens that have been tested with PT should be thoroughly cleaned after testing because:

   a. acid in the penetrant may cause severe corrosion.
   b. the oily residue from the test will severely inhibit the application of paint on aluminum alloys.
   c. a chemical reaction between the penetrant and aluminum could cause a fire.
   d. the alkalines in wet developers and most emulsifiers could cause surface pitting, particularly in moist atmospheres.

21. Which of the following is a discontinuity that might be found in rolled bar stock?

   a. Blow holes
   b. Shrinkage laps
   c. Stringers or seams
   d. Insufficient penetration

22. Anodized surfaces are usually considered poor candidates for high-sensitivity PT because the anodizing process produces a conversion layer that:

   a. is extremely smooth and slick.
   b. has a multitude of extremely small pores.
   c. may have alkaline residue that "quenches" the penetrant.
   d. cannot be cleaned by ordinary processes.

23. Which PT technique is best suited for performing inspections without electricity?

   a. Water-washable fluorescent penetrant
   b. Postemulsified fluorescent penetrant
   c. Visible dye penetrant
   d. Hydrophilic fluorescent penetrant

24. Which statement is true concerning PT of welds with rough surfaces?

   a. The postemulsified process offers advantages over the water-washable process.
   b. If the solvent-removal process is used, the best developer would be an aqueous suspension.
   c. Welds with rough surfaces cannot be successfully tested by any PT technique.
   d. Welds with rough surfaces may need to be ground smooth prior to PT.

25. A penetrant must:

   a. change viscosity in order to spread over the surface of the part.
   b. spread easily over the surface of the material.
   c. have a low flash point.
   d. be able to change color in order to fluoresce.

26. Which of the following is an example of an in-service discontinuity?

   a. Grinding cracks
   b. Porosity
   c. Stress-corrosion crack
   d. Cold lap

27. Penetrant materials are tested and qualified to what specification?

   a. AMS 2642
   b. AMS 2644
   c. AMS 2646
   d. AMS 2648

28. Which of the following penetrants can be removed with water?

   a. Method A penetrant
   b. Method B penetrant
   c. Method C penetrant
   d. Method D penetrant

29. Fluorescent dye penetrant is typically which color?

   a. Red
   b. Yellow-green
   c. Orange
   d. Blue

30. In performing field penetrant inspection of structural steel welds that have been exposed to atmospheric conditions for an extended period of time, which of the following is an important consideration when inspecting for fatigue cracks?

   a. Temperature of the weld surface
   b. The crack may be oxide filled which would prevent penetration of the penetrant.
   c. Wind conditions
   d. Orientation of the crack

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Magnetic Flux Leakage Testing

Topical Outline

2.11 Magnetic Flux Leakage Testing (MFL)
- 2.11.1 Fundamentals
  - 2.11.1.1 Magnetic field principles
  - 2.11.1.2 Magnetization by means of electric current
  - 2.11.1.3 Flux leakage
- 2.11.2 MFL
  - 2.11.2.1 Basic types of equipment and inspection materials
  - 2.11.2.2 Types of discontinuities found by MFL
  - 2.11.2.3 Sensors used in MFL
- 2.11.3 Applications
  - 2.11.3.1 Wire rope inspection
  - 2.11.3.2 Pipe body inspection
  - 2.11.3.3 Tank floor/steel plate inspection

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MFL Review Questions

1. In MFL inspection for discontinuities using an active field, the part being inspected should be magnetized:

   a. beyond saturation.
   b. near the point of maximum permeability.
   c. to saturation or near saturation.
   d. well below saturation.

2. An advantage that MFL testing has in comparison with electromagnetic testing is that MFL is:

   a. less sensitive to interferences caused by surface roughness.
   b. useful on products at temperatures above the Curie point.
   c. useful on austenitic steels.
   d. easier to use on ferromagnetic materials.

3. What type of discontinuity would NOT typically be indicated by MFL techniques?

   a. Laps
   b. Pitting accompanied by cracking.
   c. Deep-surface discontinuities
   d. Longitudinal seams

4. The strength of the magnetic field in the interior of a coil is determined by the:

   a. number of turns in the coil only.
   b. strength of applied current only.
   c. number of turns in the coil and the strength of the applied current.
   d. direction of applied current in the coil.

5. The current used for magnetization when performing an MFL inspection must be a:

   a. steady nonfluctuating current.
   b. current that reverses direction at an inconsistent rate.
   c. current that fluctuates on and off at a consistent rate.
   d. current that varies based on the thickness of the material.

6. As a general rule, hard (high-strength) ferromagnetic materials have:

   a. high coercive force and are easily demagnetized.
   b. high coercive force and are not easily demagnetized.
   c. low coercive force and are easily demagnetized.
   d. low coercive force and are not easily demagnetized.

7. In MFL, the maximum tube wall thickness for which maximum sensitivity can be maintained is:

   a. 0.08 mm (0.003 in.)
   b. 0.8 mm (0.03 in.)
   c. 8 mm (0.318 in.)
   d. 76 mm (3 in.)

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8. When examining an aboveground storage tank floor with the flux sensor on the top surface:

   a. only top surface discontinuities are detected.
   b. only bottom surface discontinuities are detected.
   c. both top and bottom surface discontinuities can be detected and can generally be distinguished from each other.
   d. both top and bottom surface discontinuities can be detected but generally cannot be distinguished from each other.

9. Materials that are weakly repelled by a magnetic field are called:

   a. diamagnetic.
   b. nonmagnetic.
   c. paramagnetic.
   d. ferromagnetic.

10. A break in the magnetic uniformity of a part (a magnetic discontinuity) is associated with a sudden change in:

    a. resistivity.
    b. inductance.
    c. permeability.
    d. capacitance.

11. A hysteresis curve describes the relation between:

    a. magnetizing force and flux density.
    b. magnetizing force and applied current.
    c. strength of magnetism and alignment of domains within material.
    d. magnetic flux density and the current generated.

12. When inspecting wire rope, a magnetic flux loop is used to monitor:

    a. broken external wires.
    b. broken internal wires.
    c. changes in inspection speed.
    d. reductions in cross-sectional area.

13. What does a Hall effect probe measure?

    a. Permeability
    b. Conductivity
    c. Flux density perpendicular to the probe surface
    d. Tangential field strength

14. The direction of magnetic lines of force are generally considered to be _____ from the direction of current flow.

    a. 45°
    b. 90°
    c. 180°
    d. 270°

15. If an MFL sensor bounces along the surface of aboveground storage tanks, it may:

    a. make it difficult to estimate the severity of an indication.
    b. generate noise in the signal.
    c. decrease the speed of the inspection.
    d. distort the magnetizing system.

16. In an MFL examination of aboveground storage tanks, which factor(s) must be considered when interpreting an indication?

    a. Signal amplitude only
    b. Signal width only
    c. Visual and ultrasonic evaluation
    d. Signal amplitude and width

17. For pipe body inspection using MFL, longitudinal magnetization is best for detection of:

    a. longitudinal discontinuities.
    b. transverse discontinuities.
    c. internal surface discontinuities.
    d. subsurface discontinuities.

18. Residual magnetization is defined as the:

    a. magnetizing force remaining after flux density has decreased.
    b. residual magnetizing force beyond the Curie point.
    c. flux density after removal of magnetizing force.
    d. flux density beyond the level of saturation.

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19. Which of the following magnetization techniques are NOT used for MFL?
    a. AC magnetization
    b. DC magnetization
    c. Permanent-magnet magnetization
    d. Eddy current magnetization

20. What is the major advantage of using a Hall effect sensor over a coil?
    a. Its element is very thin, flexible, and rugged.
    b. It requires no support or protection in industrial NDT applications.
    c. It does not require movement to detect and evaluate a leakage field.
    d. Its element is thick and rugged for continuous movement without breaking.

21. In MFL, the presence of a discontinuity results in:
    a. a local decrease in magnetic flux density.
    b. a local increase in magnetic flux density.
    c. a local increase in permeability.
    d. no variation to permeability and magnetic flux density.

Magnetic Particle Testing

Topical Outline

2.5 Magnetic Particle Testing (MT)
- 2.5.1 Fundamentals
  - 2.5.1.1 Magnetic field principles
  - 2.5.1.2 Magnetization by means of electric current
  - 2.5.1.3 Demagnetization
- 2.5.2 MT
  - 2.5.2.1 Basic types of equipment and inspection materials
  - 2.5.2.2 Test results interpretation, discontinuity indications
  - 2.5.2.3 Applications
    - 2.5.2.3.1 Welds
    - 2.5.2.3.2 Castings
    - 2.5.2.3.3 Wrought metals
    - 2.5.2.3.4 Machined parts
    - 2.5.2.3.5 Field applications

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MT Review Questions

1. Which of the following materials is considered nonferrous?
    a. Copper alloys
    b. High alloy steels
    c. Tool steels
    d. Ferritic stainless steels

2. Materials that are strongly attracted to a magnet are called:
    a. magnetized.
    b. nonmagnetic.
    c. ferromagnetic.
    d. magnetic.

3. Magnetic lines of force (magnetic fields) are oriented in what direction in relation to the direction of the magnetizing current?
    a. Parallel
    b. At right angles
    c. At a 45° angle
    d. At random angles

4. "Magnetizing flux" relates to:
    a. the direction of current flow in an electromagnet.
    b. the manner by which magnetism flows through space.
    c. the lines of force associated with a magnetic field.
    d. permanent magnets only.

5. What does subjecting a part to a magnetic field that is constantly reversing in polarity and gradually diminishing in strength accomplish?
    a. Demagnetizes the part.
    b. Magnetizes the part.
    c. Increases the residual magnetism.
    d. Locates deep-lying discontinuities.

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6. What is circular magnetization used to detect?

   a. Circumferential cracks
   b. Longitudinal cracks
   c. Cracks in cylindrical parts at right angles to the long axis of the part
   d. Deep-lying discontinuities

7. In which magnetizing method is the current passed directly through the part, thereby setting up a magnetic field at right angles to the current flow?

   a. Longitudinal magnetization
   b. Coil magnetization
   c. Central conductor magnetization
   d. Circular magnetization

8. Which of the following is a major disadvantage of using prods?

   a. The magnetic field may be applied in the wrong direction.
   b. The inspection surface may be arc burned.
   c. Magnetic saturation may occur.
   d. Control of the amperage.

9. What is it called when wet magnetic particles are applied to a part while the current is flowing?

   a. Continuous method
   b. Dry method
   c. Residual method
   d. Demagnetization method

10. How is the inside diameter of a cylinder best magnetized?

    a. By using a head shot.
    b. By using prods at either end.
    c. With a central conductor placed between contact heads.
    d. With the cylinder placed crosswise in a solenoid.

11. The amount of amperage used for MT using prods is based on the distance between the prods and the:

    a. thickness of the part.
    b. length of the prods.
    c. diameter of the prods.
    d. total length of the part.

12. When is demagnetization of a part usually necessary?

    a. If the part is made of stainless steel.
    b. If the part is a nonferromagnetic steel.
    c. If the part is high-carbon steel to be welded after inspection.
    d. If the part is to be hardened by heat treatment after inspection.

13. Which of the following is an advantage of the dry technique over the wet technique?

    a. It is more sensitive for detecting fine surface cracks.
    b. It is more capable of providing full surface coverage on irregularly shaped parts.
    c. It is easier to use for field inspection with portable equipment.
    d. It is faster when testing many small parts.

14. When are fluorescent magnetic particles used in preference over visible magnetic particles?

    a. When parts are big and bulky.
    b. When working in the field.
    c. If parts are for railroad applications.
    d. To increase the speed and reliability of detecting very small discontinuities.

15. What causes a leakage field in a steel bar?

    a. Paint on the bar surface
    b. A longitudinal crack
    c. Reversal of the magnetic field
    d. Hysteresis

16. MT techniques are recognized as superior to liquid penetrant testing techniques when:

    a. the surfaces of the test object are corroded.
    b. the surface is anodized.
    c. the parts are painted.
    d. the part is made from austenitic steel.

17. When using direct current, an indication is detected. What is the next logical step to determine if the indication results from a surface or subsurface condition?

    a. Reinspect using the surge method.
    b. Demagnetize and apply powder.
    c. Reinspect at higher amperage.
    d. Reinspect using alternating current.

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18. What should a requirement to use MT on a part include?

    a. A fabrication and service manual.
    b. A statement on the drawing that requires MT.
    c. The procedure to be used and acceptance criteria.
    d. The method of test and service conditions.

19. What is the name of the force that must be applied in the opposite direction to remove any remaining magnetism from a part after magnetization?

    a. Coercive force
    b. Demagnetizing flux
    c. Reverse saturation
    d. Direct force

20. The best type of magnetizing current for the inspection of fatigue cracks is:

    a. direct current.
    b. alternating current.
    c. half-wave rectified alternating current.
    d. full-wave current.

21. A star-shaped indication is seen on the cover pass of a weld. What type of discontinuity is indicated?

    a. Crater crack.
    b. Cooling crack.
    c. Slag inclusion.
    d. Arc burn.

22. For maximum sensitivity in MT of rough welds, what is the recommended initial surface preparation?

    a. Wire brush the weld to remove slag and scale.
    b. Use standard test weldments for comparison.
    c. Coat the weld bead with lacquer.
    d. Grind the weld bead to remove all surface irregularities.

23. Ferromagnetic materials are made up of small, polarized regions known as:

    a. photons.
    b. quarks.
    c. electrons.
    d. domains.

24. The reapplication of magnetic particle suspension after current has already been applied to a part is known as the:

    a. residual method.
    b. continuous method.
    c. magnetization method.
    d. demagnetization method.

25. Subsurface indications will have the appearance of:

    a. well-defined lines.
    b. fuzzy-looking lines.
    c. bright, solid lines.
    d. intermittent lines.

26. What are the locations on a magnetized part called where the magnetic field lines either enter or exit the part?

    a. Defects
    b. Salient points
    c. Magnetic poles
    d. Nodes

27. What is the strength of a magnetic field induced in a part is often called?

    a. Retentivity
    b. Current density
    c. Voltage
    d. Flux density

28. When inspecting steel components that have been exposed to atmospheric conditions for an extended period of time, what is an advantage of MT over PT?

    a. MT does not require a power source.
    b. Heavy rust formations do not have to be removed.
    c. MT will detect oxide-filled cracks.
    d. MT does not require removal of weld slag.

29. When performing wet MT outdoors or in a brightly lit environment, which action will best improve the visibility and contrast of magnetic particle indications?

    a. Use fluorescent particle sprays.
    b. Use a high-intensity light source.
    c. Lightly spray the subject area with a dusting of flat white paint.
    d. Shield the area from sunlight.

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Neutron Radiographic Testing

Topical Outline

2.6 Neutron Radiographic Testing (NR)

- 2.6.1 Fundamentals
  - 2.6.1.1 Sources
    - 2.6.1.1.1 Isotopic
    - 2.6.1.1.2 Neutron
  - 2.6.1.2 Detectors
    - 2.6.1.2.1 Imaging
    - 2.6.1.2.2 Nonimaging
  - 2.6.1.3 Nature of penetrating radiation and interactions with matter
  - 2.6.1.4 Essentials of safety
- 2.6.2 NR
  - 2.6.2.1 Basic imaging considerations
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    - 2.6.2.4.1 Explosives and pyrotechnic devices
    - 2.6.2.4.2 Assembled components
    - 2.6.2.4.3 Bonded components
    - 2.6.2.4.4 Corrosion detection
    - 2.6.2.4.5 Nonmetallic materials

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NR Review Questions

1. The highest intensity sources of thermal neutrons are:

   a. Cf-252 isotopes.
   b. accelerators.
   c. nuclear fission reactors.
   d. cosmic radiation.

2. Neutrons for fast-neutron radiography are obtainable from:

   a. accelerators.
   b. Co-60 or Ir-192.
   c. moderated neutrons from reactors.
   d. X-ray machines.

3. A radioactive source used for NR is:

   a. californium-252 (Cf-252).
   b. plutonium-239 (Pu-239).
   c. cobalt-60 (Co-60).
   d. cesium-137 (Cs-137).

4. The energy of the neutron is expressed in which of the following units of measurement?

   a. Becquerels (Curies)
   b. Coulombs per kilogram (Röntgen)
   c. Sieverts (rem)
   d. Electronvolts

5. What is the most important desirable feature of a thermal neutron beam for NR?

   a. background gamma radiation intensity.
   b. relatively high thermal neutron intensity.
   c. low angular divergence.
   d. relatively high thermal neutron intensity.

6. A material that slows down neutrons is called:

   a. a moderator.
   b. an accumulator.
   c. a limitor.
   d. a collimator.

7. The primary radiation mechanism for darkening radiographic film when the direct NR process is used with gadolinium screens is:

   a. alpha particles.
   b. electrons.
   c. gamma rays.
   d. light emission.

8. NR using the transfer method requires that the imaging screen must:

   a. be placed behind the film.
   b. be placed in front of the film.
   c. be very thin.
   d. become radioactive.

9. Which of the following NR converter foils CANNOT be used for transfer or indirect radiography?

   a. Dysprosium
   b. Indium
   c. Gadolinium
   d. Gold

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10. The most suitable films for producing neutron radiographs are:

    a. industrial X-ray films.
    b. red-sensitive films.
    c. instant-type films.
    d. emulsions that contain no silver halides.

11. Materials that are exposed to thermal neutron beams:

    a. must not be handled for at least 3 min after exposure has ceased.
    b. must be stored in a lead-lined room.
    c. should be monitored by means of a neutron counter.
    d. may be radioactive after exposure to neutrons has ceased.

12. Lead is:

    a. a good neutron shield.
    b. corroded by neutron exposures.
    c. a relatively poor neutron absorber.
    d. an efficient conversion screen.

13. If 2 mm (0.08 in.) of plastic attenuates a thermal neutron beam by a factor of 2, then 20 mm (0.8 in.) will attenuate it by approximately a factor of:

    a. 10
    b. 20
    c. 200
    d. 1000

14. Commonly used materials for moderating fast-neutron sources include:

    a. aluminum, magnesium, and tin.
    b. water, plastic, paraffin, and graphite.
    c. neon, argon, and xenon.
    d. tungsten, cesium, antimony, and columbium.

15. The main reason for using NR in place of X-radiography is:

    a. a lower cost.
    b. higher resolution.
    c. the ability to image objects and materials not possible with X-rays.
    d. a simpler radiographic procedure when required than X-radiography.

16. A photographic record produced by the passage of neutrons through a specimen onto a film is called:

    a. a fluoroscopic image.
    b. an isotopic reproduction.
    c. a radiograph.
    d. a track-etch photograph.

17. Many of the absorption differences between neutrons and X-rays indicate that the two techniques:

    a. cause radiation problems.
    b. complement each other.
    c. can be used interchangeably.
    d. are equally effective for imaging hydrogenous materials.

18. The penetrating ability of a thermal neutron beam is governed by:

    a. attenuating characteristics of the material being penetrated.
    b. exposure time.
    c. source-to-film distance.
    d. thickness of the converter screen.

19. The transfer exposure method is used because:

    a. it is not influenced by gamma radiation in the primary beam.
    b. it produces greater radiographic sensitivity than direct exposure using gadolinium.
    c. it is faster than the direct exposure method.
    d. the screens used in this method emit only internal conversion electrons of about 70 keV.

20. Higher resolution can be achieved in direct NR by:

    a. placing a lead intensifying screen between a gadolinium screen and the film.
    b. increasing the length-to-diameter ratio of the collimation system.
    c. increasing the exposure time.
    d. increasing the distance between the object and the film cassette.

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21. The primary advantage of using a Cf-252 source for NR is its:

a. portability.
b. low cost per unit neutron flux compared to other neutron radiographic sources.
c. high resolution.
d. long useful life.

22. The quality of results from a neutron radiographic exposure is best determined by:

a. reference standards.
b. image quality indicators.
c. neutron flux measurement.
d. densitometer readings.

23. Apparent discontinuities in the imaging screen (rather than in the test part) can be ruled out by:

a. comparing a neutron radiograph of the parts to a blank neutron radiograph of the same imaging screen with no parts in place.
b. producing a photographic copy of the original neutron radiograph using X-ray duplicating film.
c. increasing the exposure time of the radiograph.
d. decreasing the temperature of the developer solution.

24. For inspection of radioactive objects or those that emit gamma radiation when bombarded with neutrons, a preferable detection technique is the:

a. direct exposure technique.
b. transfer technique.
c. isotopic reproduction technique.
d. electrostatic-belt generator technique.

25. NR is an excellent tool for determining:

a. the coating thickness of aluminum oxide on anodized aluminum.
b. the size of voids in thick steel castings.
c. the integrity of thin plastic material within a steel housing.
d. tungsten inclusions in gas tungsten arc welding welds.

26. NR extends radiographic capability for detecting cracks in small cylinders of:

a. aluminum.
b. iron.
c. magnesium.
d. plutonium.

27. Which of the following is a preferred application of NR?

a. Detecting tungsten inclusions in titanium ingots.
b. Detecting the presence of water in the cells of stainless steel honeycomb.
c. Detecting slag inclusions in structural steel welds.
d. Determining the depth of seating of a projectile in a metallic case.

28. Common sources of neutrons for NR are:

a. electron linear accelerators.
b. isotopes of cobalt (e.g., Co-60).
c. nuclear reactors.
d. betatrons.

29. When inspecting explosive or pyrotechnic devices using NR, what is one principal advantage?

a. Clear imaging of metallic components without needing a high-energy X-ray beam.
b. The ability to visualize nonmetallic explosive fill or internal pyrotechnic material behind metallic casings.
c. A reduced need for radiation shielding due to the fast-neutron flux.
d. The simplicity of performing this technique with minimal safety precautions.

30. Radiographic detection of small amounts of explosive in thick steel will give the best contrast if what kind of neutrons are used?

a. Cold
b. Thermal
c. Epithermal
d. Fast

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Radiographic Testing

Topical Outline

2.7 Radiographic Testing (RT)
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  - 2.7.1.2 Detectors
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  - 2.7.1.4 Essentials of safety
- 2.7.2 RT
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  - 2.7.2.3 Systems factors (source/test object/detector interactions)
  - 2.7.2.4 Applications
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    - 2.7.2.4.5 Field inspections

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*Available only as part of five-volume set.

RT Review Questions

1. The penetrating ability of an X-ray beam is governed by:

   a. kilovoltage or wavelength (energy).
   b. time.
   c. milliamperage.
   d. source-to-film distance.

2. If two X-ray machines are set to the same kilovoltage and milliamperage, which of the following is true about the radiation they produce?

   a. They will always produce the same intensity and energy of radiation.
   b. They will produce the same intensity, but the energy of the radiation may differ.
   c. They will produce the same energy, but the intensity of the radiation may differ.
   d. They may produce different intensities and different energies of radiation.

3. Short wavelength electromagnetic radiation produced during the disintegration of nuclei of radioactive substances is called:

   a. X-radiation.
   b. gamma radiation.
   c. scatter radiation.
   d. beta radiation.

4. Almost all gamma radiography is performed with:

   a. natural isotopes.
   b. artificially produced isotopes.
   c. radium.
   d. Co-60.

5. The energy of gamma rays is expressed by which of the following units of measurement?

   a. Gigabecquerel (Curie)
   b. Coulomb per kilogram (Röntgen)
   c. Half-life
   d. Kiloelectronvolts (keV) or megaelectronvolts (MeV)

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6. Of the following, the source providing the most penetrating radiation is:

   a. Co-60.
   b. 220 kVp X-ray tube.
   c. 15 MeV betatron.
   d. electrons from Ir-192.

7. The difference between the densities of two areas of a radiograph is called:

   a. radiographic contrast.
   b. subject contrast.
   c. film contrast.
   d. definition.

8. The fact that gases, when bombarded by radiation, ionize and become electrical conductors makes them useful in:

   a. X-ray transformers.
   b. fluoroscopes.
   c. masks.
   d. radiation-detection equipment.

9. Why must the exposure time be increased by a factor of four when the source-to-film distance is doubled in radiography?

   a. Because the intensity of radiation decreases at an exponential as the distance increases.
   b. Because the energy of radiation is inversely proportional to the square root of the distance from the source to the film.
   c. Because the intensity of radiation is inversely proportional to the square of the distance from the source to the film.
   d. Because the effect of scattered radiation increases as the source-to-film distance increases.

10. The most important factor in X-ray absorption of a specimen is:

   a. the thickness of the specimen.
   b. the density of the specimen.
   c. Young's modulus of the material.
   d. the atomic number of the material.

11. The maximum permissible dose per calendar year is 0.05 Sv (5 rem) for:

   a. extremities.
   b. skin.
   c. whole body (total effective dose equivalent).
   d. a fetus from occupational exposure of a declared pregnant woman.

12. Exposure to small doses of X-rays or gamma rays:

   a. has a cumulative effect that must be considered when monitoring for maximum permissible dose.
   b. is beneficial because it serves to build an immunity in humans to radiation poisoning.
   c. will have no effect on human beings.
   d. will have only a short-term effect on human tissues.

13. Which of the following technique variables is most commonly used to adjust subject contrast?

   a. Source-to-film distance
   b. Milliamperage
   c. Kilovoltage
   d. Focal spot size

14. What is the basic difference between a radiograph and a fluoroscopic image?

   a. The fluoroscopic image is more sensitive.
   b. The fluoroscopic image is positive whereas the radiographic image is negative.
   c. The fluoroscopic image is brighter.
   d. There is no basic difference between the two.

15. Thin sheets of lead foil in intimate contact with X-ray film during exposure increase film density because they:

   a. fluoresce and emit visible light, which helps expose the film.
   b. absorb the scattered radiation.
   c. prevent backscattered radiation from fogging the film.
   d. emit electrons when exposed to X- and gamma radiation, which helps to darken the film.

16. When viewing a radiograph, an image of the back of the cassette superimposed on the image of the specimen is noted. What is this most likely due to?

   a. Undercut
   b. Overexposure
   c. X-ray intensity being too high
   d. Backscatter radiation

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17. What is an image quality indicator (IQI) used to measure?

 a. The size of discontinuities in a part
 b. The density of the film
 c. The quality of the radiographic technique
 d. The amount of radiation that penetrates the test object

18. In film radiography, where are IQIs typically placed to ensure accurate assessment of image quality?

 a. Between the intensifying screen and the film
 b. On the source side of the test object
 c. On the film side of the test object
 d. Between the operator and the radiation source

19. At voltages above 400 kV, the use of lead to provide protection may present serious structural problems. If this should be the case, which of the following materials would most likely be used as a substitute?

 a. Concrete
 b. Aluminum
 c. Steel
 d. Boron

20. A distinctive characteristic of megavolt radiography is that it:

 a. results in comparatively high subject contrast.
 b. results in comparatively high radiographic contrast.
 c. is applicable to comparatively thick or highly absorbing specimens.
 d. is used for stainless steels only.

21. Given the radiographic equivalency factors of 1.4 for Inconel™ and 1.0 for 304 stainless steel, what is the approximate equivalent thickness of Inconel™ to produce the same exposure as a 3.8 mm (0.15 in.) thickness of 304 stainless steel?

 a. 3 mm (0.12 in.)
 b. 9 mm (0.35 in.)
 c. 18 mm (0.7 in.)
 d. 36 mm (1.4 in.)

22. The fact that each solid crystalline substance produces its own characteristic X-ray pattern is the basis for:

 a. xeroradiography.
 b. X-ray diffraction testing.
 c. fluoroscopic testing.
 d. polymorphic testing.

23. When inspecting a light metal casting by fluoroscopy, which of the following discontinuities would most likely be detected?

 a. Copper shrinkage
 b. Microshrinkage
 c. Shrinkage
 d. Fine cracks

24. For testing a 25 mm (1 in.) steel plate 305 mm (12 in.) square for laminar discontinuities, which of the following would be most effective?

 a. 3.7 TBq (100 Ci) of Ir-192
 b. 925 GBq (25 Ci) of Co-60
 c. 250 kVp X-ray machine
 d. An ultrasonic device

25. A critical weld was made with a double vee-groove. Among those listed, which radiographic technique would provide coverage with the greatest probability for detecting the most serious discontinuities?

 a. A single exposure centered on the weld and perpendicular to the principal surface of the plate.
 b. Two exposures aligned with the vee-groove, focus ±30° off perpendicular.
 c. Two exposures, perpendicular to the plate, offset by the width of the weld bead.
 d. A single exposure centered on the weld with two films aligned ±30° off perpendicular to the principal surface of the plate.

26. A fuse assembly is radiographed so that measurements can be made on the film to determine a minimum internal clearance dimension. What should be factored into the dimension taken from the film?

 a. Projection magnification.
 b. Film latitude.
 c. Slope of the characteristic curve.
 d. IQI alignment.

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27. Miniature electronic components are to be radiographically inspected to reveal broken copper wire leads of 0.008 in. (0.2 mm) diameter. Which of the following IQIs would be most effective to use in establishing a reliable technique?

   a. A series of steel plaque-type IQIs ranging in thickness from 0.1 mm (0.005 in.) to 0.4 mm (0.015 in.), containing $1T$, $2T$, and $4T$ holes.
   b. A plastic block with the radiographic thickness equivalent of the test objects, containing precision-drilled holes ranging from 0.1 mm (0.005 in.) to 0.4 mm (0.015 in.) diameter.
   c. A plastic block with the radiographic thickness equivalent of the test objects, containing copper wires ranging from 0.1 mm (0.005 in.) to 0.4 mm (0.015 in.) diameter.
   d. A series of copper shims ranging in thickness from 0.1 mm (0.005 in.) to 0.4 mm (0.015 in.), containing $1/2T$ and $1T$ holes.

Thermal/Infrared Testing

Topical Outline

2.8 Thermal/Infrared Testing (IR)
- 2.8.1 Fundamentals
  - 2.8.1.1 Principles and theory of IR
  - 2.8.1.2 Temperature measurement principles
  - 2.8.1.3 Proper selection of IR technique
- 2.8.2 Equipment/materials
  - 2.8.2.1 Temperature measurement equipment
  - 2.8.2.2 Heat flux indicators
  - 2.8.2.3 Noncontact devices
  - 2.8.2.4 Contact temperature indicators
  - 2.8.2.5 Noncontact pyrometers
  - 2.8.2.6 Line scanners
  - 2.8.2.7 Thermal imaging
- 2.8.3 Applications
  - 2.8.3.1 Exothermic or endothermic investigations
  - 2.8.3.2 Friction investigations
  - 2.8.3.3 Fluid flow investigations
  - 2.8.3.4 Thermal resistance investigations
  - 2.8.3.5 Thermal capacitance investigations
- 2.8.4 Interpretation and evaluation

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Thermal/Infrared Testing Review Questions

1. Thermal resistance is:

   a. analogous to electrical current.
   b. a material's impedance to heat flow.
   c. proportional to the fourth power of emissivity.
   d. proportional to the rate of heat flow.

2. Conductive heat transfer can take place:

   a. across a vacuum.
   b. from a hair dryer blowing on an object.
   c. between a heat lamp and a distant object.
   d. between dissimilar metals in contact with each other.

3. The infrared/thermal energy emitted from a target surface:

   a. occurs only in a vacuum.
   b. is inversely proportional to surface emissivity.
   c. is proportional to the fourth power of the absolute surface temperature.
   d. is totally absorbed by water vapor in the air.

4. Thermal radiation reaching the surface of a thermally opaque object can be:

   a. absorbed only.
   b. absorbed and reflected only.
   c. reflected only.
   d. transmitted and absorbed only.

5. Which of the following spectral bands is included in the infrared spectrum?

   a. 0.1 to 5.5 µm
   b. 0.3 to 10.6 µm
   c. 0.4 to 20 µm
   d. 0.75 to 100 µm

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6. As a surface cools, the peak of its radiated infrared energy:
   a. shifts to longer wavelengths.
   b. shifts to shorter wavelengths.
   c. remains constant if emissivity remains constant.
   d. remains constant even if emissivity varies.

7. A graybody surface with an emissivity of 0.04 would be:
   a. transparent to infrared radiation.
   b. a fairly good emitter.
   c. almost a perfect reflector.
   d. almost a perfect emitter.

8. If a surface has an emissivity of 0.35 and a reflectivity of 0.45, its transmissivity would be:
   a. impossible to determine without additional information.
   b. 0.80.
   c. 0.10.
   d. 0.20.

9. What is the spectral band in which glass transmits infrared radiation most efficiently?
   a. 3 to 6 µm region
   b. 2 to 3 µm region
   c. 6 to 9 µm region
   d. 9 to 11 µm region

10. Infrared thermal detectors:
    a. have a broad, flat spectral response.
    b. have much faster response times than photon detectors.
    c. usually require cooling to operate properly.
    d. have much greater sensitivity than photon detectors.

11. A diffuse reflecting surface is:
    a. a polished surface that reflects incoming energy at a complementary angle.
    b. a surface that scatters reflected energy in many directions.
    c. also called a specular reflecting surface.
    d. highly transparent to infrared radiation.

12. The minimum resolvable temperature difference is a subjective measurement that depends on:
    a. the infrared imaging system's spatial resolution only.
    b. the infrared imaging system's measurement resolution only.
    c. the infrared imaging system's thermal sensitivity and spatial resolution.
    d. the infrared imaging system's minimum spot size.

13. What is the spatial resolution of an instrument related to?
    a. Thermal resolution
    b. Spectral bandwidth
    c. System responsivity
    d. Instantaneous field of view and the working distance

14. The noise equivalent temperature difference (NETD) of a thermal infrared imager tends to:
    a. improve as the target temperature increases.
    b. degrade as the target temperature increases.
    c. remain constant regardless of the target temperature.
    d. improve with increasing working distance.

15. What is the 3 to 5 µm spectral region well suited for?
    a. Inspection using a microbolometer.
    b. Measuring targets at extremely long working distances.
    c. Measuring targets warmer than 200 °C (392 °F).
    d. Operating at near ambient temperatures.

16. When measuring the temperature of glass while using a midwave (3 to 5 µm) infrared imaging system, which of the following steps is necessary?
    a. Use a 3.2 µm low-pass filter.
    b. Use a 5 µm high-pass filter.
    c. No filter is necessary if using the same emissivity setting used with long-wave imaging systems.
    d. Use a 3.9 µm bandpass filter.

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17. A line scanner is best used for applications:

   a. requiring online real-time process monitoring and control of a linear thermal process.
   b. where the material is stationary.
   c. where the process speed is no greater than 3 m/s.
   d. where the maximum temperature of the material is 300 °C (572 °F).

18. Most infrared focal plane array imagers:

   a. use more costly optics than scanning radiometers.
   b. offer better spatial resolution than scanning radiometers.
   c. offer better thermal resolution than scanning radiometers.
   d. offer less diagnostics features than scanning radiometers.

19. When measuring the temperature of a nongray target:

   a. the viewing angle is not critical.
   b. always assume a uniform emissivity.
   c. varying surface temperature differences can be ignored.
   d. errors may occur when using a variety of instruments.

20. Thermal diffusivity is:

   a. high for metals and low for porous materials.
   b. the same for all metals.
   c. low for metals and high for porous materials.
   d. the same for all porous materials.

21. What is the term used to describe a material's surface temperature response to a given energy input?

   a. Diffuse reflectivity.
   b. Thermal conductance.
   c. Thermal effusivity.
   d. Spectral transmittance.

Ultrasonic Testing

Topical Outline

2.9 Ultrasonic Testing (UT)
- 2.9.1 Fundamentals
  - 2.9.1.1 Wave propagation
    - 2.9.1.1.1 Sound fields
    - 2.9.1.1.2 Wave travel modes
    - 2.9.1.1.3 Refraction, reflection, scattering, and attenuation
  - 2.9.1.2 Transducers and sound beam coupling
- 2.9.2 UT
  - 2.9.2.1 Basic types of equipment
  - 2.9.2.2 Reference standards
  - 2.9.2.3 Test result interpretation, discontinuity indications
  - 2.9.2.4 System factors
  - 2.9.2.5 Applications
    - 2.9.2.5.1 Flaw detection and evaluation
    - 2.9.2.5.2 Thickness measurement
    - 2.9.2.5.3 Bond evaluation
    - 2.9.2.5.4 Process control
    - 2.9.2.5.5 Castings
    - 2.9.2.5.6 Weldments

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UT Review Questions

1. Ultrasonic waves propagate through test materials in the form of:

   a. electromagnetic waves.
   b. low-voltage electric fields.
   c. discontinuous radio waves.
   d. mechanical vibrations.

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2. When an ultrasonic beam passes through the interface of two dissimilar materials at an angle, a new angle of sound travel takes place in the second material due to:

    a. refraction.
    b. attenuation.
    c. rarefaction.
    d. compression.

3. What is the gradual loss of energy called as ultrasonic vibrations travel through a material?

    a. Reflection
    b. Refraction
    c. Compression
    d. Attenuation

4. Ultrasonic velocities are different for different materials. These differences are primarily caused by differences in the materials':

    a. frequency and wavelength.
    b. thickness and travel time.
    c. elasticity and density.
    d. chemistry and permeability.

5. Why is ultrasonic energy for immersion testing transmitted to the test object as a compressional wave?

    a. Compressional waves travel faster and will therefore reduce the distance of the interface signal.
    b. Liquids will only sustain compressional waves.
    c. Compressional waves are used with immersion testing only.
    d. The higher intensity of compressional waves is necessary to overcome high attenuation in liquids.

6. When inspecting coarse-grained materials, which of the following frequencies will generate a sound wave that will be most easily scattered by the grain structure?

    a. 1.0 MHz
    b. 2.25 MHz
    c. 5 MHz
    d. 10 MHz

7. In general, why are shear waves more sensitive to small discontinuities than longitudinal waves for a given frequency and in a given material?

    a. The wavelength of shear waves is shorter than the wavelength of longitudinal waves.
    b. Shear waves are not as easily dispersed in the material.
    c. The direction of particle vibration for shear waves is more sensitive to discontinuities.
    d. The wavelength of shear waves is longer than the wavelength of longitudinal waves.

8. What term describes a transducer's ability to detect echoes from small discontinuities?

    a. Resolution.
    b. Sensitivity.
    c. Definition.
    d. Gain.

9. Which of the following will create a resonance condition in a specimen?

    a. Continuous longitudinal waves.
    b. Pulsed longitudinal waves.
    c. Pulsed shear waves.
    d. Continuous shear waves.

10. The display on most basic pulse-echo ultrasonic instruments consists of:

    a. automatic read-out equipment.
    b. an A-scan presentation.
    c. a B-scan presentation.
    d. a C-scan presentation.

11. In a basic pulse-echo ultrasonic instrument, what is the component that produces the voltage that activates the search unit?

    a. An amplifier
    b. A receiver
    c. A pulser
    d. A synchronizer

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12. What is the primary purpose of reference blocks?

    a. To aid the operator in obtaining maximum back reflections.
    b. To obtain the greatest sensitivity possible from an instrument.
    c. To provide a known reflecting area in calibrating an instrument.
    d. To establish the size and orientation of a discontinuity.

13. The general use of distance-amplitude correction is to compensate for what?

    a. Attenuation, distance, and beam spread.
    b. Amplitude of noise signals.
    c. Velocity changes.
    d. Vertical nonlinearity in the ultrasonic instrument

14. In area-amplitude ultrasonic standard test blocks, the flat-bottom holes in the blocks are:

    a. all the same diameters.
    b. different in diameter, increasing in size of 1/64 in. (0.0156 in.) increments from the No. 1 block to the No. 8 block.
    c. largest in the No. 1 block and smallest in the No. 8 block.
    d. drilled to different depths from the front surface of the test block.

15. Which of the following factors has the least influence on the amount of energy reflected from a discontinuity?

    a. Size of the discontinuity.
    b. Orientation of the discontinuity.
    c. Discontinuity type.
    d. Test frequency.

16. What is the ability to locate discontinuities that are close together within the material called?

    a. Resolution
    b. Sensitivity
    c. Effectiveness
    d. Phase delay

17. Lack of parallelism between the entry surface and the back surface:

    a. may result in a screen pattern that does not contain back reflection indications.
    b. makes it difficult to locate discontinuities that lie parallel to the entry surface.
    c. usually indicates a porous condition existing in the metal.
    d. will decrease the penetrating power of the test.

18. Significant errors in ultrasonic thickness measurement can occur if:

    a. the test velocity is kept constant.
    b. the velocity of propagation deviates substantially from an assumed constant value for a given material.
    c. water is used as a couplant between the transducer and the part being measured.
    d. longitudinal waves are used.

19. How can shear waves be induced in the test material during contact UT?

    a. By placing an X-cut quartz crystal directly on the surface of the material and coupling through a film of oil.
    b. By using two transducers on opposite sides of the test specimen.
    c. By using an angle-beam transducer with the transducer mounted on a plastic wedge so that sound enters the part at an angle.
    d By placing a spherical acoustic lens on the face of the transducer.

20. The most commonly used method of producing shear waves in a test part when inspecting by the immersion technique is by:

    a. transmitting longitudinal waves into a part in a direction perpendicular to its front surface.
    b. using two crystals vibrating at different frequencies.
    c. using a low-frequency transducer.
    d. angulating the transducer to the proper angle with respect to the entry surface of the test part.

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21. In immersion testing, proof that the search unit is normal (perpendicular) to a flat entry surface is indicated by what?

    a. Maximum reflection amplitude from the entry surface.
    b. Elimination of water multiples.
    c. Maximum reflection amplitude from the back surface.
    d. Maximum amplitude of the initial pulse.

22. In immersion testing, the water distance between the search unit and the test piece:

    a. should be as small as possible.
    b. will have no effect on the test.
    c. should be the same as the water distance used during calibration.
    d. should be as great as possible.

23. Generally, the best UT technique for detecting discontinuities oriented along the fusion zone in a welded plate is:

    a. an angle-beam contact method employing surface waves.
    b. an immersion test using surface waves.
    c. a resonance technique.
    d. an angle-beam method using shear waves.

24. When inspecting thin sheet for laminar discontinuities with the ultrasonic wave directed perpendicular to the surface, what should be observed?

    a. The amplitude of the front surface reflection.
    b. The multiple reflection pattern.
    c. The amplitude of the initial pulse.
    d. Signals that "walk" or move along the time base as the transducer is scanned over the sheet.

25. Ultrasonic inspection of castings is occasionally impractical because of:

    a. extremely small grain structure typical in castings.
    b. coarse grain structure.
    c. uniform flow lines.
    d. random orientation of discontinuities.

26. Angle-beam testing of plate will often miss:

    a. cracks that are perpendicular to the sound wave.
    b. inclusions that are randomly oriented.
    c. laminations that are parallel to the front surface.
    d. a series of small discontinuities.

27. In addition to a weld area being inspected by the angle-beam technique, which additional test is typically performed to detect laminations in the base metal?

    a. Through-transmission testing
    b. Guided wave testing
    c. Straight-beam testing
    d. Surface-wave testing

28. An ultrasonic test using a straight-beam contact search unit is being conducted through the thickness of a flat part, such as plate. What should this test detect?

    a. Laminar-type discontinuities with major dimensions parallel to the plane of the rolled surface.
    b. Transverse-type discontinuities with major dimensions at right angles to the plane of the rolled surface.
    c. Radial discontinuities with major dimensions along the length but radially oriented to the rolled surface.
    d. Rounded discontinuities at the edges of the rolled plate.

29. UT techniques are useful in testing laminate and sandwich construction test objects for:

    a. paint thickness.
    b. bond integrity.
    c. leakage.
    d. surface roughness.

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30. UT techniques are frequently used in online automatic process control applications to measure and control:

  a. moisture content in materials.
  b. surface roughness of turbine blade castings.
  c. the thickness of cold-rolled strips, sheets, and plates.
  d. chemical activity in chemical etching processes.

31. Which of the following statements about field inspection applications of UT is true?

  a. Manual and automatic systems can be used for field inspections.
  b. Because the equipment is large and bulky, field inspections are difficult, at best.
  c. Aircraft and other field maintenance inspections usually require three persons: one to manipulate the transducer, one to monitor the instrument, and one to record results.
  d. Digital displays must be used for outdoor inspection because of the limited brightness of screen displays.

32. Which of the following waves is able to follow a surface around a curve?

  a. Longitudinal wave
  b. Shear wave
  c. Surface wave
  d. Lamb wave

33. The metallurgical nature of wrought iron plate or pipe is that it contains many large, flat inclusions oriented parallel to the surface. When performing thickness testing of a wrought iron pipe, which of the following may occur?

  a. The sound wave will be scattered by the inclusions.
  b. The sound wave will be reflected by large inclusions resulting in a false thickness reading.
  c. The sound wave will pass through the inclusions and an accurate thickness measurement will be attained.
  d. The sound wave will be absorbed by the inclusions.

Visual Testing

Topical Outline

2.10 Visual Testing (VT)

- 2.10.1 Fundamentals
  - 2.10.1.1 Principles and theory of VT
  - 2.10.1.2 Selection of correct visual technique
  - 2.10.1.3 Equipment and materials
- 2.10.2 Specific applications
  - 2.10.2.1 Metal joining processes
  - 2.10.2.2 Pressure vessels
  - 2.10.2.3 Pumps
  - 2.10.2.4 Valves
  - 2.10.2.5 Bolting
  - 2.10.2.6 Castings
  - 2.10.2.7 Forgings
  - 2.10.2.8 Extrusions
  - 2.10.2.9 Microcircuits
- 2.10.3 Interpretation and evaluation
  - 2.10.3.1 Codes and standards
  - 2.10.3.2 Environmental factors

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Visual Testing Review Questions

1. VT could be best described as:

  a. detection of near surface anomalies and various color variations.
  b. optical detection of surface anomalies and checking conformance to specification.
  c. evaluation of metallurgical conditions via electronic microscope.
  d. examination for a wide variety of discontinuities open to or just below the surface.

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2. What element of the eye is analogous to a digital camera, converting a light pattern into electronic signals?

   a. Eye muscle
   b. Iris
   c. Lens
   d. Retina

3. Illumination varies inversely as the square of the distance between the source and the point on the surface increases. What is this law called?

   a. Inverse square law
   b. Cosine law
   c. Generation of light law
   d. Lambert's law

4. When measuring surface roughness, R_a is defined as what?

   a. Average distance between the highest and lowest points.
   b. Average waviness from crest to trough.
   c. Average distance of the profile to the mean line.
   d. Parameter of friction between contact surfaces.

5. What is the minimum luminance recommended by the Illuminating Engineering Society (IES) for task lighting with medium contrast and small size detection desired?

   a. 100 lx
   b. 200 lx
   c. 500 lx
   d. 2000 lx

6. What is the direction of view called in a borescope or videoscope when viewing 45° off the straight-ahead direction of the probe?

   a. Forward slant
   b. Forward oblique
   c. Retrospective
   d. Angulated

7. How many bundles are there in a fiberscope and what are they called?

   a. One; monochromatic bundle.
   b. Two; light guide and image guide.
   c. Two; light guide and CCD cable bundle.
   d. Three; light guide, image guide, and fiber sheath.

8. What is a commonly evident surface discontinuity visible to the unaided eye following forming?

   a. Forging bursts at the center of the billet.
   b. Laminations at end preparations of plate for welding.
   c. Edge breaks in temper-rolled sheets of steel.
   d. Segregation between pours.

9. What type of cracking occurs at the termination point of a weld made by tungsten inert gas welding?

   a. Crater crack
   b. Hot tearing
   c. Cold cracking
   d. Hot cracking

10. What is pillowing corrosion on aircraft lap joints, typically on aircraft skins, usually attributed to?

    a. Misalignment corrosion of fasteners used to attach the skins to the airframe.
    b. Stretching of the skins beyond their yield points over time.
    c. Expansion of corrosion products under the skins.
    d. Twisting of the airframe during tight turns and similar maneuvers.

11. When inspecting welds for discontinuities located by VT, which of the following discontinuities is usually judged the least detrimental, depending on its depth?

    a. Undercut
    b. Cracks
    c. Lack of fusion
    d. Incomplete penetration

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12. How can the heat-affected zone (HAZ) of carbon steel welds be made visible?

    a. Etchants to enhance the visibility of the microstructure.
    b. High-frequency ultrasonic microscopic means.
    c. Color-contrast penetrating liquids.
    d. Arrays of temperature-sensitive markings.

13. Reflectance is a measure of:

    a. the reactance of a coating.
    b. the mirror-like reflectance of a surface.
    c. the amount of artificial light at the inspection surface.
    d. the flatness of a finish coating.

14. A datum:

    a. is a theoretical point identified on the blueprint.
    b. is a theoretical point, axis, or plane, derived from actual part features.
    c. is a feature controlled by two separate datums.
    d. has form only in terms of itself.

15. Color order systems describe color by:

    a. value and brightness saturation.
    b. primary and secondary colors.
    c. value, hue, and saturation.
    d. value of the primary color constituents.

16. How can a concave mirror be helpful during visual inspection?

    a. By enlarging the image viewed.
    b. By diverging and optically reversing small images.
    c. By converging and optically reversing small images.
    d. By increasing the light returned to the inspector.

17. Which weld discontinuity would NOT be visually evident when examining a completed fillet weld?

    a. Weld profile
    b. Crater crack
    c. Incomplete penetration
    d. Undercut

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ANSWERS TO REVIEW QUESTIONS

Acoustic Emission Testing

1b	2d	3d	4c	5a	6c	7a	8b	9d	10a

Electromagnetic Testing

1c	2a	3b	4c	5a	6c	7d	8c	9c	10d	11a	12a	13c	14b
15a	16c	17b	18d	19c	20b	21a	22a	23b	24d	25c	26a	27c	28c

Leak Testing

1d	2b	3d	4c	5b	6c	7a	8c	9d	10a	11c	12d	13b	14c
15b													
16c	17a	18a											

Liquid Penetrant Testing

1b	2d	3a	4d	5d	6b	7b	8a	9d	10a	11c	12b	13c	14c
15b	16b	17a	18a	19c	20d	21c	22b	23c	24a	25b	26c	27b	28a
29b	30b												

Magnetic Flux Leakage

1c	2d	3c	4c	5a	6b	7c	8d	9a	10c	11a	12d	13c	14b
15b	16c	17b	18c	19d	20c	21b							

Magnetic Particle Testing

1a	2c	3b	4c	5a	6b	7d	8b	9a	10c	11a	12d	13c	14d
15b	16c	17d	18c	19a	20b	21a	22a	23d	24a	25b	26c	27d	28c
29d													

Neutron Radiography

1c	2a	3a	4d	5b	6a	7b	8d	9c	10a	11d	12c	13d	14b
15c	16c	17b	18a	19a	20b	21a	22b	23a	24b	25c	26d	27b	28c
29b	30a												

Radiographic Testing

1a	2d	3b	4b	5d	6c	7a	8d	9c	10d	11c	12a	13c	14b
15d	16d	17c	18b	19a	20c	21a	22b	23c	24d	25b	26a	27c	

Thermal/Infrared

1b	2d	3c	4b	5d	6a	7c	8d	9b	10a	11b	12c	13d	14a
15c	16d	17a	18b	19d	20a	21c							

Ultrasonic Testing

1d	2a	3d	4c	5b	6d	7a	8b	9a	10b	11c	12c	13a	14b
15d	16a	17a	18b	19c	20d	21a	22c	23d	24d	25b	26c	27c	28a
29b	30c	31a	32d	33b									

Visual Testing

1b	2d	3a	4c	5c	6b	7b	8c	9a	10c	11a	12a	13b	14b
15c	16a	17c											

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CHAPTER 5

MATERIALS AND PROCESSES

Overview

Commensurate with the need for NDT Level III personnel to have basic knowledge of materials and processes, books have been written specifically for the purpose of presenting the fundamentals of these subjects from an NDT perspective. Clearly, not all NDT Level III personnel need to specialize in depth in more than a few facets of industrial materials and manufacturing processes. However, the relationships between NDT technology and materials and processing are pervasive. NDT personnel charged with the responsibilities of selecting appropriate NDT methods, developing techniques and procedures, and directing the efforts of others in providing meaningful and reliable NDT must have a fundamental and broad knowledge of the origins, nature, behavior, and application of materials and the processes by which they are shaped into products.

The importance of the relationships between NDT and materials and processing technology is characterized in the ASNT Level III Basic examination. Approximately one-third of the Basic examination covers fundamentals of basic materials, fabrication, and product technology. Listed on the following pages are sample questions typical of those that have importance to NDT personnel. Note that most have an underlying basis that ties their content to decision points in NDT.

Some NDT Level III personnel are not placed by their employers in work assignments that require significant knowledge of materials and processing technology. However, progression in NDT technology from Level II to Level III in most industrial and commercial situations does require the individual to respond to more complex technical questions, not only about the details of NDT but also about issues related to causes and effects. Imagine being called upon to select NDT methods, develop techniques and procedures, and instruct others in carrying out an inspection of a critical component without knowing whether the component was a casting, forging, or weldment. Even if a Level III is aware of a component's processing history, without a thorough understanding of the potential issues that may arise during those processes, it is unreasonable to expect them to make informed decisions regarding which NDT methods or techniques should be applied and how they should be implemented.

So fundamental is this relationship that the book *Materials and Processes for NDT Technology* was published in 1981 and revised in 2016. The questions here follow the chapters in this second, updated edition of the book.

As a suggestion in the use of this section of the study guide, try to answer the questions about a particular chapter before proceeding to the next chapter. If the subject matter is unfamiliar, or if the questions are difficult to answer, read the chapter, especially the section(s) referenced by the question. The material in the book is presented specifically with the NDT technologist in mind. It is concise and well-illustrated, and it does not require an engineering or scientific background to be understood. Complex theories and unimportant details are not included.

The Basic Examination Level III Topical Outline for 3.0 Basic Materials, Fabrication, and Product Technology in ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel* (2024) is presented below for your reference:

3.0 Basic Materials, Fabrication, and Product Technology
- 3.1 Fundamentals of material technology
  - 3.1.1 Properties of materials
    - 3.1.1.1 Strength and elastic properties
    - 3.1.1.2 Physical properties
    - 3.1.1.3 Material properties testing
  - 3.1.2 Origin of discontinuities and failure modes
    - 3.1.2.1 Inherent discontinuities
    - 3.1.2.2 Process-induced discontinuities
    - 3.1.2.3 Service-induced discontinuities
    - 3.1.2.4 Failures in metallic materials
    - 3.1.2.5 Failures in nonmetallic materials
  - 3.1.3 Statistical nature of detecting and characterizing discontinuities
- 3.2 Fundamentals of fabrication and product technology
  - 3.2.1 Raw materials processing
  - 3.2.2 Metals processing
    - 3.2.2.1 Primary metals
      - 3.2.2.1.1 Metal ingot production
      - 3.2.2.1.2 Wrought primary metals
    - 3.2.2.2 Castings
      - 3.2.2.2.1 Green sand molded
      - 3.2.2.2.2 Metal molded
      - 3.2.2.2.3 Investment molded
    - 3.2.2.3 Welding
      - 3.2.2.3.1 Common processes
      - 3.2.2.3.2 Hard-surfacing
      - 3.2.2.3.3 Solid-state

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3.2.2.4 Brazing
3.2.2.5 Soldering
3.2.2.6 Machining and material removal
- 3.2.2.6.1 Turning, boring, and drilling
- 3.2.2.6.2 Milling
- 3.2.2.6.3 Grinding
- 3.2.2.6.4 Electrochemical
- 3.2.2.6.5 Chemical

3.2.2.7 Forming
- 3.2.2.7.1 Cold-working processes
- 3.2.2.7.2 Hot-working processes

3.2.2.8 Powdered metal processes
3.2.2.9 Heat treatment
3.2.2.10 Surface finishing and corrosion protection
- 3.2.2.10.1 Shot peening and grit blasting
- 3.2.2.10.2 Painting
- 3.2.2.10.3 Plating
- 3.2.2.10.4 Chemical conversion coatings

3.2.2.11 Adhesive joining

3.2.3 Nonmetals and composite materials processing
- 3.2.3.1 Basic materials processing and process control
- 3.2.3.2 Nonmetals and composites fabrication
- 3.2.3.3 Adhesive joining

3.2.4 Dimensional metrology
- 3.2.4.1 Fundamental units and standards
- 3.2.4.2 Gauging
- 3.2.4.3 Interferometry

The questions that follow are referenced in *Materials and Processes for NDT Technology*, second edition, published in 2016. References are keyed in brackets. The first number refers to the chapter and subsequent numbers to sections and subsections within the chapter. Some chapters are more heavily weighted than others to better reflect the kinds of questions appearing on a Basic Level III examination. Although very few questions derive from Chapter 12 in this unit, this chapter in particular provides a good overview of various NDT methods that would enhance the study of Section II of this study guide.

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REVIEW QUESTIONS

Chapter 1: Manufacturing and Materials

1. Most solid metals and plastics that have reasonable strength at room temperature are called:

   a. composite materials.
   b. manufacturing materials.
   c. allotropic materials.
   d. engineering materials.

[1.1]

2. "Manufacturing" refers to processing that starts with raw material in a bulk form and is concerned mainly with processing the raw material in a manner that changes its:

   a. shape.
   b. chemical form.
   c. mechanical properties.
   d. physical properties.

[1.5]

3. Frequently, dimensions are permitted to vary within specified limits. These variations are called:

   a. variances.
   b. fudge factors.
   c. tolerances.
   d. factors of safety.

[1.6.2]

4. Design engineers are responsible for establishing the function, appearance, quality, and cost of a product. Regarding the role of nondestructive testing (NDT) in product design, which of the following is true?

   a. As a group, designers (by their training and education) are adequately informed about NDT to establish NDT procedures and acceptance criteria.
   b. When NDT appears necessary in a design, the designer should properly select the methods and techniques to be used by reference to NDT handbooks.
   c. Designers should depend solely upon NDT personnel to establish acceptance criteria.
   d. Designers should seek input from NDT personnel to ensure all required inspections can be performed.

[1.6.3]

Chapter 2: Classification, Structure, and Solidification of Materials

1. In general, metals exist as:

   a. amorphous solids.
   b. mixtures and compounds of iron and carbon.
   c. crystalline solids.
   d. face-centered cubic lattices.

[2.4]

2. The terms "body-centered cubic," "face-centered cubic," and "hexagonal close-packed" all refer to the:

   a. different sized grains that can exist at the same time in a metallic structure.
   b. sequence of crystalline growth in a typical mild steel.
   c. lattice structures that make up unit cells in a solid metallic structure.
   d. change in a metallic structure as it undergoes plastic deformation.

[2.4]

3. Which of the following materials is NOT typically used in the as-cast state?

   a. Aluminum
   b. Pure iron
   c. Zinc
   d. Magnesium

[2.5.4]

4. Processes called "austenitizing," "annealing," and "normalizing" are:

   a. approximate equilibrium heat-treatment processes.
   b. performed only on nonferrous metals.
   c. cold-working processes.
   d. age-hardening processes.

[2.6.3]

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5. Annealing is usually performed to:

   a. increase hardness.
   b. increase strength.
   c. remove strain hardening.
   d. increase impact toughness.

[2.6.3]

6. The term "precipitation hardening" is often used interchangeably with the term:

   a. age hardening.
   b. recrystallization.
   c. annealing.
   d. work hardening.

[2.6.4]

7. In a tensile test on a cylindrical specimen, the strain measured on the specimen gauge length is used to calculate:

   a. age hardening.
   b. hardness.
   c. the modulus of elasticity.
   d. instability.

[2.6.5]

8. Which of the following material properties are of most concern if corrosion resistance is essential?

   a. Processing properties
   b. Mechanical properties
   c. Physical properties
   d. Electrochemical properties

[2.7.2]

9. Attacks on metals by direct chemical action and/or electrolysis are called:

   a. corrosion.
   b. erosion.
   c. austenitic transformations.
   d. galvanization.

[2.7.3]

10. Which of the following groups prevent corrosion?

   a. Anodizing, plating, and painting
   b. Insulating, jacketing, and strapping
   c. Annealing, normalizing, and hardening
   d. Ductility, enlarged grains, and increased hardness

[2.7.4]

11. The process of returning ductility to a cold-worked low-carbon steel is called:

   a. precipitation.
   b. recrystallization.
   c. allotropic change.
   d. austenitization.

[2.8.6]

12. The primary goal of alloying for engineering materials is:

   a. to introduce substitutional or interstitial impurities.
   b. to increase the cost of manufacturing.
   c. to create point or line defects in the crystal lattice.
   d. to increase the strength of the nucleus.

[2.8.7]

Chapter 3: Properties of Materials

1. A common industrial example of atomic diffusion is:

   a. carburization of low-carbon steel.
   b. electron beam melting of a nickel-based super alloy.
   c. structural adhesive used in a demanding application.
   d. plasma cleaning operation prior to wire-bonding.

[3.1]

2. Which of the following statements is true regarding the electrical conductivity of aluminum alloys?

   a. Most aluminum alloys are in the range of 70% to 96% IACS.
   b. Clad aluminum takes on the conductivity of the base metal.
   c. Each basic wrought aluminum alloy has a conductivity distinct from any other.
   d. The conductivity of an aluminum alloy is lower than that of pure aluminum.

[3.2.3]

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3. Which of the following requires permanent deformation?

a. Load below the yield strength point
b. Low-frequency dynamic loading
c. Strain hardening
d. Elongation within the elastic range

[3.3]

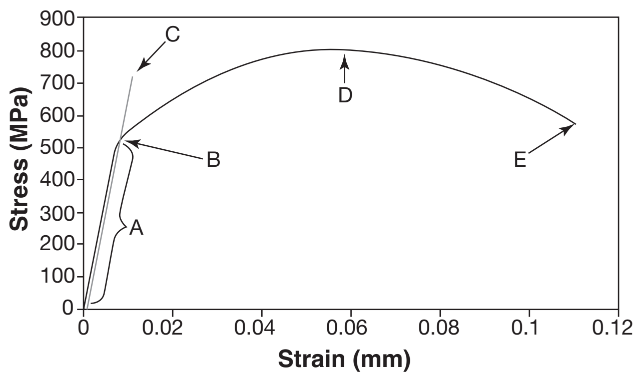

Figure 1

4. In Figure 1, point B is called the:

a. linear-elastic region.
b. yield strength point.
c. ultimate tensile strength.
d. modulus of elasticity.

[3.3.1]

5. Tensile tests are conducted on specimens from a newly developed alloy in order to determine the ultimate tensile strength of the material. Such tests are referred to as:

a. indirect tests.
b. physical properties tests.
c. destructive tests.
d. acoustic emission tests.

[3.3.2]

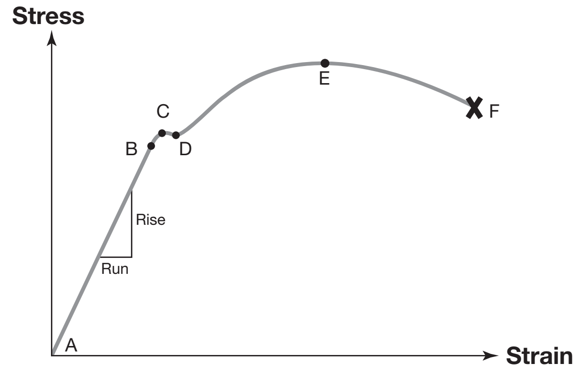

Figure 2

6. In Figure 2, which of the following ranges indicates the effect of work hardening (to its maximum) caused by plastic flow of the material during a tensile load?

a. A–B
b. B–C
c. C–D
d. D–E

[3.3.3]

7. In Figure 2, the points represented by E and F would be closer together if the material being tested were:

a. less ductile.
b. loaded in tension.
c. loaded in lapshear.
d. more ductile.

[3.3.3]

8. The modulus of elasticity, or Young's modulus, is the quotient of strength divided by strain up to the:

a. yield strength.
b. tensile strength.
c. compressive strength.
d. resistance to stress.

[3.3.3]

9. Direct hardness tests provide a measure of a material's ability to resist:

a. bending.
b. permanent deformation or penetration.
c. tensile stresses.
d. elongation.

[3.3.5]

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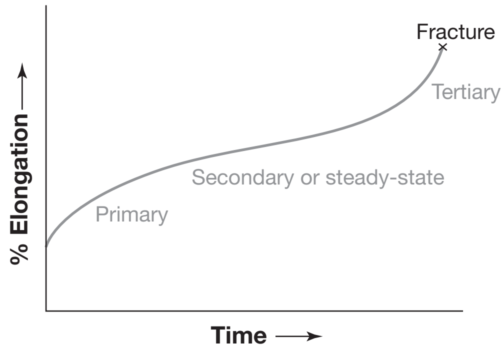

Figure 3

10. What does Figure 3 indicate?
    a. Endurance limit.
    b. Toughness related to strength.
    c. Number of fatigue cycles.
    d. Creep rate at a specific temperature and load.

[3.3.8]

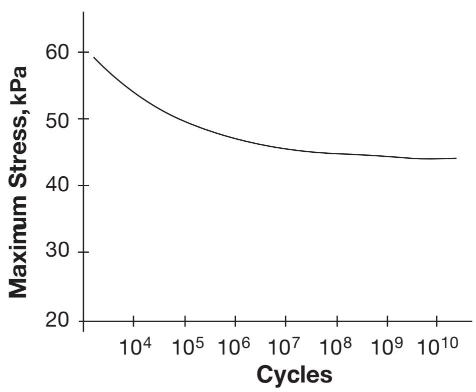

Figure 4

11. Figure 4 typifies:
    a. an S-N plot.
    b. a creep test curve.
    c. a stress-strain diagram.
    d. a true stress-strain diagram.

Chapter 4: Production and Properties of Common Metals

1. The reduction of iron ore, by mixing with coke, limestone, and oxygen for combustion of the coke, is accomplished in:
    a. a blast furnace.
    b. an open-hearth furnace.
    c. a Bessemer converter.
    d. a basic oxygen furnace.

[4.1.2]

2. In the iron- and steelmaking process, “pig iron” refers to:
    a. the waste material that contains high concentrations of impurities and slag and is either discarded or used as a byproduct.
    b. a high-carbon, low-ductility metal produced in the blast furnace that can be used to make subsequent types of iron and steel.
    c. the molten metal from the blast furnace that is not usable and is poured off into a series of crude castings called “pigs.”
    d. low-cost metal used in large production factories.

[4.1.2]

3. An undesirable byproduct of the steelmaking process is:
    a. coke.
    b. low-carbon steel.
    c. low-alloy steel.
    d. slag.

[4.1.2]

4. Which of the following techniques is often used to speed up the steelmaking process?
    a. Adding large amounts of carbon to the molten metal.
    b. Reducing the amount of scrap steel that is often added to the molten metal.
    c. Adding oxygen to the molten metal.
    d. Converting the old open-hearth furnaces into electric furnaces.

[4.1.3]

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5. Typically, the highest quality of steel is produced in:

   a. an electric furnace.
   b. an open-hearth furnace.
   c. a Bessemer furnace.
   d. a basic oxygen furnace.

[4.1.3]

6. By which of the following processes is most of the world's steel produced?

   a. Bessemer converter
   b. Electric furnace
   c. Open hearth
   d. Basic oxygen furnace

[4.1.3]

7. What percentage of carbon is found in steel?

   a. Between 3 and 4%
   b. Between 2 and 3%
   c. Less than 0.2%
   d. Less than 2%

[4.1.6]

8. A steel with 40 points of carbon contains:

   a. 40% carbon.
   b. 4% carbon.
   c. 0.4% carbon.
   d. 0.04% carbon.

[4.1.6]

9. Low-carbon steel contains approximately:

   a. 0.6 to 2.5% carbon.
   b. 0.06 to 0.25% carbon.
   c. 0.5 to 1.6% carbon.
   d. 5 to 16% carbon.

[4.1.6]

10. Which of the following is true relative to the comparison of the properties of aluminum-based alloys and iron-based alloys?

   a. Iron has a lower melting point than aluminum.
   b. Iron can exist in several different crystalline structures, and its properties can be controlled by heat treatment.
   c. Iron can be alloyed to increase its strength, whereas aluminum is strongest in its pure state.
   d. Iron is preferred in load-carrying designs, but it should not be used for any deformation-type of manufacturing process.

[4.1.8]

11. Corrosion-resistant steels having relatively high percentages of nickel and chromium are called:

   a. wrought iron.
   b. low-alloy steels.
   c. stainless steels.
   d. nonferrous steels.

[4.1.8]

12. Austenitic stainless steels are paramagnetic; this means that:

   a. alternating current must be used when using the magnetic particle testing method.
   b. the steel is very dense and, relative to other steels, difficult to penetrate with X-rays.
   c. ultrasonic testing is the logical NDT method to choose because of the coarse-grained nature of paramagnetic material.
   d. the material has a very low magnetic permeability.

[4.1.8]

13. Which of the following is an advantage of cast steel over wrought steels?

   a. Cast steels usually have higher mechanical properties than wrought steels.
   b. Cast steels have more isotropic properties than wrought steels.
   c. Cast steels are more corrosion-resistant than wrought steels.
   d. Cast steels cannot be heat treated and are thus less expensive to produce than wrought steels.

[4.1.10]

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14. Which of the following nonferrous metals is the most important structural material?

 a. Copper alloys
 b. Nickel alloys
 c. Zinc alloys
 d. Aluminum alloys

[4.2.1]

15. Many metals exhibit an increase in strength caused by plastic flow beyond the elastic limit. This effect is called:

 a. twinning.
 b. plastic deformation.
 c. work hardening.
 d. age hardening.

[4.2.2]

16. The heat treatment of aluminum for the purpose of hardening and strengthening:

 a. is not possible with aluminum alloys because they contain no carbon and cannot undergo allotropic changes.
 b. can produce tensile strengths equivalent to some carbon steels.
 c. requires the use of special furnaces and is rarely done as a practical application.
 d. requires that iron and carbon be alloyed for the best results.

[4.2.4]

17. Brass and bronze are alloys of zinc, tin, and a large percentage of:

 a. beryllium.
 b. copper.
 c. lead.
 d. nickel.

[4.3.1]

18. Monel™ and Inconel™ are:

 a. nickel alloys.
 b. steel alloys.
 c. magnesium alloys.
 d. aluminum alloys.

[4.4.3; Table 4.12]

19. Which of the following metals has a density approximately two-thirds that of aluminum?

 a. Magnesium
 b. Iron
 c. Copper
 d. Nickel

[4.7.1]

20. A high-strength, low-density, corrosion-resistant metal alloy of significance in the aircraft, marine, and chemical processing industries is:

 a. tungsten.
 b. zinc.
 c. titanium.
 d. magnesium.

[4.8]

Chapter 5: Polymers, Ceramics, and Composites

1. Which of the following statements is true concerning plastics following their initial polymerization?

 a. Thermoplastics can be endlessly reshaped.
 b. Thermosetting plastics do not soften, but char and deteriorate when reheated.
 c. All plastics are synthetic and contain no natural materials.
 d. Plastics have a complex molecular structure, making it expensive to bind with other materials.

[5.1.1.2]

2. Based on the strength-to-weight ratio:

 a. no plastic materials can compare with metals.
 b. plastics, as a group, are superior in strength to most ferrous metals.
 c. some plastics, including nylon, may have strengths greater than some steels.
 d. plastics, being chemically inert, retain their strength longer than carbon steels in corrosive environments.

[5.1.2]

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3. Which of the following statements is true regarding plastics processing?

   a. Unlike metals, plastics must be processed without the addition of heat.
   b. All plastic-molding processes use liquid-state materials introduced into the mold cavity.
   c. Injection molding can be done only with thermosetting materials.
   d. Both thermoplastics and thermosetting plastics may be processed by molding, casting, and extrusion.

[5.1.3]

4. Reinforced plastic molding involves the use of:

   a. thermosetting plastics and fibrous reinforcement materials.
   b. thermosetting plastics and metallic powder reinforcement.
   c. thermoplastics and wood fiber reinforcement materials.
   d. thermoplastics and metallic powder reinforcement.

[5.1.3.5]

5. The major difference between materials classified as "composites" and those classified as "mixtures" is that:

   a. composites contain metallic constituents and mixtures are nonmetallic.
   b. mixtures start as liquids blended together and composites start as solids.
   c. mixtures are elastomeric, whereas composites are characterized as having at least one plastic component.
   d. mixtures are a type of composite with random orientation and shape of the constituents.

[5.2.1]

6. Which of the following statements is true concerning composite materials?

   a. Composite materials are engineered from one or more reinforcing agents and a matrix to increase strength and reduce weight.
   b. When composite materials are cured, the constituents lose their original identity and form chemical compounds with one another.
   c. A unique feature of composite materials is that their tensile strength frequently exceeds the strength of the strongest constituent.
   d. Composites are usually formed into complex three-dimensional shapes with each dimension approximately equal to the other two.

[5.3]

7. Which of the following materials is typically considered when the application requires high compressive strength?

   a. Glass fibers
   b. Aramid fibers
   c. Carbon fibers
   d. Ceramic fibers

[5.3.3.2]

8. Which of the following statements is true concerning honeycomb?

   a. The function of a honeycomb core is to lighten, stiffen, and strengthen by utilizing the "sandwich principle."
   b. In honeycomb, the walls of the cellular core material are aligned parallel with the plane of the face sheets.
   c. Honeycomb containing nonmetallic elements can be bonded by adhesives, brazing, or diffusion welding.
   d. Honeycomb combining metallic and nonmetallic elements cannot be used in cryogenic service due to the permeability of the nonmetallic elements.

[5.3.5]

9. The mechanism of adhesion may combine mechanical interlocking with:

   a. stickiness of the adhesive.
   b. roughness of the adherends.
   c. dynamic mechanical forces.
   d. cohesion.

[5.3.6.2]

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10. Which of the following tests uses a pendulum to break a specimen that is notched and supported on both ends, with the result of measuring energy absorption?

 a. Creep test
 b. Charpy test
 c. Fatigue test
 d. Transverse rupture test

[5.3.7.2]

Chapter 6: Casting

1. The design of the casting is important because the quality of the finished product can be adversely affected by:

 a. liquid metal pour volume, gate locations, and nonuniform casting sections.
 b. mold temperature, slow postcasting cooling, and magnetization.
 c. riser baffles, pour cup angle, and mold parting lines.
 d. alloy element segregation, change in metal phase, and sprue spacing.

[6.1.1.1]

2. The part of the casting where the gate or riser attaches:

 a. is the area used to establish reference standards for cast materials.
 b. provides the best quality material because of rapid cooling in this area.
 c. may provide a concentration point for discontinuities.
 d. is designed to create nonuniform section thicknesses.

[6.1.1.3]

3. Risers, feeders, or feed heads in castings serve to provide sources of molten metal to compensate for:

 a. misruns.
 b. cold shuts.
 c. dendritic grain growth.
 d. shrinkage.

[6.1.1.4]

4. Which of the following may cause a discontinuity even though its intended purpose is to prevent shrinkage cavities by absorbing heat from the molten metal in the center of the casting?

 a. Riser
 b. Internal chill
 c. Core
 d. Chaplet

[6.1.1.5]

5. In a casting, shrinkage occurs:

 a. only after the transformation from liquid to solid.
 b. only during the transformation from liquid to solid.
 c. before, during, and after the transformation from liquid to solid.
 d. only when the metal is in the liquid state.

[6.1.2.2]

6. Large voids or porosity in a casting result from:

 a. turbulent flow of the molten metal during pouring.
 b. alloy element segregation.
 c. molten metal boiling because of superheat.
 d. gas evolution before and during solidification.

[6.1.2.2]

7. During the solidification of a casting, the shrinkage that occurs:

 a. may cause cavities that are enlarged by the evolution of gases.
 b. may cause porosity and shrinkage cavities primarily in the outer surfaces where the metal cools first.
 c. requires that the pattern used be slightly smaller than the desired dimension of the finished casting.
 d. may be eliminated by investment casting.

[6.1.2.2]

8. A casting process used to produce elongated shapes by drawing solidified metal from a water-cooled mold backed by molten metal is:

 a. centrifugal casting.
 b. continuous casting.
 c. draw casting.
 d. extrusion.

[6.2.1]

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9. Green sand-casting molds include:

   a. sand, clay, and water.
   b. sand, wax, and solvent.
   c. sand, refractory metals, and water.
   d. sand, carbon, and green clay.

[6.2.2.2]

10. Which of the following NDT methods can be commonly used to inspect castings for unfused chaplets and to determine that all the core materials have been removed?

   a. Ultrasonic testing
   b. Magnetic particle testing
   c. Radiographic testing
   d. Electromagnetic testing

[6.2.2.6]

11. Mold material in the form of inserts that exclude metal flow and thus form internal surfaces or passages in a casting are called:

   a. chills.
   b. chaplets.
   c. cores.
   d. patterns.

[6.2.2.6]

12. Small metal supports used to support and position cores become part of a casting by fusing with the molten metal. Such devices are called:

   a. core hangers.
   b. chills.
   c. risers.
   d. chaplets.

[6.2.2.6]

13. Casting molds made by covering a heated metal pattern with sand that is mixed with particles of thermosetting plastic are called:

   a. green sand molds.
   b. shell molds.
   c. die casting molds.
   d. permanent molds.

[6.2.2.10]

14. Another term for "precision casting" and the "lost-wax process" is:

   a. investment casting.
   b. die casting.
   c. metal mold casting.
   d. shell mold casting.

[6.2.4]

15. Permanent molds are most frequently made of:

   a. ceramics.
   b. fused sand and plastic.
   c. metal.
   d. plaster.

[6.2.6]

16. A casting process used to produce hollow products like large pipes and hollow shafts is:

   a. investment casting.
   b. blow casting.
   c. core casting.
   d. centrifugal casting.

[6.2.7]

17. Which of the following metals has low strength and high corrosion resistance, and is used largely in die-casting operations?

   a. Zinc
   b. Aluminum
   c. Magnesium
   d. Manganese

[6.2.8]

18. Which of the following is true regarding solidification of molten metal in a casting mold?

   a. The metal cools at a constant rate, thus providing fine equiaxed grains throughout.
   b. Cooling takes place in phases having different rates that produce different types of grain structure in different sections of the casting.
   c. Solidification occurs at a constant rate, beginning at the interior of the casting and progressing outward.
   d. Thick sections tend to cool more rapidly than thin sections because thin sections consist mostly of fine equiaxed grains.

[6.3.2]

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19. The term used to describe a discontinuity in a casting that occurs when molten metal interfaces with already solidified metal with failure to fuse at the interface is:
   a. hot tear.
   b. cold shut.
   c. inclusion.
   d. segregation.

[6.4.3]

Chapter 7: Metal Forming

1. The advantages of deformation of metals as a manufacturing process are:
   a. consistent duplication, fast production, low cost for high volumes.
   b. low tooling cost, less NDT, doesn't require heat treatment.
   c. easier to recycle, better grain structure, doesn't require special materials.
   d. more labor, only makes simple shapes, holds finishes well.

[7.2]

2. Which of the following would have the least ductility?
   a. Cold-rolled steel plate
   b. Hot-rolled steel plate
   c. Cast iron
   d. Hot-rolled aluminum plate

[7.2]

3. Forged products invariably exhibit:
   a. high susceptibility to corrosion.
   b. lower strength than their cast counterparts.
   c. directional properties.
   d. poor weldability.

[7.2.1]

4. The manufacturing process used most to form metals into three-dimensional shapes is:
   a. casting.
   b. machining.
   c. welding.
   d. forging.

[7.2.1]

5. Which of the following product forms is generally selected for high strength and controlled property directionality?
   a. Castings
   b. Forgings
   c. Extrusions
   d. Hot-rolled flat stock

[7.2.1]

6. NDT is often used on products intended for secondary operations to:
   a. ensure that further operations are not performed on material that contains discontinuities that could cause rejection of the manufactured part.
   b. determine that discontinuities do not exist in the material that could damage the rolling mills and other equipment.
   c. determine the ductility of the material after the rolling operation is complete.
   d. accurately determine the compressive strength of the material after it passes through the rolling mill.

[7.2.1.1]

7. An NDT method best suited to locating discontinuities caused by inclusions rolled into steel plate is:
   a. radiographic testing.
   b. ultrasonic testing.
   c. visual testing.
   d. magnetic particle testing.

[7.2.1.1]

8. Slabs, blooms, and billets are:
   a. the shapes that an ingot is rolled into prior to a variety of secondary operations.
   b. the three consecutive stages that the metal goes through during the production of products such as angle iron and channel iron.
   c. types of discontinuities that occur during the hot rolling of steel.
   d. the three different shapes produced during typical cold-rolling operations.

[7.2.2.1]

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9. Before cold-finishing operations can be done on hot-rolled materials, cleaning is often performed by immersing the hot-rolled material in acid baths in a process called:

   a. degreasing.
   b. descaling.
   c. pickling.
   d. anodizing.

[7.2.2.1]

10. During the steelmaking process, a large number of discontinuities such as slag, porosity, and shrinkage cavities exist in the top of the ingot. These discontinuities are:

    a. mostly eliminated in subsequent hot working due to the pressure that "welds" the void shut.
    b. located with NDT at later stages of production.
    c. almost nonexistent with modern steelmaking processes.
    d. removed by cropping up to one-third off the top of the ingot.

[7.2.2.1]

11. Discontinuities with their origin in the original ingot can be reduced in severity by the closing and welding of voids and the breaking up and elongation of inclusions by which of the following processes?

    a. Cold working
    b. Hot rolling
    c. Heat treatment
    d. Welding

[7.2.2.1]

12. Cold-rolling sheet steel usually begins with a material that:

    a. has been completely inspected with an automated radiographic system.
    b. has been previously hot rolled to dimensions close to the size of the finished product.
    c. has less ductility and greater hardness than typical hot-rolled steel.
    d. will have a lower yield and tensile strength after cold working.

[7.2.2.2]

13. Which of the following statements is true concerning deformation processes?

    a. Hot working usually follows cold working.
    b. Hot working must be followed by heat treatment.
    c. Cold working usually follows hot working.
    d. Cold working renders brittle material more ductile.

[7.2.2.2]

14. Machinability and fatigue resistance are improved in most metals that have been:

    a. hot worked.
    b. cold worked.
    c. heat treated.
    d. cast.

[7.2.2.2]

15. The manufacturing process performed principally on flat products and bars that improves hardness, strength, surface finish, and dimensional accuracy is:

    a. cold rolling.
    b. hot rolling.
    c. forging.
    d. sintering.

[7.2.2.2]

16. Most seamless tubing made without welds is processed by:

    a. casting.
    b. piercing.
    c. cold rolling.
    d. brazing.

[7.2.2.2]

17. With flat products such as cold-rolled strip and sheet, ultrasonic and radiation gauges may be used to provide an accurate measurement of:

    a. strain rate.
    b. surface roughness.
    c. thermal properties.
    d. thickness.

[7.2.2.2]

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18. Most steel pipe is produced by forming and:

  a. drawing.
  b. welding.
  c. forging.
  d. pressing.

[7.2.2.3]

19. A process that requires the use of large, powerful equipment that forms ductile material into a wide variety of long-length, uniform, cross-sectional shapes best describes:

  a. forging.
  b. powder metallurgy.
  c. extrusion.
  d. die casting.

[7.2.3]

20. Among other factors, the advantageous effects of recrystallization depend upon the:

  a. rate of heating.
  b. temperature at which deformation takes place.
  c. presence of carbon in excess of 25% for steels.
  d. presence of silicon in excess of 0.1% for steels.

[7.2.4]

21. In drawing and deep drawing, the final shape often can be completed in a series of draws, each successively deeper. What process performed between draws might effectively reduce the number of draws required?

  a. Recrystallization
  b. Pickling
  c. Etching
  d. Hardening heat treatment

[7.2.4]

22. Spinning can be used to form:

  a. spherical deep-drawn shapes.
  b. cemented carbide cutting tools.
  c. rectangular sheet metal tanks.
  d. solid spheres.

[7.2.4]

23. Deep drawing, blanking, punching, and shearing are operations most commonly applied to:

  a. castings.
  b. slabs or blooms.
  c. extracted ore.
  d. sheet metal.

[7.3.2]

24. Most new developments in sheet-metal forming typically use nonconventional energy sources. What is a common feature of these processes?

  a. The use of lasers for controlled heat input.
  b. The use of cryogenics to super-cool the metal prior to forming.
  c. The use of energy sources that release large amounts of energy in a very short time.
  d. The use of large autoclaves that contain both the tooling and the metal being formed.

[7.3.4]

25. Chemical catalysts, filter elements, and bearings are commonly made by:

  a. powder metallurgy.
  b. galvannealing.
  c. vacuum metallization.
  d. hot isostatic pressing.

[7.4]

26. Powder metallurgy provides two unique advantages in metals processing. One is the capability to produce shapes and objects of refractory metals that are extremely difficult or impractical to melt; the other is to:

  a. economically produce metals with extremely low melting temperatures.
  b. produce metal shapes with controlled porosity.
  c. produce metals that can be easily machined by electrochemical processes.
  d. produce metals that are corrosion resistant.

[7.4.1]

27. A major purpose of pressing the metal powders during powder metallurgy processing is to:

  a. squeeze out excess moisture.
  b. further refine the grains.
  c. compact the powders into mechanical and atomic closeness.
  d. decrease the contact area.

[7.4.2]

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28. Powder metallurgy sintering:

a. is performed above the material's melting temperature.
b. allows parts to be hot worked, heat treated, or machined afterward.
c. results in a maximum of 70% theoretical density.
d. leaves parts in a fragile state subject to handling damage.

[7.4.2]

29. In the powder metallurgy process, sintering is:

a. in most cases a fully solid-state process.
b. never a fully solid-state process.
c. principally done at room temperature.
d. always done at elevated temperature and high pressure.

[7.4.3]

Chapter 8: Joining and Fastening

1. An assembly that has been created by joining two or more parts by one or more welds is called a:

a. joint.
b. bonded structure.
c. weld.
d. weldment.

[8.1]

2. A general definition of welding describes the joining of two surfaces:

a. with a filler metal that has a higher melting point than the base metal.
b. with a filler material that is different from the base material.
c. in a permanent union established by atom-to-atom bonds.
d. where both heat and pressure are necessary for permanent bonding.

[8.1]

3. Melting of faying surfaces, proximity of surfaces, and cleanliness are requirements for:

a. soldering.
b. adhesive bonding.
b. fastening.
d. fusion bonding.

[8.1.2.1]

4. Metallurgical effects in a weld, such as grain size variation and shrinkage, are similar to those that occur in:

a. forgings.
b. castings.
c. extrusions.
d. hot-rolled plates.

[8.1.2.1]

5. Pressure welding can be accomplished with pressure alone, but what else is usually added?

a. Heat
b. Filler material
c. Oxides
d. Adhesives

[8.1.2.2]

6. In pressure bonding, heat has the effect of:

a. increasing malleability.
b. age hardening.
c. reducing the grain size.
d. causing a phase change.

[8.1.2.2]

7. Soldering, brazing, and braze welding all:

a. have the same strength characteristics.
b. use a process where only the filler metal is actually melted.
c. are fusion-type weldments.
d. use a process where both the base metal and filler metal are melted.

[8.1.2.3]

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8. Melting of only the filler material, proximity of surfaces, and cleanliness are requirements for:

   a. fusion bonding.
   b. brazing.
   c. diffusion bonding.
   d. friction stir welding.

[8.1.3.2]

9. In the process of diffusion welding, often called “diffusion bonding,” the base metal is joined by:

   a. melting the weld joint area with strip heaters.
   b. using high-temperature adhesives.
   c. putting it under pressure at temperatures below the melting point.
   d. the heat of frictional movement between the surfaces to be joined.

[8.1.4]

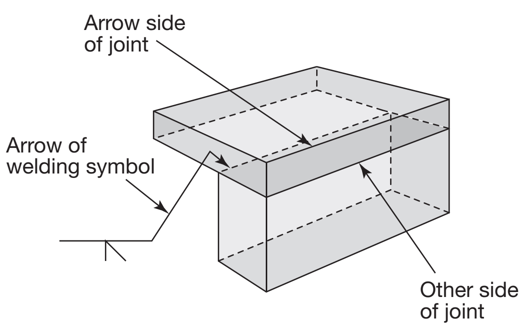

Figure 5

10. What type of weld joint is depicted in Figure 5?

   a. Corner joint
   b. Butt joint
   c. Tee joint
   d. Edge joint

[8.2.1.1]

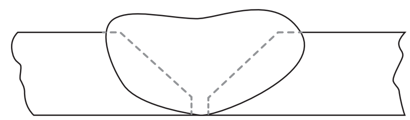

Figure 6

11. What type of weld joint preparation is shown in Figure 6?

   a. J-groove.
   b. Double J-groove.
   c. V-groove.
   d. Square groove.

[8.2.1.1]

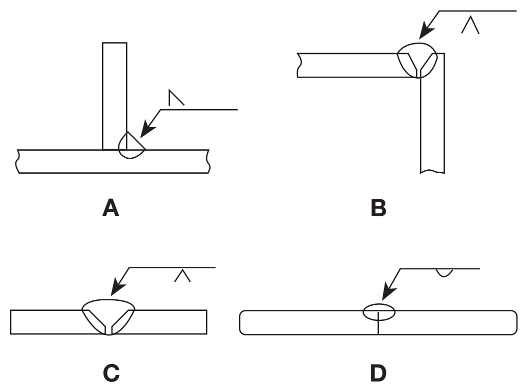

Figure 7

12. In Figure 7, which weld symbol needs to be changed to match the weld condition shown?

   a. A
   b. B
   c. C
   d. D

[8.2.1.1]

13. The uneven shrinkage and brittle structures that occur due to the rapid cooling of a weld can often be reduced by:

   a. preheating the weldment prior to welding.
   b. using a filler metal with a higher carbon content than the base metal.
   c. clamping the weldment in a rigid fixture.
   d. overdesigning the size of the weldment to prevent shrinkage.

[8.2.2]

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14. Welds and weldments have been known to develop cracks long after cooling but prior to being used in service. What is the principal cause for such cracks?

a. Accelerated corrosion at high temperature
b. Scattered porosity in the weld
c. Improper selection of base material
d. Excessive residual stresses

[8.2.2]

15. The principal purpose of preheat treatment and postheat treatment in welds is to:

a. reduce the probability of formation of porosity in the weld.
b. neutralize residual stresses and geometric distortion.
c. create grains in the weld that are the same as those in the base metal.
d. cause the weld ripple and reinforcement to blend into the base metal.

[8.2.2]

16. In welding, the most obvious discontinuities are those associated with structural anomalies in the weld itself. Which conditions are welding process discontinuities?

a. Part fit-up, laminations, incorrect base metal
b. Weld dimension, location, distortion
c. Correct filler metal, corrosion, sequence
d. Insulation, paint, metallic coatings

[8.2.2]

17. Thermal conductivity of a metal is an important factor to consider in making quality weldments because:

a. some metals, such as aluminum, have a low conductivity, which results in weld discontinuities due to localized heat buildup.
b. some metals, such as stainless steel, have a high conductivity, which results in lack of fusion discontinuities as the heat is quickly removed from the weld zone.
c. in some metals, such as aluminum, very high temperature gradients are produced, causing stresses during cooling.
d. some metals, such as stainless steel, have low conductivity, which results in weld discontinuities caused by localized overheating.

[8.2.2]

18. In arc welding, the electric arc is usually sustained between an electrode and the:

a. welding machine.
b. workpiece.
c. coating on the electrode.
d. shielding gas.

[8.2.5]

19. Which of the following gases are most frequently used as shielding to provide an inert atmosphere in the vicinity of the weld?

a. Argon, helium, and carbon dioxide
b. Neon, tritium, and helium
c. Sulphur dioxide, argon, and oxygen
d. Argon, nitrogen, and hydrogen

[8.2.5; 8.2.5.4]

20. The burn-off rate and amount of spattering during the arc welding process can often be controlled by:

a. proper postheating of the entire weldment.
b. frequent changing of the tungsten electrode.
c. maintaining the longest arc length possible to reduce the heat in the weld zone.
d. selecting the proper electrode polarity.

[8.2.5.1]

21. Due to high temperatures and rapid rate of cooling, the filler material used in fusion welds:

a. is coated with an oxide to help reduce weld discontinuities.
b. contains alloys that will help compensate for properties lost during the welding process.
c. is alloyed with nickel, copper, and carbon to eliminate cracking.
d. should be as close as possible to the same alloy content as the base material.

[8.2.6]

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22. When molten metal is transferred from the electrode to the weld zone, which of the following can be used to shield the molten metal from the atmosphere?

 a. Base metal, filler metal, supplemental powder
 b. Tungsten, solid wire, spray powder
 c. Electrode coating, shielding gas, granular flux
 d. Controlled heat input, clean surfaces, qualified weld procedure

[8.2.6]

23. Which of the following welding processes uses a nonconsumable electrode with the arc maintained in an atmosphere of inert gas?

 a. Gas tungsten arc welding
 b. Submerged arc welding
 c. Gas metal arc welding
 d. Electroslag welding

[8.2.6]

24. Shielding in the submerged arc welding process is provided by:

 a. gases.
 b. a flux-coated welding rod.
 c. granular flux that completely surrounds the arc.
 d. chopped glass fibers.

[8.2.6]

25. A welding process that is most frequently carried out in a vacuum chamber is:

 a. plasma arc welding.
 b. electron beam welding.
 c. electroslag welding.
 d. friction welding.

[8.2.7]

26. The welding process capable of very high intensity and rate of heat transfer is:

 a. braze welding.
 b. diffusion welding.
 c. soldering.
 d. plasma arc welding.

[8.2.9]

27. The welding process in which the arc is extinguished after melting a slag cover and in which the base metal and copper slides form a sort of moving mold is called:

 a. submerged arc welding.
 b. electroslag welding.
 c. electron beam welding.
 d. slag mold welding.

[8.2.14]

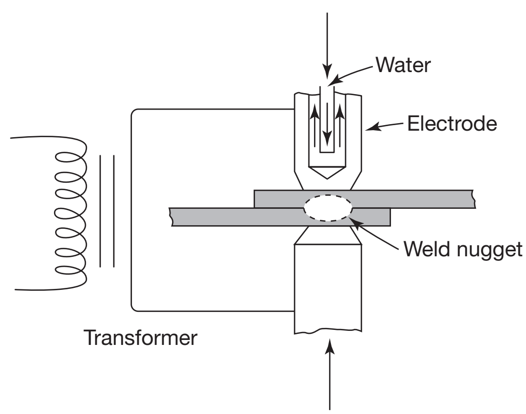

Figure 8

28. Which welding process is depicted in Figure 8?

 a. Electron beam welding
 b. Plasma arc welding
 c. Resistance spot welding
 d. Friction welding

[8.2.15]

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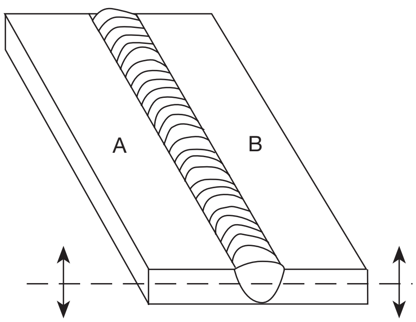

Figure 9

29. If the V-groove weld shown in Figure 9 was made in multiple passes and NOT clamped or restrained, typical warping would take place in which direction?

    a. Edges A and B would be raised due to the contraction of the weld metal.
    b. Edges A and B would be lowered due to the expansion of the weld metal in the weld zone.
    c. In a multipass weld, there would be little if any warpage.
    d. Expansion and contraction would be equal in a vee-groove weld as shown.

[8.3.2]

30. Cracks in the weld metal are primarily of which three types?

    a. Shallow, deep, and intermittent
    b. Longitudinal, transverse, and crater
    c. Laminar, through, and oblique
    d. Longitudinal, laminar, and intermittent

[8.3.3]

31. A slag inclusion can result from which of the following?

    a. Small pieces of tungsten being dislodged from the electrode in the gas tungsten arc process
    b. Excessive overlap on intermediate passes in a multipass weldment
    c. Insufficient cleaning of successive passes in a multipass weldment
    d. Contaminants in the welding flux

[8.3.3]

32. Undercut on a weld pass is usually caused by:

    a. poor operator technique.
    b. a rate of travel that is too slow, which causes the base metal to become too hot.
    c. the use of an electrode that is too large for the current capacity of the welding machine.
    d. welding in the vertical position.

[8.3.3]

33. Crater cracks may take the form of a single crack or star-shaped cracks and will usually be found:

    a. by magnetic particle techniques since crater cracks are always subsurface.
    b. anywhere along a weld where the welding was stopped and restarted.
    c. in the natural crater formed between the two plates in a typical fillet weld.
    d. in the root area of a multiple-pass weld where the weld metal failed to flow completely into the root opening.

[8.3.3]

34. Weldments subject to restraint during welding can develop high residual stresses. Unrestrained weldments can develop:

    a. geometric distortion.
    b. high residual stresses.
    c. cracking after the weld has cooled.
    d. fatigue cracking.

[8.3.4.1]

35. After welding, many steel weldments are heat treated to obtain more uniform properties between the weld and base metal and to relieve stress. Which heat-treatment method is often used following welding?

    a. Tempering
    b. Martensitic aging
    c. Normalizing
    d. Spheroidizing

[8.3.4.1]

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36. When steel has been quench hardened and then reheated to some point below the lower transformation temperature for the purpose of reducing brittleness, this is called:

a. austenitization.
b. thermal slip deformation.
c. allotropic change.
d. tempering.

[8.3.4.2]

Chapter 9: Material Removal Processes

1. In describing machinability, three different measurements are generally considered on a relative, if not quantitative, basis. These are:

a. shear, tensile, and impact strength of the material being machined.
b. surface finish of the material achievable, power consumption required to remove a given volume of material, and expected tool life.
c. softness of the material, sharpness of the cutting tool, and type of machine used to remove the material.
d. volume of material before machining, volume of material after machining, and time required to remove that volume.

[9.1.4]

2. Equipment that aids in material removal from a workpiece by establishing a suitable set of motions and maintaining known positions are:

a. millworking machines.
b. factory machines.
c. machine tools.
d. metal-cutting machines.

[9.2.1]

3. A cutting operation that has the ability to cut through thicknesses of more than 0.9 m (3 ft) of steel and is commonly used to remove surface discontinuities on castings and forgings by "scarfing" is called:

a. oxyacetylene cutting.
b. friction cutting.
c. ultrasonic cutting.
d. plasma arc cutting.

[9.2.2.1]

4. The process used for shaping metals by chemical dissolution only, with selective removal accomplished by masking areas where metal is NOT to be removed, is called:

a. electrical discharge machining.
b. chemical milling.
c. electrochemical machining.
d. electroforming.

[9.4.2]

5. The acronym EDM refers to:

a. engineering design materials.
b. energy discharge machines.
c. electrodynamic machining.
d. electrical discharge machining.

[9.4.3]

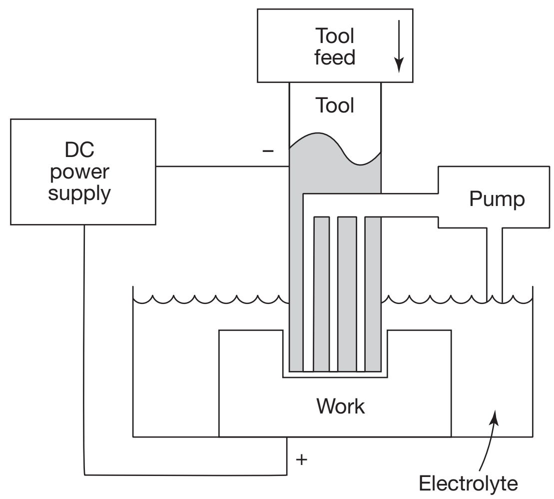

Figure 10

6. Figure 10 illustrates:

a. chemical milling.
b. electrochemical machining.
c. ultrasonic machining.
d. electrolytic grinding.

[9.4.4]

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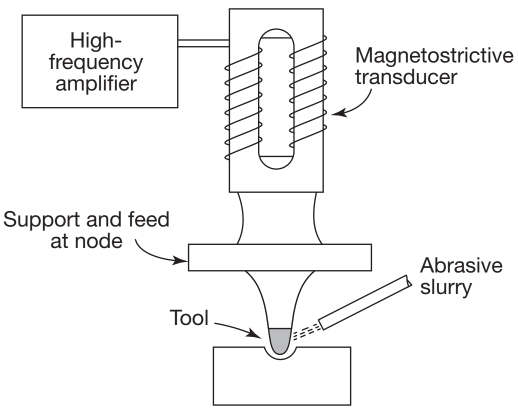

Figure 11

7. The process illustrated in Figure 11 is called:

   a. ultrasonic machining.
   b. electrical discharge machining.
   c. electrochemical machining.
   d. abrasive drilling.

[9.4.5]

8. When used with respect to machine tools, the acronym N/C means:

   a. nonmetal cutters.
   b. noncorrosive coolant.
   c. numerical control.
   d. negative clearance.

[9.5]

9. Which of the following can result in economical setup and reduced machine time with an increase in repeatability and accuracy for a variety of machining operations?

   a. The use of NDT to evaluate the finished product.
   b. The use of electrical discharge machining to replace the conventional lathes and surface grinders.
   c. The use of modern ultrasonic machining operations.
   d. The use of numerical control systems applied to conventional types of machining operations.

[9.5]

Chapter 10: Surface Treatments and Coatings

1. Which of these are properties of surface coatings?

   a. Weld strength, profile, and surface finish
   b. Corrosion protection, appearance, and change surface friction
   c. Surface hardness, discontinuity covering, and fluid tight
   d. Light protection, flux removal, and acoustic emission reduction

[10.1]

2. A pickling bath would be used in the manufacturing of metallic components to:

   a. apply a corrosion-resistant layer.
   b. remove iron-based oxides.
   c. produce an attractive green patina.
   d. grow a wear-resistant layer of chromium or nickel.

[10.1]

3. Carburizing and flame hardening are examples of:

   a. annealing processes.
   b. case-hardening processes.
   c. processes that produce ductile surfaces.
   d. electrochemical processes.

[10.2]

4. Which of the following nondestructive tests would provide the best results in measuring the case depth on a case-hardened part?

   a. Ultrasonic immersion testing using a very low-frequency probe
   b. Radiographic testing
   c. Electromagnetic testing
   d. Magnetic particle testing

[10.2.1]

5. The best and most economical cleaners used for removal of oils and greases are:

   a. pickling baths.
   b. deionized water sprays.
   c. wire brushes and cloth buffers.
   d. petroleum solvents.

[10.3.2]

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6. Coatings are often applied to protect a material; their thicknesses can frequently be determined nondestructively by:

   a. acoustic emission testing.
   b. electromagnetic testing.
   c. surface-wave ultrasonic techniques.
   d. optical holography.

[10.4.2]

7. A process that is the reverse of electrochemical machining and that involves the deposition of metals on other metals or nonmetals is called:

   a. chemical milling.
   b. electrical discharge machining.
   c. electroplating.
   d. magnetoforming.

[10.4.2]

8. Metals commonly applied to other metals by electroplating are:

   a. nickel, chromium, and cadmium.
   b. tin, zinc, and tungsten.
   c. silver, gold, and carbon.
   d. copper, aluminum, and magnesium.

[10.4.2]

9. Which coating process causes paint particles to be directly attracted to a substrate to efficiently form an even coating?

   a. Chemical conversion coating
   b. Electrostatic spraying
   c. Electroplating
   d. Thermal spraying

[10.4.4]

10. Some materials, such as aluminum, are corrosion resistant:

   a. by virtue of the immediate oxidation of newly exposed surfaces.
   b. only if anodized.
   c. because the material itself will not readily combine with oxygen.
   d. against all types of corrosive atmospheres.

[10.4.5]

11. A process that converts the base metal surface to an oxidized barrier layer of very small porous cells is called:

   a. galvanizing.
   b. anodizing.
   c. plating.
   d. metallizing.

[10.4.5]

12. The anodized surface on aluminum:

   a. can produce a high background during a penetrant test.
   b. is very dense and makes X-ray penetration difficult.
   c. can produce cracks that are easily detected by electromagnetic testing techniques.
   d. must be removed before performing ultrasonic tests.

[10.4.5]

13. A corrosion protection material commonly applied to steel by hot dipping and galvanizing is:

   a. porcelain.
   b. paint.
   c. zinc.
   d. chromic acid.

[10.4.6.2]

Chapter 11: Introduction to Nondestructive Testing

1. Which of the following statements best differentiates a "defect" from a "discontinuity"?

   a. Discontinuities can propagate and become defects.
   b. All discontinuities are defects.
   c. All defects will lead to failure if undetected; discontinuities are harmless.
   d. Discontinuities are external natural boundaries only; defects are internal flaws originating from errors in processing.

[11.1]

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2. NDT is often differentiated from other measurement or inspection techniques in that:

   a. NDT is a measurement of dimensions, geometry, and appearance.
   b. NDT uses electronic instruments to identify, evaluate, and locate discontinuities.
   c. NDT involves indirect tests related to some other quality or characteristic of the material.
   d. NDT is an inspection tool used to confirm the findings of the many other quality assurance techniques.

[11.1]

3. Which of the following are a function of NDT?

   a. Testing product to failure point, product application, and cost reduction
   b. Tensile tests, Charpy tests, and Vickers hardness testing
   c. Identifying material, checking for discontinuities, and testing without destroying the product
   d. Checking dimensions, geometry, and appearance

[11.2]

4. A false indication is one formed by a:

   a. discontinuity larger than accept/reject criteria.
   b. discontinuity smaller than accept/reject criteria.
   c. fatigue crack.
   d. factor unrelated to a discontinuity.

[11.2]

5. An important basis for the success of NDT design procedures is:

   a. the need to ensure that unexpected discontinuities of some critical size are not present when the component enters service.
   b. that all discontinuities are detected by NDT or proof testing before the component enters service.
   c. in the use of large factors of safety.
   d. in the use of a value of strength that the material used in the design is presumed to possess.

[11.2]

6. If properly used, NDT can assist in determining whether a test specimen is functioning as designed by:

   a. accurately measuring the tensile strength of design materials.
   b. predicting the time it will take a given size discontinuity to grow to a critical size.
   c. determining the corrosion rate.
   d. providing an accurate evaluation of the number and type of discontinuities that exist in a material.

[11.2]

7. Even at the early stages of product planning, NDT should be considered because:

   a. codes and specifications are too restrictive.
   b. the design of the part should permit easy access to critical areas for later inspection.
   c. the NDT method may strengthen the product.
   d. the NDT processes are generally more time-intensive than other processes.

[11.2]

8. Unexpected early fatigue failure can often be prevented by using NDT to:

   a. verify the cyclic loading on a component.
   b. detect surface discontinuities that could be stress risers.
   c. measure the endurance limit of a part undergoing cyclic stresses.
   d. determine percent elongation of a material before it is placed in service.

[11.2]

9. What is it called when only a certain percentage of manufactured parts are chosen for inspection?

   a. Standard deviation
   b. Natural selection
   c. Analysis of variance
   d. Sampling

[11.2]

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10. Implied in sampling inspection is:
    a. that the chosen plan will produce precise numbers of acceptable parts.
    b. the need for a sample size of 100 units or multiples thereof.
    c. an understanding that defective products may be present and untested.
    d. the need to collect data in the form of variables instead of attributes.

[11.2]

11. A vernier line measurement in a visual acuity test permits verification of:
    a. letter recognition.
    b. trichromatic vision.
    c. code interpretation.
    d. hyperacuity.

[11.3.2]

12. Factors of safety are often in the range of 2 to 4. These factors:
    a. are provided for engineering mistakes.
    b. are added as a corrosion allowance.
    c. could possibly be reduced with the assurance of NDT techniques that the material was free of discontinuities.
    d. provide allowances for poor welding techniques.

[11.3.3]

Chapter 12: Nondestructive Testing Methods

1. In visual testing, which of the following measurement tools uses the principle of light wave interference to check surface flatness?
    a. Optical comparator
    b. Vernier caliper
    c. Sine bar
    d. Optical flat

[12.1.2.1]

2. Devices that show magnified, reflected, or profile images of the workpiece on a frosted glass screen are called optical:
    a. comparators.
    b. flats.
    c. projectors.
    d. micrometers.

[12.1.2.1]

3. An NDT method that has the ability to measure changes in electrical conductivity caused by the effects of heat treatment is:
    a. magnetic particle testing.
    b. acoustic emission testing.
    c. electromagnetic testing.
    d. immersion ultrasonic testing.

[12.8.1]

Chapter 13: NDT Applications

1. Process control based on the means and ranges of measurements taken on periodic samples requires the measurements to be taken of:
    a. attributes.
    b. variables or parameters.
    c. either attributes or variables.
    d. neither attributes nor variables.

[13.1]

2. Monitoring temperature for industrial process control is an example of:
    a. parameter-based measurement.
    b. geometric dimensioning and tolerancing.
    c. the direct comparison technique.
    d. lateral measurement.

[13.1]

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3. Metal corrosion that is accelerated when the metal is under load is called:

a. pitting corrosion.
b. galvanic corrosion.
c. intergranular corrosion.
d. stress corrosion.

[13.4.2]

Chapter 14: NDT and Engineering

1. A statement that a particular experiment produced a 0.9 probability of detection with a 95% confidence level means that:

a. there is a 90% likelihood that the probability of detection is overstated.
b. there is a 95% likelihood that the probability of detection is overstated.
c. on average, 90% of discontinuities will be detected 95% of the time.
d. on average, 95% of discontinuities will be detected 90% of the time.

[14.2]

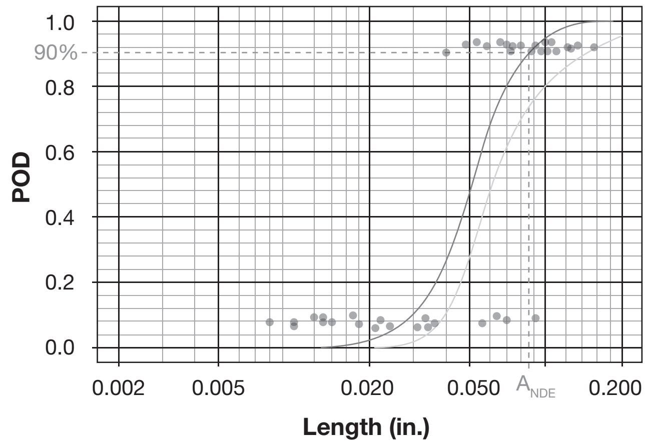

Figure 12

2. On the probability of detection (POD) curve shown in Figure 12:

a. the A_{NDE} line indicates the length of discontinuity that is undetectable in the given test.
b. the Y axis is set for a 90% confidence level that detectable discontinuities will be detected.
c. the Y axis is set for a 90% probability level that detectable discontinuities will be detected.
d. only "hits" are recorded, not "misses."

[14.2]

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ANSWERS TO REVIEW QUESTIONS

Chapter 1: Manufacturing and Materials

1d 2a 3c 4d

Chapter 2: Classification, Structure, and Solidification of Materials

1c 2c 3b 4a 5c 6a 7c 8d 9a 10a 11b 12a

Chapter 3: Properties of Materials

1a 2d 3c 4b 5c 6d 7a 8a 9b 10d 11a

Chapter 4: Production and Properties of Common Metals

1a 2b 3d 4c 5a 6d 7d 8c 9b 10b 11c 12d 13b 14d
15c 16b 17b 18a 19a 20c

Chapter 5: Polymers, Ceramics, and Composites

1b 2c 3d 4a 5d 6a 7c 8a 9d 10b

Chapter 6: Casting

1a 2b 3d 4b 5c 6d 7a 8b 9a 10c 11c 12d 13b 14a
15c 16d 17a 18b 19b

Chapter 7: Metal Forming

1a 2c 3c 4d 5b 6a 7b 8a 9c 10d 11b 12b 13c 14b
15a 16b 17d 18b 19c 20b 21a 22a 23d 24c 25a 26b 27c 28b
29a

Chapter 8: Joining and Fastening

1d 2c 3d 4b 5a 6a 7b 8b 9c 10a 11c 12a 13a 14d
15b 16b 17d 18b 19a 20d 21b 22c 23a 24c 25b 26d 27b 28c
29a 30b 31c 32a 33b 34a 35c 36d

Chapter 9: Material Removal Processes

1b 2c 3a 4b 5d 6b 7a 8c 9d

Chapter 10: Surface Treatments and Coatings

1b 2b 3b 4c 5d 6b 7c 8a 9b 10a 11b 12a 13c

Chapter 11: Introduction to Nondestructive Testing

1a 2c 3c 4d 5a 6d 7b 8b 9d 10c 11d 12c

Chapter 12: Nondestructive Testing Methods

1d 2a 3c

Chapter 13: NDT Applications

1b 2a 3d

Chapter 14: NDT and Engineering

1c 2c

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APPENDIX A

CODE OF ETHICS FOR LEVEL III PERSONNEL CERTIFIED BY ASNT CERTIFICATION SERVICES

The “Code of Ethics for Level III NDT Personnel Certified by ASNT” is part of the ASNT Level III Certification Application for the Basic examination and all method examinations, including the PdM Basic and IR method examinations. By signing the application, the applicant agrees to abide by the Code of Ethics for as long as he or she is certified. The ASNT Level III Certification Application may be downloaded from the ASNT website at asnt.org under the Certification drop-down menu. The text of the Code of Ethics is reprinted below in full:

1.0 Preamble

1.1 In order to safeguard the life, health, property, and welfare of the public, to maintain integrity and high standards of skills and practices in the profession of nondestructive testing, the following rules of professional conduct shall be binding upon every person issued a certificate by ASNT Certification Services as a Level III.

1.1.1 The Level III who holds a certificate is charged with having knowledge of the existence of the reasonable rules and regulations hereinafter provided for his/her conduct as ASNT Level III, and also shall be familiar with their provisions and understand them. Such knowledge shall encompass the understanding that the practice of nondestructive testing under this certification is a privilege, as opposed to a right, and the Level III shall be forthright and candid in statements or written responses to the Ethics Committee of the Certification Management Committee.

1.1.2 “The Level III” as referred to herein is that individual who has been issued a certificate by the ASNT Certification Services pursuant to its heretofore published requirements, rules, and procedures for such certification. This Code of Ethics is binding upon all individuals so certified.

2.0 Integrity

2.1 The Level III is obligated to act with complete integrity in professional matters for each client or employer as a faithful agent or trustee; shall be honest and impartial; and shall serve the public, clients, and employer with devotion.

2.2 The Level III shall make claims regarding certification only with respect to the scope for which certification has been granted.

2.3 The Level III shall not use their certification in a misleading manner or in such a manner as to bring ASNT Certification Services into disrepute. The Level III shall not make any statement regarding the certification that ASNT Certification Services may consider misleading or unauthorized.

3.0 Responsibility of ASNT Certification Services

The Level III shall:

3.1 Immediately report to ASNT Certification Services any perceived violation(s) of this Code of Ethics or any attempt to pressure or force a certified individual to violate this Code of Ethics.

3.2 Not attempt to cheat on ASNT Certification Services examinations; not attempt to bribe or threaten ASNT Certification Services, Pearson Vue, or other third-party testing personnel; not falsify documents; not falsely claim, misrepresent, or permit misrepresentation or misuse of their own or others’ professional qualifications, knowledge, training, experience, work responsibilities, or certifications.

3.3 Inform employer/client in the event that certification is suspended, cancelled, or withdrawn and return to ASNT the Level III certificate and wallet card immediately; and

3.3.1 immediately discontinue the use of the ASNT Certification Services logo. Due to ASNT Certification Services’ trademark copyright, ASNT Certification Services logo is not to be used by any individual or entity without the explicit written consent of ASNT Certification Services.

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4.0 Responsibility to the Public

The Level III shall:

4.1 Protect the safety, health, and welfare of the public in the performance of professional duties. Should the case arise where the Level III faces a situation where the safety, health, and welfare of the public are not protected, he/she shall:

4.1.1 apprise the proper authority if it is evident that the safety, health, and welfare of the public are not being protected; and

4.1.2 refuse to accept responsibility for the design, report, or statement involved; and

4.1.3 if necessary, sever the relationship with the employer or client; and

4.1.4 be completely objective in any professional report, statement, or testimony, avoiding any omission which would, or reasonably could, lead to fallacious inference, finding, or misrepresentation; and

4.1.5 express an opinion as a technical witness before any court, commission, or other tribunal, only when such opinion is founded upon adequate knowledge of the facts in issue, upon a background of technical competence in the subject matter, and upon an honest conviction of the accuracy or propriety of the testimony; and

4.1.6 undertake to perform assignments only when qualified by training and experience in the specific technical fields involved. In the event a question arises as to the competence of a Level III to perform an assignment in a field of specific discipline which cannot be otherwise resolved to the Ethics Committee's satisfaction, the Ethics Committee, either upon request of the Level III, or by its own volition, may require him/her to submit to an appropriate inquiry by or on behalf of the Ethics Committee.

5.0 Public Statements

5.1 The Level III will issue no statements, criticisms, or arguments on nondestructive testing matters connected with public policy which are inspired or paid for by an interested party, or parties, unless he/she has prefaced the remark(s) by explicitly identifying himself/herself, by disclosing the identities of the party on whose behalf he/she is speaking, and by revealing the existence of any pecuniary interest he/she may have in these matters.

5.2 The Level III will publicly express no opinion on a nondestructive testing matter unless it is founded upon adequate knowledge of the facts in issue, upon a background of technical competence in the subject matter, and upon honest conviction of the accuracy and propriety of the testimony.

5.3 The Level III shall show professional and appropriate behavior, including, but not limited to, online and social media. The term "social media" is used within this Code to describe dynamic and socially interactive networked information and communication technologies by which personal information or opinions can be presented for public consumption via the internet.

6.0 Conflict of Interest

6.1 The Level III shall conscientiously avoid conflict of interest(s) with the employer or client, but when avoidable, shall disclose the circumstances to the employer or client.

6.2 The Level III shall promptly inform the client or employer of any business associations, interests, or circumstances which could influence his/her judgment or the quality of services to the client or employer.

6.3 The Level III shall not accept compensation, financial or otherwise, from more than one party for services on the same project, or for services pertaining to the same project, unless the circumstances are fully disclosed to, and agreed to, by all interested parties or their duly authorized agents.

6.4 The Level III shall not solicit or accept financial or other valuable consideration from material or equipment suppliers for specifying their products.

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6.5 The Level III shall not solicit or accept gratuities, directly or indirectly, from contractors, their agents, or other parties dealing with the client or employer in connection with work for which he/she is responsible.

6.6 As an elected, retained, or employed public official, the Level III (in their capacity as a public official) shall not review or approve work that was performed by himself/herself, or under his/her direction, on behalf of another employer or client.

7.0 Solicitation of Employment

7.1 The Level III shall not pay, solicit, nor offer, directly or indirectly, any bribe or commission for professional employment with the exception of payment of the usual commission for securing salaried positions through licensed employment agencies.

7.2 Level III shall seek professional employment on the basis of qualification and competence for proper accomplishment of the work.

7.3 The Level III shall not falsify or permit misrepresentation of his/her, or his/her associates', academic or professional qualification. He/she shall not misrepresent or exaggerate the degree of responsibility in or for the subject matter of prior assignments.

7.4 Brochures or other presentations incident to the solicitation of employment shall not misrepresent pertinent facts concerning employers, employees, associates, joint ventures, or past accomplishments with the intent and purpose of enhancing qualifications and work.

8.0 Improper Conduct

8.1 The Level III shall not sign documents for work for which he/she does not have personal professional knowledge and direct technical supervisory control and responsibility.

8.2 The Level III shall not knowingly associate with, or permit the use of, his/her name or firm name in a business venture by any person or firm which he/she knows, or has reason to believe is engaging in business or professional practices of a fraudulent or dishonest nature.

8.3 The Level III shall conduct themselves in an honest and ethical manner. It is expected that Level III's observe all laws applicable to our business, including but not limited to international, federal, state/provincial, and local laws.

8.4 While this code addresses many ethical issues, it cannot address every issue that a Level III may encounter. As such, if a situation arises in which a Level III is unsure if an action would be deemed unethical, the Level III may consult ASNT Certification Services.

9.0 Unauthorized Practice

9.1 Any violation of this code shall be deemed to be an unauthorized practice and upon proper complaint, investigation, due process hearing and ruling of the Ethics Committee of the ASNT Certification Services Certification Management Committee in accordance with procedures heretofore established and published, sanctions may be applied to the individual(s) in violation.

9.1 If the applied sanction is suspension or revocation of certification, the certificate holder agrees to discontinue all claims of ASNT Certification Services certification and must return all certificates and wallet cards issued by ASNT Certification Services.

10.0 Rulings of Other Jurisdictions

Conviction of an NDT-related felony while ASNT Certification Services certification is valid or the revocation or suspension of a Professional Engineer's License by another jurisdiction or similar rulings by other professional associations may be grounds for a charge of violation of this Code.

I agree to abide by this Code of Ethics.

____

Name (Please print)

____

Signature

____

Date (mm/dd/yyyy)

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APPENDIX B

ASNT CERTIFICATION SERVICES NDT LEVEL III PROGRAM DOCUMENT

Foreword

This document establishes the requirements for the ASNT Certification Services (ASNT CS) NDT Level III Program.

1.0 Scope

1.1 The ASNT NDT Level III program provides third-party certification for nondestructive testing (NDT) or predictive maintenance (PdM) personnel whose specific jobs require appropriate knowledge of the technical principles underlying the tests they perform, witness, monitor, or evaluate. This document establishes the system for ASNT NDT Level III Certification in NDT and PdM. Certification under this system results in the issuance of an ASNT CS certificate (digital badging) attesting to the fact that the holder has met the published guidelines for the Basic and Method Examinations.

2.0 Terms and Definitions

2.1 **ASNT Certification Services Certification:** Written testimony of qualification whereby ASNT CS certifies that an individual has met the Basic and Method Examination guidelines.

2.2 **ASNT NDT Level III:** An individual who, having passed ASNT CS administered Basic and Method Examinations, holds a current, valid ASNT NDT Level III certification in at least one method.

2.3 **Certificate:** Document in the form of a letter, card, or other medium (e.g., digital badge), issued by ASNT CS under the provisions of this document, indicating that the named person has fulfilled the certification requirements.

2.4 **Employer:** The corporate, private, or public entity which employs personnel directly or indirectly for wages, salary, fees, or other considerations. This includes employers who obtain their qualified supplemental workforce personnel through third-party agencies, provided the use and certification of those supplemental employees is addressed in the employer's written practice.

2.5 **Examination-NDT Basic:** A Level III written examination covering knowledge of materials science and processes technology, NDT personnel qualification and certification, and the basic principles of NDT methods.

2.6 **Examination-PdM Basic:** A Level III written examination covering knowledge of machinery technology, PdM personnel qualification and certification, and the basic principles of PdM methods.

2.7 **Examination-Method:** A Level III written examination on the principles, theory, techniques, and applications within an NDT/PdM method.

2.8 **Experience:** Work activities accomplished in the applicable NDT/PdM method under the direction of qualified supervision, including the performance of the method and related activities but not including time spent in organized training programs.

2.9 **Qualification:** Demonstration or possession of skills, training, knowledge, and experience required for personnel to properly perform NDT to a level as specified in this document.

2.10 Recertification: The process of extending one's certification after the initial period of validity and maintaining certification for a specified period thereafter.

2.11 **Renewal:** The process of extending one's certification after the initial period of validity and maintaining certification for a specified period by application.

2.12 **SNT-TC-1A:** The NDT Personnel Qualification and Certification in Nondestructive Testing recommended practice providing the guidelines for the establishment of qualification and certification programs for NDT.

2.13 **Training:** An organized program developed to impart the knowledge and skills necessary for qualification.

3.0 Certification Outcome

3.1 An ASNT NDT Level III certificate holder shall have the skills and knowledge to establish techniques; to interpret codes, standards, and specifications; and to prepare or approve procedures and instructions.

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3.2 An ASNT NDT Level III shall also have general familiarity with other NDT methods, shall be capable of conducting or directing the training and examination of testing personnel in the methods for which the ASNT NDT Level III is qualified, and shall have knowledge of materials, fabrication, and product technology in order to establish techniques and to assist in establishing acceptance criteria when none are otherwise available.

3.3 An ASNT PdM Level III shall also have general familiarity with other PdM methods, shall be capable of conducting or directing the training and examination of testing personnel in the methods for which the ASNT NDT Level III is qualified, and shall have knowledge of applicable machinery technology in order to establish techniques and to assist in establishing acceptance criteria when none are otherwise available.

4.0 Eligibility for Examination

4.1 ASNT CS is committed to equal opportunity for all applicants in its certifications regardless of race, religion, national origin, marital status, sex, sexual orientation, gender identity, or disability where a person is otherwise qualified or could be with reasonable accommodation.

4.2 Candidates shall have a combination of education and experience in accordance with the guidelines in SNT-TC-1A.

4.3 When applying for examination(s), the candidate shall document the validity of the personal information requested, including education and experience needed to establish eligibility.

4.4 The candidate shall agree to abide by the code of ethics for ASNT NDT Level III personnel.

4.5 A maximum of three (3) retakes of any failed exams in a Method or the Basic Examination is permitted within a two (2) year period from the initial examination date.

4.5.1 A candidate who fails the initial Basic or Method Examination shall not be permitted to retake the examination for thirty (30) days from the date of failure.

4.5.2 A candidate who fails the Basic or Method Examination for a second time shall not be permitted to retake the examination for 90 days from the date of failure.

4.5.3 A candidate who fails the Basic or Method Examination for a third time shall not be permitted to retake the examination for one (1) year from the date of failure and must reapply as a new candidate. In no case, shall an individual be permitted to sit for an examination for the same method more than three (3) times in a two (2) year period.

5.0 Qualification Examinations

5.1 All ASNT NDT Level III Candidates must successfully complete both:

5.1.1 a Basic Examination, regardless of the method(s) for which the candidate seeks certification; and

5.1.2 a Method Examination, which is given at the candidate's request, in one or more of the different methods.

5.2 Examination Descriptions

5.2.1 NDT Basic Examination

5.2.1.1 This written examination consists of 135 questions that assess the candidate's knowledge of:

5.2.1.1.1 other NDT methods as required for ASNT NDT Level II personnel,

5.2.1.1.2 materials, fabrication, and product technology, and

5.2.1.1.3 qualification and certification according to ASNT SNT-TC-1A and ANSI/ASNT CP-189, latest edition.

5.2.2 Method Examination

5.2.2.1 This written examination consists of a minimum of 90 questions that assess the candidate's knowledge and application of fundamentals, principles, and techniques for that method in which certification is sought. The Method Examination may include a portion that examines the candidate's ability to comprehend a specification and apply its requirements.

5.3 Employer Responsibility

5.3.1 Additional examinations required by the employer's program document should be administered by the employer.

5.4 Examination Grading

5.4.1 The grading of certification examinations shall be done by the ASNT CS Certification Department in accordance with nationally accepted psychometric principles.

6.0 Examination Results

6.1 Candidates who successfully pass all required examinations for ASNT CS certification shall be issued certification documents as described in the Certification Section.

6.2 Notification of result(s) will be sent to each candidate.

6.3 Candidates who do not pass all required examinations for ASNT CS certification may reapply to retake the failed examinations in accordance with paragraph 4.5.

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7.0 Certification

7.1 Based on successful completion of the necessary examinations, ASNT CS will issue the candidate an ASNT CS certificate.

7.2 ASNT CS certification attests to an individual having satisfied the requirements for the Basic and Method Examinations as detailed in SNT-TC-1A. This does not constitute license or authorization to perform NDT, as the employer has the sole responsibility for authorizing employees to perform NDT. The employer, through a Level III or other designated person as denoted in the employer's written practice, should review the individual's qualification records for satisfactory completeness prior to authorizing the individual to perform NDT. These records shall be retained as specified in SNT-TC-1A.

7.3 ASNT NDT Level III certificates remain the property of ASNT CS and shall be surrendered to ASNT CS on demand.

8.0 Validity

8.1 Initial certification shall be valid for a period of five (5) years, with the certification period starting on the date that a candidate successfully passes both the Basic and Method Examinations.

8.1.1 When the Basic Examination is passed, that result shall remain valid for a period of five (5) years from the examination date. If the Method Examination is not passed within that period, the Basic Examination must be retaken.

8.1.2 When the Method Examination is passed, that result shall remain valid for a period of two (2) years from the examination date. If the Basic Examination is not passed within that period, the Method Examination must be retaken.

8.2 Certification shall be revoked if ASNT CS determines that an ASNT CS certificate holder has violated the ASNT CS NDT Level III Code of Ethics.

9.0 Renewal and Recertification

9.1 Recertification is required at five (5)-year intervals to extend certification beyond the initial or previous validity period.

9.2 ASNT NDT Level III certifications may be renewed by one (1) of the following methods:

9.2.1 Renewal by application

9.2.1.1. Candidates may submit a renewal application and fees to ASNT CS between 180 and 60 days prior to the certification expiration date. The application shall include the employer's signed attestation that the applicant has satisfactorily performed Level III NDT functions during the current certification period without significant interruption.

9.2.1.2 By application through points, demonstrating a combination of active employment and continuing participation/ education in NDT/PdM.

9.2.2 Recertification by examination

9.2.2.1 The applicant must pass an ASNT NDT Level III examination for each test method. The recertification examinations must be successfully passed prior to the current certification expiration date, or the certificate holder must reapply as an initial applicant. Personnel who fail a recertification examination may not renew by application.

9.2.3 By examination, prior to expiration of certification in the applicable method. The Basic Examination does not have to be taken again as long as certification is continuously maintained.

10.0 Applicant Rights

10.1 An appeals process exists for the resolution of appeals, complaints, and disputes received from candidates, certified persons, their employers, and other parties regarding the certification process, qualification criteria, or the performance of certified persons.

10.2 Information gained in the course of the certification process shall not be disclosed to any third party except as required by law.

11.0 Program Changes

11.1 Changes to the ASNT NDT Level III Program will be posted on the ASNT CS website asntcertification.org.

12.0 Accommodation for Disabilities

12.1 ASNT CS will provide the appropriate accommodations for persons with documented disabilities. Candidates should contact ASNT CS prior to examination dates to arrange special accommodations.

13.0 Revision History

Revision	Date	Changes
00 through 04		Information not recorded before 08-13-2024
05	08-13-2024	LLC changes made along with bringing the document to the current practice

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APPENDIX C

MEASUREMENT UNITS FOR NONDESTRUCTIVE TESTING

In 1960, the General Conference on Weights and Measures established the International System of Units (SI). SI units are designed so that a single set of measurement units can be used by all branches of science, engineering, and the general public. SI units are the modern version of the metric system and end the division between metric units used by scientists and metric units used by engineers and the public. For example, scientists have given up their units based on centimeter and gram, and engineers have abandoned the kilogram-force in favor of the newton. Electrical engineers have retained the ampere, volt, and ohm, but changed all units related to magnetism.

Table 1 lists the seven base SI units. Table 2 lists derived units with special names. In SI units, the unit of time is the second (s); the hour (h) is also recognized for common use. Traditional or imperial units should not be used in science and engineering. Table 3 gives some conversions to SI units.

In science and engineering, very large or very small numbers with units are expressed by using multipliers, prefixes of 10^3 or 10^{-3} intervals (Table 4). The multiplier becomes a part of the SI symbol. For example, a millimeter (mm) is 0.001 meter (m). The volume unit cubic centimeter (cm^3) is $(0.01\ m)^3$. Unit submultiples such as the centi, deci, and hecto are less common in scientific and technical uses of SI units because of their variance from the convenient 10^3 or 10^{-3} intervals that make equations easy to manipulate.

In SI, the distinction between upper and lower case letters is meaningful and should be observed. For example, the meanings of the prefix m (milli) and the prefix M (mega) differ by nine orders of magnitude.

For more information, the reader is referred to the information available through national standards organizations and specialized information compiled by technical organizations.

TABLE 1

SI base units

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

TABLE 2

SI derived units with special names[a]

Quantity	Units	Symbol	Relation to Other SI Units[b]
Capacitance	farad	F	$C \cdot V^{-1}$
Catalytic activity	katal	kat	$s^{-1} \cdot mol$
Conductance	siemens	S	$A \cdot V^{-1}$
Energy	joule	J	$N \cdot m$
Frequency (periodic)	hertz	Hz	$1 \cdot s^{-1}$
Force	newton	N	$kg \cdot m \cdot s^{-2}$
Inductance	henry	H	$Wb \cdot A^{-1}$
Illuminance	lux	lx	$lm \cdot m^{-2}$
Luminous flux	lumen	lm	$cd \cdot sr$
Electric charge	coulomb	C	$A \cdot s$
Electric potential[c]	volt	V	$W \cdot A^{-1}$
Electric resistance	ohm	W	$V \cdot A^{-1}$
Magnetic flux	weber	Wb	$V \cdot s$
Magnetic flux density	tesla	T	$Wb \cdot m^{-2}$
Plane angle	radian	rad	1
Power	watt	W	$J \cdot s^{-1}$
Pressure (stress)	pascal	Pa	$N \cdot m^{-2}$
Radiation absorbed dose	gray	Gy	$J \cdot kg^{-1}$
Radiation dose equivalent	sievert	Sv	$J \cdot kg^{-1}$
Radioactivity	becquerel	Bq	$1 \cdot s^{-1}$
Solid angle	steradian	sr	1
Temperature	degree Celsius	°C	K
Time[a]	hour	h	3600 s
Volume[a]	liter	L	dm^3

a. Hour and liter are not SI units but are accepted for use with SI.
b. Number one (1) expresses a dimensionless relationship.
c. Electromotive force.

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TABLE 3

Examples of conversions to SI units.

Quantity	Measurement in Non-SI Unit	Multiply by	To Get Measurement in SI Unit
Angle	minute (min)	$2.908\,882 \times 10^{-4}$	radian (rad)
	degree (deg)	$1.745\,329 \times 10^{-2}$	radian (rad)
Area	square inch (in.^{2})	645	square millimeter (mm^2)
Distance	angstrom (Å)	0.1	nanometer (nm)
	inch (in.)	25.4	millimeter (mm)
Energy	British thermal unit (BTU)	1.055	kilojoule (kJ)
	calorie (cal), thermochemical	4.184	joule (J)
Power	British thermal unit per hour (BTU · h^{-1})	0.293	watt (W)
Specific heat	British thermal unit per pound degree Fahrenheit (BTU·lb$_m$$^{-1}$ · °F^{-1})	4.19	kilojoule per kilogram per kelvin (kJ·kg^{-1} · K^{-1})
Force	pound force	4.448	newton (N)
Torque (couple)	foot-pound (ft - lbf)	1.36	newton meter (N·m)
Pressure	pound force per square inch (lbf · in.$^{-2}$)	6.89	kilopascal (kPa)
Frequency (cycle)	cycle per minute	60^{-1}	hertz (Hz)
Illuminance	footcandle (ftc)	10.76	lux (lx)
	phot (ph)	10 000	lux (lx)
Luminance	candela per square foot (cd · ft^{-2})	10.76	candela per square meter (cd · m^{-2})
	candela per square inch (cd · in.$^{-2}$)	$1.550\,003 \times 10^{-3}$	candela per square meter (cd · m^{-2})
	footlambert (ftl)	3.426	candela per square meter (cd · m^{-2})
	lambert	$3.183\,099 \times 10^{-3}$	candela per square meter (cd · m^{-2})
	nit (nt)	1	candela per square meter (cd · m^{-2})
	stilb (sb)	10 000	candela per square meter (cd · m^{-2})
Radioactivity	curie (Ci)	37	gigabecquerel (GBq)
Ionizing radiation exposure	roentgen (R)	0.258	millicoulomb per kilogram (mC · kg^{-1})
Mass	pound (lb$_m$)	0.454	kilogram (kg)
Temperature (increment)	degree Fahrenheit (°F)	0.556	kelvin (K) or degree Celsius (°C)
Temperature (scale)	degree Fahrenheit (°F)	(°F – 32) ÷ 1.8	degree Celsius (°C)
Temperature (scale)	degree Fahrenheit (°F)	(°F – 32) ÷ 1.8 + 273.15	kelvin (K)

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TABLE 4

SI prefixes and multipliers

Prefix	Symbol	Multiplier
yotta	Y	10^{24}
zetta	Z	10^{21}
exa	E	10^{18}
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
hecto[a]	h	10^{2}
deka[a]	da	10
deci[a]	d	10^{-1}
centi[a]	c	10^{-2}
milli	m	10^{-3}
micro	µ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}
zepto	z	10^{-21}
yocto	y	10^{-24}

a. Avoid these prefixes (except in dm^3 and cm^3) for science and engineering.

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ASNT LEVEL III STUDY GUIDE

BASIC

FIFTH EDITION

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